

IMPERIAL

ADVANCED CLASSICAL PHYSICS

IMPERIAL COLLEGE LONDON

DEPARTMENT OF PHYSICS

Comprehensive Lecture Summaries on Advanced Classical Physics

Based on Lectures by Prof. Andrew Tolley

Author:

Eugene (Yoogeun) Song

Date: May 4, 2025

These summaries draw upon the lecturer's outstanding presentations and are further enhanced by my own insights and carefully curated materials from respected sources, including:

- Classical Mechanics by Herbert Goldstein, Charles P. Poole, and John L. Safko

I hope these compiled summaries prove beneficial and insightful for students who aspire to learn advanced classical physics.

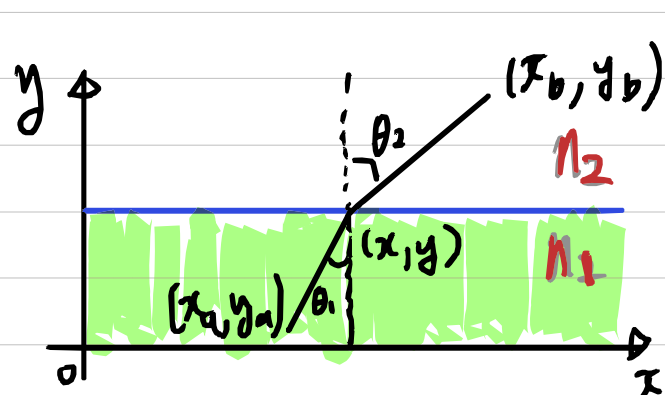
– Yoogeun Song.

§1. Lagrangian Mech.

§1.1. Action Principle

§1.1.1. Fermat's principle

"Principle of least action"



Snell's law of refraction
for light passing thru
media of diff. indices
of refraction.

We consider a lgt ray from pt. (x_a, y_a)
to (x_b, y_b) , the optical path length follows

$$T(x) = \frac{n_1}{c} \sqrt{(x_a - x)^2 + (y_a - y)^2} + \frac{n_2}{c} \sqrt{(x_b - x)^2 + (y_b - y)^2}$$

$$\begin{aligned} \frac{\partial T(x)}{\partial x} &= \frac{n_1}{c} \frac{x - x_a}{\sqrt{(x_a - x)^2 + (y_a - y)^2}} + \frac{n_2}{c} \frac{x - x_b}{\sqrt{(x_b - x)^2 + (y_b - y)^2}} \\ &= -\frac{n_1}{c} \sin \theta_1 + \frac{n_2}{c} \sin \theta_2 = 0 \end{aligned}$$

$$\hookrightarrow \frac{\sin \theta_2}{\sin \theta_1} = \frac{n_1}{n_2}$$

What happens if the refractive index is a fn.
of $n(x, y)$. Fermat's action

$$T = \int dT = \int \frac{n(x, y)}{c} \sqrt{dx^2 + dy^2}$$

$$\underline{c}(\lambda) = (x(\lambda), y(\lambda))$$

$$\text{s.t. } \begin{aligned} x(\lambda_a) &= x_a, & x(\lambda_b) &= x_b \\ y(\lambda_a) &= y_a, & y(\lambda_b) &= y_b \end{aligned}$$

$\therefore$ The Fermat's action:

$$T = \int_{\lambda_a}^{\lambda_b} d\lambda \frac{n(x(\lambda), y(\lambda))}{c} \sqrt{\left(\frac{dx}{d\lambda}\right)^2 + \left(\frac{dy}{d\lambda}\right)^2}$$

§1.1.2. E-LEs

In configuration space (the space of coords
& vel.s), the action S is defined as an
integral over a fn. $L(x, \dot{x}, t)$, known as
the Lagrangian, as

$$S[x(t)] = \int_{t_a}^{t_b} dt L(x(t), \dot{x}(t), t)$$

The Lagrangian $\rightarrow$ simply the difference of the
KE T & the PE V , i.e.,

$$L = T - V.$$

The actual physical trajectory is the fn. x
that minimizes the action subj to B.C.s

$$x(t_a) = x_a \quad \& \quad x(t_b) = x_b.$$

$$\delta S[x(t)] = \int_{t_a}^{t_b} dt \left[\frac{\partial L}{\partial x} \delta x(t) + \frac{\partial L}{\partial \dot{x}} \delta \dot{x}(t) \right]$$

$$\frac{\partial L}{\partial x} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) = 0 \quad \text{E-L-E}$$

Goldstein

(2)

#2. Min. surface of revol.

$$2\pi x ds = 2\pi x \sqrt{1+\dot{y}^2} dx$$

$$2\pi \int_1^2 dx x \sqrt{1+\dot{y}^2}$$

$$f = x \sqrt{1+\dot{y}^2}$$

$$\frac{\partial f}{\partial y} = 0 \quad \& \quad \frac{\partial f}{\partial \dot{y}} = \frac{x \dot{y}}{\sqrt{1+\dot{y}^2}}$$

$$\frac{d}{dx} \left(\frac{x \dot{y}}{\sqrt{1+\dot{y}^2}} \right) = 0 = \frac{d}{dx} \left(\frac{\partial f}{\partial \dot{y}} \right)$$

$$\frac{x \dot{y}}{\sqrt{1+\dot{y}^2}} = \text{const.} = \frac{\partial f}{\partial \dot{y}} = C_1$$

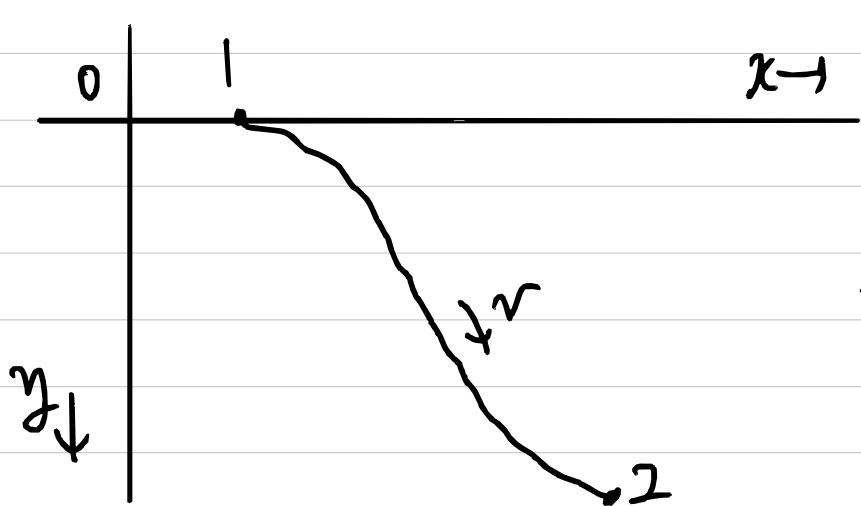
$$\dot{y}^2 (x^2 - C_1^2) = C_1^2 \rightarrow \frac{dy}{dx} = \frac{C_1}{\sqrt{x^2 - C_1^2}}$$

y =

$$\int dy = \int \frac{C_1}{\sqrt{x^2 - C_1^2}} dx + C_2 = C_1 \cosh^{-1} \frac{x}{C_1} + C_2$$

$$x = C_1 \cosh \frac{y - C_2}{C_1}$$

#3. Brachistochrone prb.



$$t_{12} = \int_1^2 \frac{ds}{v}$$

$$KE = PE \rightarrow \frac{1}{2} mv^2 = mgy$$

E conservation

$$v = \sqrt{2gy}$$

$$t_{12} = \int_1^2 \sqrt{\frac{1+\dot{y}^2}{2gy}} dx$$

$$ds = \sqrt{dx^2 + dy^2} = \sqrt{1+y'^2} dx$$

$$\frac{y}{a} = 1 - \cos \left[\frac{x + \sqrt{y(2a-y)}}{a} \right]$$

$$\frac{dF}{dx} = \cancel{\frac{\partial F}{\partial x}} + \frac{\partial F}{\partial y} \frac{dy}{dx} + \frac{\partial F}{\partial y'} \frac{dy'}{dx}$$

$$= \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \frac{dy}{dx} + \frac{\partial F}{\partial y'} \frac{dy'}{dx}$$

$$= \frac{d}{dx} \left(\frac{\partial F}{\partial y'} y' \right)$$

$$\frac{d}{dx} \left(\frac{\partial F}{\partial y'} y' - F \right) = 0$$

$$\frac{\partial}{\partial y'} \left(\sqrt{\frac{1+y'^2}{2gy}} \right) y' - \sqrt{\frac{1+y'^2}{2gy}} = C$$

$$\cancel{\frac{1}{\sqrt{2gy}}} \frac{y'}{\sqrt{1+y'^2}} y' - \sqrt{\frac{1+y'^2}{2gy}} = C \sqrt{2gy}$$

$$\cancel{y'^2} - (1+y'^2) = C \sqrt{1+y'^2} \sqrt{2gy}$$

$$= -1 = C \sqrt{1+y'^2} \sqrt{2gy}$$

sqring

$$1 = C^2 (1+y'^2) (2gy)$$

$$1+y'^2 = \frac{1}{2gy C^2}$$

$$y' = \frac{1}{2gy C^2} - 1$$

(3)

$$\frac{dy}{dx} = \sqrt{\frac{1-2gyC^2}{2gyC^2}} \rightarrow \sqrt{\frac{2gyC^2}{1-2gyC^2}} dy = dx$$

$$\int \sqrt{\frac{2gyC^2}{1-2gyC^2}} dy = \int dx$$

$$\text{Let } y = a \sin^2 \theta \rightarrow a = \frac{1}{2gC^2}$$

$$dy = 2a \sin \theta \cos \theta d\theta$$

$$a = \frac{1}{2gC^2} \rightarrow 2ga = \frac{1}{C^2}$$

$$\rightarrow 2gyC^2 = \sin^2 \theta$$

$$1-2gyC^2 = \cos^2 \theta$$

$$\frac{dx}{dy} = \sqrt{\frac{\cos^2 \theta}{\sin^2 \theta}} = \frac{\cos \theta}{\sin \theta}$$

$$dx = \frac{\cos \theta}{\sin \theta} dy = \frac{\cos \theta}{\sin \theta} 2a \sin \theta \cos \theta d\theta$$

$$= 2a \cos^2 \theta d\theta \rightarrow \text{Use } \cos^2 \theta = \frac{1+\cos(2\theta)}{2}$$

$$x = \int 2a \frac{1+\cos(2\theta)}{2} d\theta$$

$$= a \int (1+\cos(2\theta)) d\theta$$

$$= a\theta + \frac{a}{2} \sin 2\theta + \cancel{b} \rightarrow 0$$

$$\text{Since } y = a \sin^2 \theta$$

$$\rightarrow \sin \theta = \sqrt{\frac{y}{a}} \quad \& \quad \theta = \arcsin\left(\sqrt{\frac{y}{a}}\right)$$

$$\& \sin(2\theta) = 2 \sin \theta \cos \theta$$

$$\& \cos \theta = \sqrt{1-\frac{y}{a}}$$

$$\sin(2\theta) = 2 \sqrt{\frac{y}{a}} \sqrt{1-\frac{y}{a}} = 2 \frac{\sqrt{y(a-y)}}{a}$$

$$\frac{x}{a} = \arcsin\left(\sqrt{\frac{y}{a}}\right) + \frac{\sqrt{y(a-y)}}{a}$$

$$\cos(2\theta) = 1-2\sin^2 \theta$$

$$\sin^2 \theta = \frac{1-\cos(2\theta)}{2}$$

(4)

$$\int \sqrt{\frac{2gy c^2}{1-2gy c^2}} dy = \int dx$$

Let $y = A \cosh^2 \theta$. & $A = \frac{1}{2gc^2}$

$$dy = 2A \cosh \theta \sinh \theta d\theta$$

$$\begin{aligned} 2gc^2 y &= \cosh^2 \theta \\ &= \frac{y}{A} \end{aligned}$$

$$\int \frac{\sqrt{\frac{A-y}{A}}}{\sqrt{\frac{y}{A}}} dy = \int \sqrt{\frac{A-y}{y}} dy$$

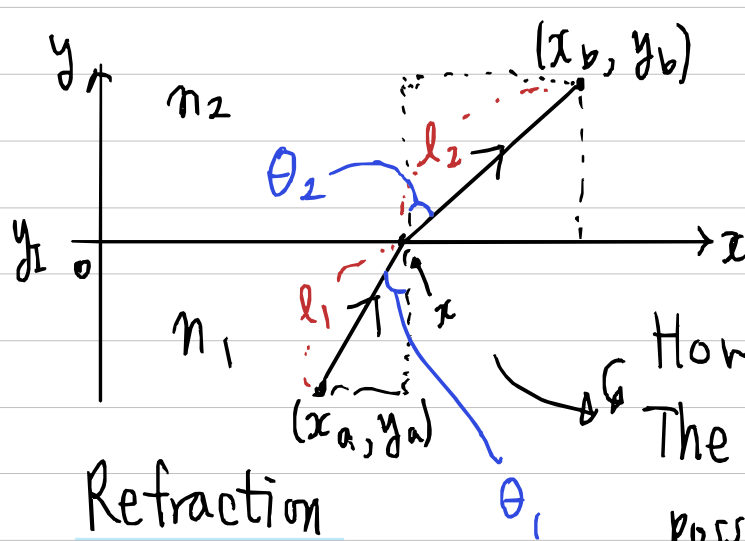
$$dy = 2A \cosh \theta \sinh \theta d\theta$$

$$A - y = A - A \cosh^2 \theta = A \sinh^2 \theta$$

$$\int \sqrt{\frac{A}{A}} \frac{\sinh^2 \theta}{\cosh^2 \theta} (2A \cosh \theta \sinh \theta d\theta)$$

$$= \int 2A \sinh^2 \theta d\theta$$

$$= x$$

Fermat's Principle of least time.

$$v = \frac{c}{n} = \text{phase vel.}$$

$n = \text{refrac. index}$

How light travels?

The time taken is the smallest possible.

- Fermat's principle

$$t = \frac{d}{v} = \frac{n d}{c}$$

$$l_1 = \sqrt{(x - x_a)^2 + (y_I - y_a)^2}$$

$$l_2 = \sqrt{(x_b - x)^2 + (y_I - y_b)^2}$$

$$T(x) = \frac{n_1 \sqrt{(x - x_a)^2 + (y_I - y_a)^2}}{c} + \frac{n_2 \sqrt{(x_b - x)^2 + (y_I - y_b)^2}}{c}$$

Find minimum time,

$$\frac{\partial T(x)}{\partial x} = 0$$

$$\frac{\partial T(x)}{\partial x} = \frac{n_1}{c} \frac{(x - x_a)}{\sqrt{(x - x_a)^2 + (y_I - y_a)^2}} - \frac{n_2}{c} \frac{x_b - x}{\sqrt{(x - x_b)^2 + (y_I - y_b)^2}}$$

= 0 MINIMUM TIME

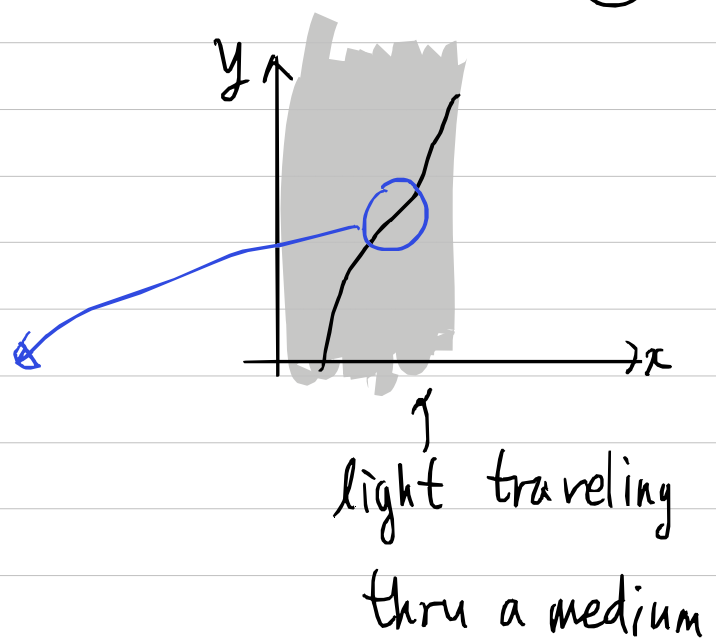
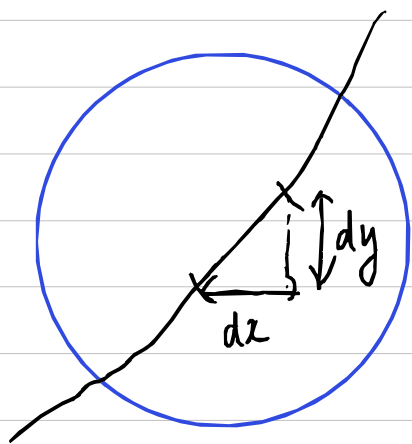
$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

Snell's Law

$$\sin \theta_2 = \frac{x_b - x}{l_2}$$

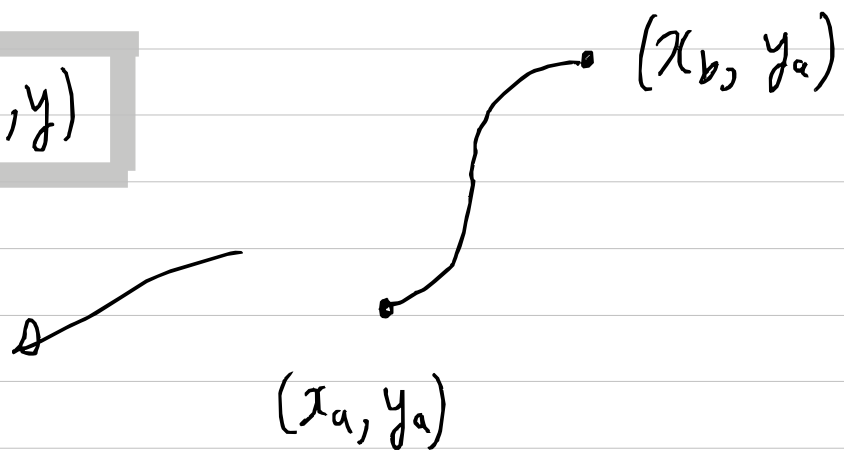
$$\sin \theta_1 = \frac{x - x_a}{l_1}$$

(2)

 $D=2$ $n(x, y)$ 

$$dl^2 = dx^2 + dy^2$$

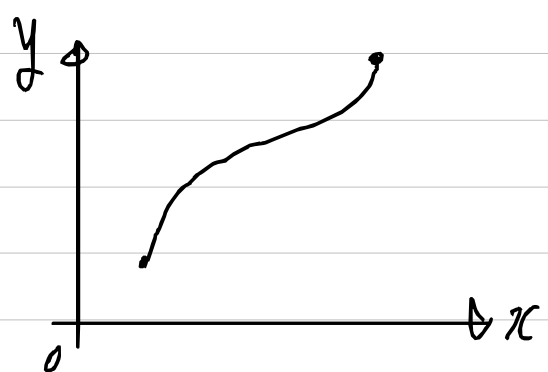
$$dT = \frac{dl}{c} n(x, y)$$



$$T = \int_{t_a}^{t_b} dT = \int \frac{dl}{c} n(x, y) = \int \frac{1}{c} n(x, y) \sqrt{dx^2 + dy^2}$$

★ Parameterized Curve

$x(\lambda), y(\lambda)$



(EX1) $y = x^3 + x^2$

$$x(\lambda) = \lambda + 3 \quad \& \quad y(\lambda) = (\lambda + 3)^3 + (\lambda + 2)^3$$

(EX2) $x(\lambda) = \lambda \rightarrow y(\lambda) = \lambda^3 + \lambda^2$

Then,

$$dx = \frac{dx}{d\lambda} d\lambda \quad \& \quad dy = \frac{dy}{d\lambda} d\lambda$$

$$ds = \sqrt{dx^2 + dy^2} = \sqrt{\left(\frac{dx}{d\lambda}\right)^2 (d\lambda)^2 + \left(\frac{dy}{d\lambda}\right)^2 (d\lambda)^2} = \sqrt{\left(\frac{dx}{d\lambda}\right)^2 + \left(\frac{dy}{d\lambda}\right)^2} d\lambda$$

$$T = \int_{\lambda_a}^{\lambda_b} d\lambda \frac{n(x, y)}{c} \sqrt{\left(\frac{dx}{d\lambda}\right)^2 + \left(\frac{dy}{d\lambda}\right)^2}$$

$$T = \int_{\lambda'_a}^{\lambda'_b} d\lambda' \frac{n(x, y)}{c} \sqrt{\left(\frac{dx}{d\lambda'}\right)^2 + \left(\frac{dy}{d\lambda'}\right)^2}$$

$$\lambda \rightarrow f(\lambda) = \lambda'$$

Gauge Symmetry $\rightarrow$

Reparameterization
Invariance

In GR, diffeomorphism invariance.

★(EX)

Choose $\lambda = y$.

"Gauge choice"

$$T = \int_{y_a}^{y_b} dy \frac{n(x, y)}{c} \sqrt{\left(\frac{dx}{dy}\right)^2 + 1}$$

↑
we chose $x = x(y)$

* (EX) Choose $\lambda = x$.

(3)

$$T = \int_{x_a}^{x_b} dx \frac{n(x, y)}{c} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$\lambda = x + y^2$$

* (EX) Choose $\lambda = t$.

$$x(t), y(t)$$

$$T = \int_{t_1}^{t_2} dt \frac{n(x, y)}{c} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = t_2 - t_1$$

Specify factor $x(y)$

→ Functional $T[x(y)]$

Find the fn. $x(y)$ for which $T[x(y)]$ is an extremum.

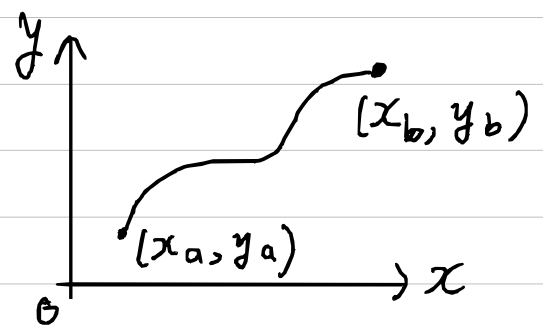
$$T[x(y) + \delta x(y)] \approx T[x(y)]$$

$$f(x + \varepsilon) \approx f(x) \approx f(x) + \underbrace{\varepsilon f'(x)}_{=0} + \mathcal{O}(\varepsilon^2)$$

$$f'(x) = 0$$

* Fermat Prb

Minimize/Extremize



$$T = \int_{y_a}^{y_b} dy \frac{n(x, y)}{c} \sqrt{1 + \left(\frac{dx}{dy}\right)^2} \quad (\lambda = y)$$

* Action $S = \int p dx$

§1.13. E-L E

Time
Switch $T \Rightarrow S$ "action"

$y \Rightarrow t$

$$S = \int_{t_a}^{t_b} dt L$$

"Lagrangian"

$$L\left(x, \frac{dx}{dt}, \frac{d^2x}{dt^2}, \frac{d^3x}{dt^3}, \dots\right)$$

disastrous instabilities

$$L\left(x(t), \frac{dx}{dt}(t)\right)$$

$$S = \int_{t_a}^{t_b} dt L(x(t), \dot{x}(t))$$

$$\left(\dot{x} = \frac{dx}{dt}\right)$$

$$E = -\frac{1}{2} \dot{x}^2$$

* Principle of least action
(extremized action)

• Variation of action.

$$\begin{aligned} \delta S &= S[x + \delta x] - S[x] \\ &= S[x(t) + \delta x(t)] - S[x(t)] \end{aligned}$$

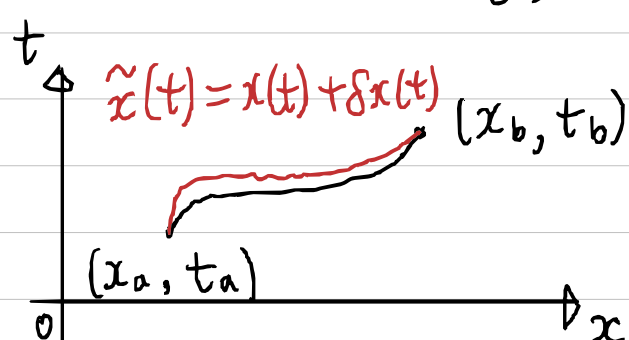


$$\frac{\partial f(x)}{\partial x} = 0$$

MIN & MAX $\rightarrow$ deriv. ≈ 0

• Fermat $\underline{f(x + \delta x) \approx f(x)}$

$$\begin{aligned} &= f(x) + \delta x \frac{\partial f}{\partial x} + \mathcal{O}(\delta x^2) \\ &\approx f(x) \end{aligned}$$



$$S[x(t)] = \int_{t_a}^{t_b} dt L(x(t), \dot{x}(t))$$

(2)

$$S[x(t) + \delta x(t)] = \int_{t_a}^{t_b} dt \, L(x(t) + \delta x(t), \dot{x}(t) + \delta \dot{x}(t))$$

$$0 = \delta f = f(x + \delta x) - f(x)$$

$$\delta S = \int_{t_a}^{t_b} dt \left[L(x(t) + \delta x(t), \dot{x}(t) + \delta \dot{x}(t)) - L(x(t), \dot{x}(t)) \right]$$

~~$+ O(\delta^2)$~~
perturbed

$$\stackrel{\text{Taylor Expansion}}{=} \int_{t_a}^{t_b} dt \left[\frac{\partial L}{\partial x} \delta x(t) + \frac{\partial L}{\partial \dot{x}} \delta \dot{x}(t) \right]$$

Taylor Expansion

$$L(x(t) + \delta x(t), \dot{x}(t) + \delta \dot{x}(t))$$

$$= L(x(t), \dot{x}(t)) + \delta x \frac{\partial L}{\partial x} + \delta \dot{x} \frac{\partial L}{\partial \dot{x}} + \cancel{O(\delta^2)}$$

Integration by parts for the 2nd term:

$$\int_{t_a}^{t_b} \delta \dot{x}(t) \frac{\partial L}{\partial \dot{x}} dt = \int_{t_a}^{t_b} \frac{\partial L}{\partial \dot{x}} \frac{d}{dt} (\delta x(t)) dt$$

$$\left(\int u dv = uv - \int v du \right) \text{ w/ } u = \frac{\partial L}{\partial \dot{x}} \text{ \& } v = \delta x(t)$$

$$= \left[\frac{\partial L}{\partial \dot{x}} \delta x(t) \right]_{t_a}^{t_b} - \int_{t_a}^{t_b} \delta x(t) \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) dt$$

$$\delta x(t_b) - \delta x(t_a) = 0$$

Boundary term $\rightarrow 0$

* The infinitesimal part $\rightarrow$ We're imagining δx as infinitesimal.
(δx)

(3)

* Principle of least action $\delta S = 0$

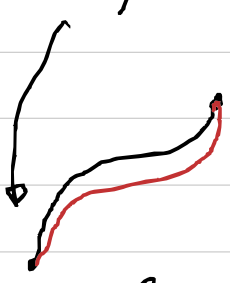
$\delta S = 0$ regardless of the infinitesimal variation.

$\delta S = 0$ even when $\delta x \neq 0$.

$$\delta S = \underbrace{\left[\delta x \frac{\partial L}{\partial \dot{x}} \right]_{t_a}^{t_b}}_{\text{"Boundary" term}} + \underbrace{\int_{t_a}^{t_b} dt \left[\delta x \left(\frac{\partial L}{\partial x} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) \right) \right]}_{\text{"Bulk" contribution}}$$

Demanding $\left[\delta x \frac{\partial L}{\partial \dot{x}} \right]_{t_a}^{t_b} = 0 \rightarrow$ Easy
 $\hookrightarrow$ a constraint we impose.

The paths start & end at the same (x_a, t_a) & (x_b, t_b) , respectively.



$$\therefore \delta x(t_a) = \delta x(t_b) = 0$$

$$\delta S = \int_{t_a}^{t_b} dt \delta x(t) \left[\frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} \right] = 0$$

for any $\delta x(t) = 0$ at $t = t_a, t_b$.

For δS to be 0, $\frac{\partial L}{\partial x} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) = 0 : E-L E$

Only needed

1. Taylor Expansion

2. Integrating by parts

The E-L E determines the trajectories or paths, which extremize the action.

In the path integral formulation,

the extrema of the action implies "constructive interference" $\rightarrow$ a phase factor $e^{iS/\hbar}$.

"Constructive interference" occurs when multiple paths have similar action values, meaning their phase factors $e^{iS/\hbar}$ are nearly the same.

∫ Their contributions add up, reinforcing each other.

(4)

§1.1.5. Back to Fermat

$$t \rightarrow y \quad \& \quad S \rightarrow T$$

$$L\left(x, \frac{dx}{dy}\right) = \frac{n(x(y), y)}{c} \sqrt{1 + \left(\frac{d(x(y))}{dy}\right)^2}$$

$$\frac{d}{dy} \left(\frac{\partial L}{\partial \left(\frac{dx}{dy}\right)} \right) = \frac{\partial L}{\partial x} \quad (t \leftrightarrow y)$$

$$\frac{d}{dy} \left(\frac{n(x(y), y) \frac{dx}{dy}}{c \sqrt{1 + \left(\frac{dx}{dy}\right)^2}} \right) = \frac{1}{c} \frac{\partial n(x(y), y)}{\partial x} \sqrt{1 + \left(\frac{dx}{dy}\right)^2}$$

* A non-rel. particle in 1-sp. dim.

$$L(x, \dot{x}) = T - V = \frac{1}{2} m \dot{x}^2 - V(x)$$

$$\bullet \text{ E-L E: } \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) = \frac{\partial L}{\partial x}$$

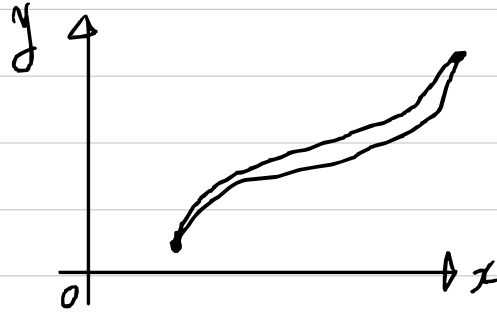
Newton's 2nd Law

$$\frac{dp}{dt} = \frac{d}{dt} (m\dot{x}) = - \frac{\partial V}{\partial x} \quad \circ (NII)$$

$$F = - \frac{\partial V}{\partial x}$$

* Hamilton Principle / Principle of least action

$$\delta S = 0$$



* General CM.

• State q_i ($i=1, \dots, N$)

$$\{q_i\} = q_1, q_2, \dots, q_N$$

• Dynamics $q_i(t)$

• Local / Causal $\rightarrow$ locality & causality

$$S = \int dt \quad L(q_i(t), \dot{q}_i(t))$$

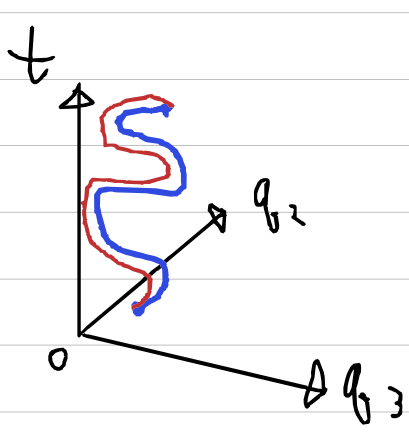
• Non-Local theory

$$S = \int dt \quad L(q_i(t), \dot{q}_i(t-3s))$$

3 seconds ago,
i.e., the past.

$$= \int dt \int dt' \quad L(q_i(t), q_i(t'), \dots)$$

$$\{q_i(t)\} = \{q_1(t), q_2(t), \dots, q_N(t)\}$$

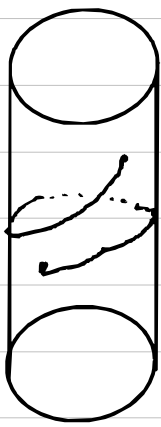


$$q_i(t) + \delta q_i(t)$$

$$= \{q_1(t) + \delta q_1(t), q_2(t) + \delta q_2(t), \dots, q_N(t) + \delta q_N(t)\}$$

$(i=1, \dots, N)$

(EX)



Where there are many extrema,

QM'ally, the correct ans. is the

linear superposition of the wv. fn.s associated.

* We vary the action:

$$\delta S = S[q_i(t) + \delta q_i(t)] - S[q_i(t)]$$

$$\delta S = \int_{t_1}^{t_2} dt \left[L(q_i(t) + \delta q_i(t), \dot{q}_i(t) + \delta \dot{q}_i(t)) - L(q_i(t), \dot{q}_i(t)) \right]$$



Taylor Expansion

$$L(q_i + \delta q_i, \dot{q}_i + \delta \dot{q}_i) = L(q_i, \dot{q}_i)$$

$$+ \sum_{i=1}^N \delta q_i \frac{\partial L}{\partial q_i} + \sum_{i=1}^N \delta \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} + O(\delta^2)$$

$$L(q_1, q_2, \dots, q_N, \dot{q}_1, \dot{q}_2, \dots, \dot{q}_N)$$

$$\delta S = \int_{t_1}^{t_2} dt \left(\sum_{i=1}^N \delta q_i \frac{\partial L}{\partial q_i} + \sum_{i=1}^N \delta \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} \right)$$

* I b P.

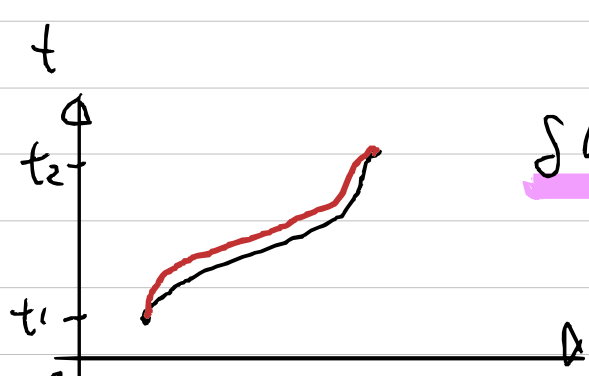
$$\int_{t_1}^{t_2} dt \left(\frac{du}{dt} v \right) = \left[uv \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} dt \left(u \frac{dv}{dt} \right)$$

(2)

$$\delta \dot{q}_i = \frac{d}{dt} (\delta q_i)$$

$$\delta S = \int_{t_1}^{t_2} \left(\sum_{i=1}^N \delta q_i \left(\frac{\partial L}{\partial q_i} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) \right) \right) dt$$

$$+ \left[\sum_{i=1}^N \delta q_i \frac{\partial L}{\partial \dot{q}_i} \right]_{t_1}^{t_2}$$



$$\delta q_i(t_1) = \delta q_i(t_2) = 0$$

$\delta S = 0$ for any $\delta q_i(t)$ satisfy

$$\delta q_i(t_1) = \delta q_i(t_2) = 0,$$

For each individual i , the only way to δS be always zero is if $\left(\frac{\partial L}{\partial q_i} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) \right) = 0$,

for each individual δq_i .

$$\therefore \frac{\partial L}{\partial q_i} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) = 0 \quad \forall i = 1, \dots, N$$

E-L E

q_i : Generalized Coord.s

& $p_i = \frac{\partial L}{\partial \dot{q}_i}$: Generalized momenta

$p_i = \begin{cases} \vec{p} & \text{for Cartesian coord.s.} \\ \vec{L} & \text{for Angular coord.s.} \end{cases}$

$$\frac{dp_i}{dt} = \frac{\partial L}{\partial q_i} \quad : \text{N I L} \quad (\text{Newton's 2nd Law})$$

$$q_i \rightarrow q_i' = q_i'(q_1, \dots, q_n)$$

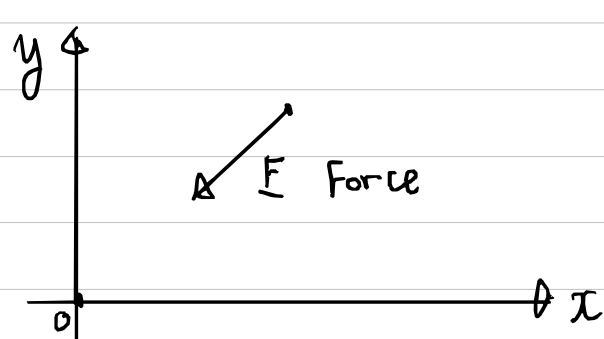
$$L(q_i, \dot{q}_i) = L'(q_i', \dot{q}_i')$$

$$\frac{\partial L^i}{\partial q_i'} - \frac{d}{dt} \left(\frac{\partial L^i}{\partial \dot{q}_i'} \right) = 0 \quad \forall i = 1, \dots, N$$

EX.

NR particle in 2 sp.-dim.s in a central potl.

$$\text{Cartesian} \quad S = \int dt \left[\frac{1}{2} m \dot{x}^2 + \frac{1}{2} m \dot{y}^2 - V(\sqrt{x^2 + y^2}) \right]$$



Cartesian Newton's Law

x-EOM:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) = \frac{\partial L}{\partial x} \quad \& \quad \frac{d}{dt} (m \dot{x}) = -V' \frac{x}{\sqrt{x^2 + y^2}}$$

$$V'(x) = \frac{dV}{dx}$$

y-EOM:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{y}} \right) = \frac{\partial L}{\partial y} \quad \& \quad \frac{d}{dt} (m \dot{y}) = -V' \frac{y}{\sqrt{x^2 + y^2}}$$

$$V'(y) = \frac{dV}{dy}$$

Polar coord.s (2D)

③

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$\dot{x}^2 + \dot{y}^2 = \dot{r}^2 + r^2 \dot{\theta}^2$$

$$\dot{x} = \dot{r} \cos \theta - (\sin \theta) r \dot{\theta}$$

$$\dot{y} = \dot{r} \sin \theta + (\cos \theta) r \dot{\theta}$$

$$S = \int dt \left[\frac{1}{2} m \dot{r}^2 + \frac{1}{2} m r^2 \dot{\theta}^2 - V(r) \right]$$

E-L E:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}} \right) = \frac{\partial L}{\partial r} \rightarrow \frac{d}{dt} (m \dot{r}) = -V' + \underline{m r^2 \dot{\theta}^2}$$

↑
Inertial
Forces
(fake
force)

$$\delta S = 0$$

$$S = \int dt L(q_i, \dot{q}_i)$$

$$(i=1, \dots, N)$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) = \frac{\partial L}{\partial q_i} \quad : \quad E - L = E$$

$$L(q_1, \dots, q_N, \dot{q}_1, \dots, \dot{q}_N)$$

$$p_i = \frac{\partial L}{\partial \dot{q}_i} \quad : \quad \text{conj. mtm.}$$

(conjugate mtm.)

$$\frac{dp_i}{dt} = \frac{\partial L}{\partial q_i}$$

specific value

If it happens, L is indep. of q_J

$$\frac{\partial L}{\partial q_J} = 0 \quad \& \quad \frac{dp_J}{dt} = 0 \quad \& \quad p_J = \text{conserved quantity}$$

$$\text{e.g., } S = \int dt \left[\frac{1}{2} m \dot{x}^2 + \frac{1}{2} m \dot{y}^2 - \frac{1}{2} k y^2 \right]$$

$$\frac{\partial L}{\partial x} = 0 \quad \& \quad p_x = \frac{\partial L}{\partial \dot{x}} = m \dot{x}$$

$$\frac{dp_x}{dt} = 0$$

(2)

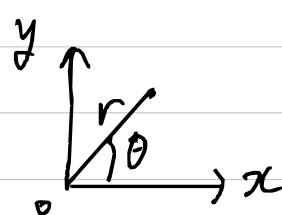
* In polar coord.s, w/ the central potl. $V(r)$,

$$S = \int dt \left[\frac{1}{2} m \dot{r}^2 + \frac{1}{2} m r^2 \dot{\theta}^2 - V(r) \right]$$

$$\frac{\partial L}{\partial \theta} = 0 \quad \& \quad p_{\theta} = \frac{\partial L}{\partial \dot{\theta}}$$

$$= m r^2 \dot{\theta}$$

= ang. mtn.



* Another conserved quantity.

$$H = \sum_{i=1}^N p_i \dot{q}_i - L(q, \dot{q})$$

$$\frac{\partial L}{\partial t} = 0$$

$$\frac{dH}{dt} = \sum_{i=1}^N \dot{p}_i \dot{q}_i + \sum_{i=1}^N p_i \ddot{q}_i - \sum_{i=1}^N \underbrace{\frac{\partial L}{\partial q_i}}_{=p_i} \dot{q}_i - \sum_{i=1}^N \underbrace{\frac{\partial L}{\partial \dot{q}_i}}_{=p_i} \ddot{q}_i$$

$$\frac{df}{dt}(x(t), y(t)) = \frac{dx}{dt} \frac{\partial f}{\partial x} + \frac{dy}{dt} \frac{\partial f}{\partial y}$$

$$\frac{dH}{dt} = 0 \quad (\text{if Lagr. is indep. of time } t)$$

$$H = \text{Hamil.} \equiv E_n.$$

$$L = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) - V(x, y)$$

$$p_x = \frac{\partial L}{\partial \dot{x}} = m \dot{x}, \quad p_y = \frac{\partial L}{\partial \dot{y}} = m \dot{y}$$

$$H = p_x \dot{x} + p_y \dot{y} - L$$

$$= 2T - (T - V) = T + V = E_n.$$

$$= \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) + V(x, y)$$

H is conserved.

Lagr. L doesn't really have a phys. meaning in itself.

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} (q, \dot{q}, t) \right) \quad \swarrow \quad \bar{i} \text{ is one val. here.}$$

$$= \underbrace{\frac{\partial}{\partial t}}_{\text{EXPLICIT TIME}} \left(\frac{\partial L}{\partial \dot{q}_i} \right) + \sum_{j=1}^N \ddot{q}_j \frac{\partial}{\partial \dot{q}_j} \frac{\partial L}{\partial \dot{q}_i} + \sum_{j=1}^N \dot{q}_j \frac{\partial}{\partial q_j} \frac{\partial L}{\partial \dot{q}_i}$$

EXPLICIT TIME

DEPENDENCE w/ Partial Deriv...

$$\sum_{j=1}^N Z_{ij} \ddot{q}_j = \frac{\partial L}{\partial q_i} - \frac{\partial}{\partial t} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \sum_{j=1}^N \dot{q}_j \frac{\partial^2 L}{\partial q_j \partial \dot{q}_i}$$

$$Z_{ij} = \frac{\partial^2 L}{\partial \dot{q}_i \partial \dot{q}_j} = \text{Kinetic matrix.}$$

$$= \begin{pmatrix} Z_{11} & Z_{12} & \dots & Z_{1n} \\ Z_{21} & \dots & \dots & \dots \\ \vdots & & & \\ Z_{n1} & & & Z_{nn} \end{pmatrix}$$

(3)

$$\underline{q} = \begin{pmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{pmatrix} \quad \& \quad \underline{\underline{Z}} = \begin{pmatrix} Z_{11} & Z_{12} & \dots \\ Z_{21} & Z_{22} & \dots \\ Z_{31} & Z_{32} & \dots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

$$\underline{\underline{Z}} \ddot{\underline{q}} = \begin{pmatrix} Z_{11} \ddot{q}_1 + Z_{12} \ddot{q}_2 + Z_{13} \ddot{q}_3 + \dots \\ Z_{21} \ddot{q}_1 + Z_{22} \ddot{q}_2 + \dots \\ \vdots \end{pmatrix}$$

$$\underline{q} = \begin{pmatrix} \sum_{j=1}^N Z_{1j} \ddot{q}_j \\ \sum_{j=1}^N Z_{2j} \ddot{q}_j \\ \vdots \end{pmatrix} = \sum_{j=1}^N Z_{ij} \ddot{q}_j = -F_i$$

$$\underline{\underline{Z}} \ddot{\underline{q}} = -\underline{F} \quad \& \quad \underline{F} = \begin{pmatrix} f_1 \\ f_2 \\ \vdots \end{pmatrix}$$

(force)

If $\text{Det}(\underline{\underline{Z}}) \neq 0$, $\underline{\underline{Z}}$ is invertible.

$$\ddot{\underline{q}} = -\underline{\underline{Z}}^{-1} \underline{F}$$

$$\ddot{q}_i = -\sum_{j=1}^N (\underline{\underline{Z}}^{-1})_{ij} F_j$$

$$\sum_{j=1}^N Z_{ij} (\underline{\underline{Z}}^{-1})_{jk} = (1)_{ik} = \delta_{ik} = \text{Identity}$$

$$\underline{\underline{1}} = \begin{pmatrix} 1 & & 0 \\ 0 & 1 & \\ & & \ddots \end{pmatrix}$$

$$\delta_{ik} = \begin{cases} 0 & , \quad i \neq k \\ 1 & , \quad i = k \end{cases}$$

EX. $N=3$, $(q_1, q_2, q_3) = (x, y, z)$

$$S = \int dt \left[\frac{1}{2} m (\dot{x}^2 + \dot{y}^2) - \frac{1}{2} k (x^2 + y^2) - \frac{1}{2} \alpha z^2 + \beta zx \right]$$

$$Z_{ij} = \frac{\partial^2 L}{\partial \dot{q}_i \partial \dot{q}_j} = \begin{pmatrix} Z_{11} & Z_{12} & Z_{13} \\ Z_{21} & Z_{22} & Z_{23} \\ Z_{31} & Z_{32} & Z_{33} \end{pmatrix}$$

$$\underline{\underline{Z}} = \begin{pmatrix} \frac{\partial^2 L}{\partial \dot{x}^2} & \frac{\partial^2 L}{\partial \dot{x} \partial \dot{y}} & \frac{\partial^2 L}{\partial \dot{x} \partial \dot{z}} \\ \frac{\partial^2 L}{\partial \dot{y} \partial \dot{x}} & \frac{\partial^2 L}{\partial \dot{y}^2} & \frac{\partial^2 L}{\partial \dot{y} \partial \dot{z}} \\ \frac{\partial^2 L}{\partial \dot{z} \partial \dot{x}} & \frac{\partial^2 L}{\partial \dot{z} \partial \dot{y}} & \frac{\partial^2 L}{\partial \dot{z}^2} \end{pmatrix}$$

Using $\frac{\partial^2}{\partial \dot{x}^2} (\dot{x}^2) = 2$

Let

$$\underline{\underline{Z}} = \begin{pmatrix} m & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \& \quad \det(\underline{\underline{Z}}) = 0$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) = \frac{\partial L}{\partial x} \rightarrow \frac{d}{dt} (m \dot{x}) = -kx + \beta z$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{y}} \right) = \frac{\partial L}{\partial y} \rightarrow \frac{d}{dt} (m \dot{y}) = -ky$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{z}} \right) = \frac{\partial L}{\partial z} \rightarrow \frac{d}{dt} (0) = 0 = -\alpha z + \beta x$$

If $\alpha \neq 0$, solve: $\frac{\partial L}{\partial z} = 0$ for z .

$$z = \frac{\beta x}{\alpha} = \text{auxiliary variable}$$

plug it back into $m \ddot{x} = -kx + \beta z$

$$x\text{-EOM: } \begin{cases} m \ddot{x} = -kx + \frac{\beta^2 x}{\alpha} \\ m \ddot{y} = -ky \end{cases}$$

& the action also changes:

$$S = \int dt \left[\frac{1}{2} m (\dot{x}^2 + \dot{y}^2) - \frac{1}{2} k (x^2 + y^2) - \frac{1}{2} \frac{\beta^2 x^2}{\alpha} + \frac{\beta^2 x^2}{\alpha} \right]$$

$$= \int dt \left[\frac{1}{2} m (\dot{x}^2 + \dot{y}^2) - \frac{1}{2} k (x^2 + y^2) + \frac{1}{2} \frac{\beta^2 x^2}{\alpha} \right]$$

(2)

$$\alpha=0 : \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{z}} \right) = \frac{\partial L}{\partial z} = 0 = \beta x$$

($\beta \neq 0$)

z : a Lagrange multiplier. $\downarrow$

$$\begin{cases} x=0 \\ z=0 \end{cases} \rightarrow \text{(Holonomic) Constraint}$$

3 coord.s $\rightarrow$ 1 propagating DOF.
(y)

$$C_\alpha[q_i] = 0, \quad \alpha = 1, \dots, N_{\text{constraints}}$$

$$S_{\text{constraints}} = \int dt \left[L(q_i, \dot{q}_i) + \sum_{\alpha=1}^N \lambda_\alpha C_\alpha[q_i] \right]$$

$= L_{\text{extended}}$

Extend coord.s: $N + N_{\text{constraint}}$,
($q_1, \dots, q_N, \lambda_1, \dots, \lambda_N$)

$$\delta S_{\text{extended}} = 0$$

$$\frac{d}{dt} \left(\frac{\partial L_{\text{ext}}}{\partial \dot{\lambda}_\alpha} \right) = \frac{\partial L_{\text{ext}}}{\partial \lambda_\alpha} = 0 = C_\alpha$$

$$\frac{d}{dt} \left(\frac{\partial L_{\text{extended}}}{\partial \dot{q}_i} \right) = \frac{\partial L_{\text{ext}}}{\partial q_i}$$

$$\frac{dp_i}{dt} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) = \frac{\partial L}{\partial q_i} + \sum_{\alpha=1}^{N_{\text{constraint}}} \lambda_\alpha \frac{\partial C_\alpha}{\partial q_i}$$

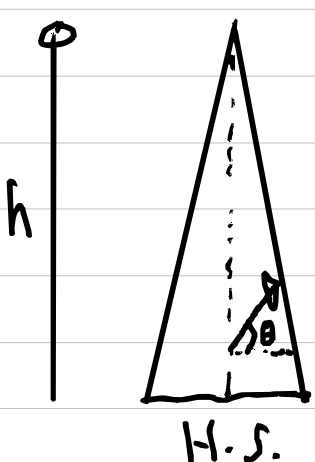
(EX) Helter Skelter

Unconstrained. in the absence of friction.

only force in the game $\rightarrow$ Gravity

$$V = mgz$$

$$S = \int dt \left[\frac{1}{2} m \dot{x}^2 + \frac{1}{2} m \dot{y}^2 + \frac{1}{2} m \dot{z}^2 - mgz \right]$$



$$z = h - \alpha \theta$$

$$= \int dt \left[\frac{1}{2} m \dot{z}^2 + \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m r^2 \dot{\theta}^2 - mgz \right]$$

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \left. \vphantom{\begin{matrix} x \\ y \end{matrix}} \right\} \text{Polar}$$

(3)

For this H.S.,

• Constraint 1 : $z = h - \alpha \theta$.

• Constraint 2 : $r = \rho \theta$
 $\hookrightarrow$ the radius increases linearly as the kid goes down.

$$C_1 = z - h + \alpha \theta$$



$$C_2 = r - \rho \theta$$

* Extended sys. by introducing Lagrange multipliers :

$$\begin{aligned} S_{\text{ext}} &= \int dt \left[\frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2 + \dot{z}^2) - mgz + \lambda_1 C_1 + \lambda_2 C_2 \right] \\ &= \int dt \left[\frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2 + \dot{z}^2) - mgz + \lambda_1 (z - h + \alpha \theta) \right. \\ &\quad \left. + \lambda_2 (r - \rho \theta) \right] \end{aligned}$$

* The reduced action.

• Solve constraints + sub 'em back in.

$$\begin{aligned} C_2 \rightarrow r = \rho \theta \rightarrow \dot{r} = \rho \dot{\theta} \rightarrow z = h - \alpha \theta \\ \downarrow \\ \dot{z} = -\alpha \dot{\theta} \end{aligned}$$

Sub them into unconstrained action :

$$\begin{aligned} S_{\text{reduced}} &= \int dt \left[\frac{1}{2} m (\alpha^2 \dot{\theta}^2 + \rho^2 \dot{\theta}^2 + \rho^2 \theta^2 \dot{\theta}^2) - mg(h - \alpha \theta) \right] \\ &= \int dt \left[\frac{1}{2} m (\alpha^2 + \rho^2 + \rho^2 \theta^2) \dot{\theta}^2 - mgh + mg\alpha \theta \right] \end{aligned}$$

$$\hookrightarrow \theta - E - LE : \frac{d}{dt} \left[\frac{\partial L_{\text{reduced}}}{\partial \dot{\theta}} \right] = \frac{\partial L_{\text{reduced}}}{\partial \theta}$$

$$\frac{d}{dt} \left[m (\alpha^2 + \rho^2 + \rho^2 \theta^2) \dot{\theta} \right] = mg\alpha + m\rho^2 \theta \dot{\theta}^2$$

: EOM for this sys. (H.S.)

④

The beauty of Lagr. is that we're able to change coord.s.

From

$$S_{\text{ext}} = \int dt \left[\frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2 + \dot{z}^2) - mgz + \lambda_1 (z - h + \alpha \theta) + \lambda_2 (r - \beta \theta) \right]$$

we get 5 EOMs.

$$z\text{-EOM: } m \ddot{z} = -mg + \lambda_1$$

$$r\text{-EOM: } m \ddot{r} = mr \dot{\theta}^2 + \lambda_2$$

$$\theta\text{-EOM: } m \frac{d}{dt} (r^2 \dot{\theta}) = \lambda_1 \alpha - \lambda_2 \beta$$

$$\text{Constraint 1 Eq.: } z = h - \alpha \theta$$

$$\text{Constraint 2 Eq.: } r = \beta \theta$$

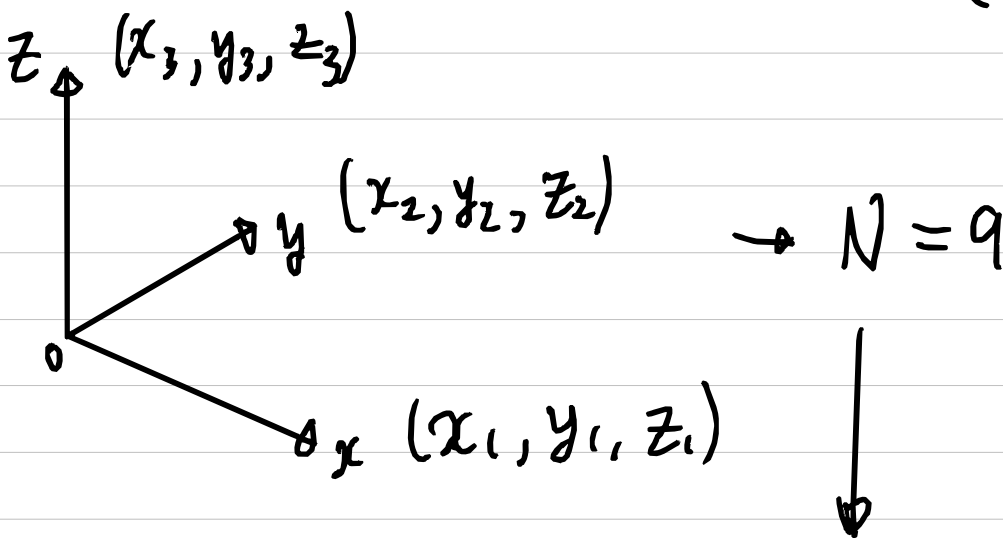
5 eqns
in this
case.

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) = \frac{\partial L}{\partial q_i}$$

* Index Notation

$$q_i \quad (i = 1, 2, \dots, N)$$

$$(EX) \quad N=3 \quad : \quad (q_1, q_2, q_3) = (x, y, z) \\ \text{or } (r, \theta, \phi)$$



$$(q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9) \\ = (x_1, x_2, x_3, y_1, y_2, y_3, z_1, z_2, z_3)$$

$$S = \int dt \left[\frac{1}{2} m \left(\dot{x}_1^2 + \dot{y}_1^2 + \dot{z}_1^2 + \dot{x}_2^2 + \dot{y}_2^2 + \dot{z}_2^2 + \dot{x}_3^2 + \dot{y}_3^2 + \dot{z}_3^2 \right) - V(x_1, x_2, x_3, y_1, y_2, y_3, z_1, z_2, z_3) \right]$$

$$A_i = B_i + C_i \quad \rightarrow \text{Index Notation}$$

N eqns

$$A_1 = B_1 + C_1$$

$$A_2 = B_2 + C_2$$

$\vdots$

$$\left. \begin{array}{l} A_1 = B_1 + C_1 \\ A_2 = B_2 + C_2 \\ \vdots \end{array} \right\} \rightarrow \underline{A} = \underline{B} + \underline{C}$$

$$\begin{pmatrix} A_1 \\ \vdots \\ A_q \end{pmatrix} = \begin{pmatrix} B_1 + C_1 \\ \vdots \\ B_q + C_q \end{pmatrix}$$

(2)

* Matrices

$$(N=3) \quad \underline{\underline{M}} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{pmatrix} \end{matrix}$$

$$\underline{B} = \underline{\underline{M}} \underline{A}$$

$$\begin{pmatrix} B_1 \\ B_2 \\ B_3 \end{pmatrix} = \begin{pmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix}$$

$$= \begin{pmatrix} M_{11}A_1 + M_{12}A_2 + M_{13}A_3 \\ M_{21}A_1 + M_{22}A_2 + M_{23}A_3 \\ M_{31}A_1 + M_{32}A_2 + M_{33}A_3 \end{pmatrix}$$

$$B_i = \sum_{j=1}^N M_{ij} A_j$$

Repeated Index



Implicit Summation

$$A_{ij} B_{jk} C_{kl} D_{lm} = (\underline{\underline{A}} \underline{\underline{B}} \underline{\underline{C}} \underline{\underline{D}})_{im}$$

$$= \sum_{j=1}^N \sum_{k=1}^N \sum_{l=1}^N A_{ij} B_{jk} C_{kl} D_{lm}$$

$i, m \rightarrow$ only one of them are there.

$A_{ijk} \nrightarrow$ 3D matrix (Rubik's cube)

A_{ijkl} e.g., $R^{\mu}_{\nu\rho\sigma}$ (Riemann Tnsr)

$\hookrightarrow$ curvature
& space time

$$V(q_1(t), \dots, q_N(t), t)$$

$$\frac{dV}{dt} = \frac{\partial V}{\partial t} + \sum_{i=1}^N \frac{dq_i}{dt} \frac{\partial V}{\partial q_i}$$

(3)

* Taylor Expansion

$$\begin{aligned}
 f(x) &= f(x_* + (x - x_*)) \\
 &= f(x_*) + \frac{df}{dx}(x_*)(x - x_*) + \frac{1}{2} \frac{d^2 f}{dx^2}(x_*)(x - x_*)^2 \\
 &\quad + \dots \\
 &= \sum_{n=0}^{\infty} \frac{1}{n!} \frac{d^n f}{dx^n}(x_*)(x - x_*)^n
 \end{aligned}$$

$$\begin{aligned}
 f(q) &= f(q_i^* + (q - q_i^*)) \\
 &= f(q^*) + \sum_{i=1}^N \frac{\partial f}{\partial q_i}(q_i^*)(q_i - q_i^*) \\
 &\quad + \frac{1}{2} \sum_{i,j=1}^N \frac{\partial^2 f(q^*)}{\partial q_i \partial q_j} (q_i - q_i^*)(q_j - q_j^*) + \mathcal{O}((q - q^*)^3)
 \end{aligned}$$

$$A_{ij} = \frac{\partial^2 f}{\partial q_i \partial q_j}(q^*)$$

$$V_i = q_i - q_i^*$$

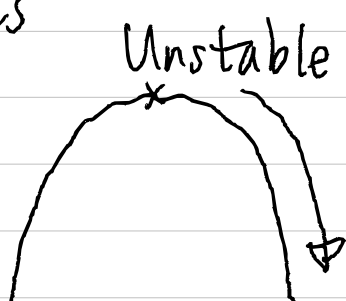
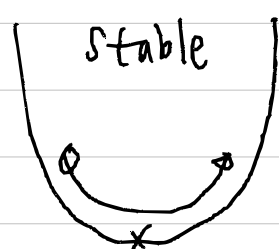
$$A_{ij} V_i V_j = \sum_{i=1}^N \sum_{j=1}^N V_i A_{ij} V_j$$

$$\begin{aligned}
 &= V_1 A_{11} V_1 + V_1 A_{12} V_2 + V_1 A_{13} V_3 \\
 &\quad + V_2 A_{21} V_1 + V_2 A_{22} V_2 + V_2 A_{23} V_3 \\
 &\quad + V_3 A_{31} V_1 + V_3 A_{32} V_2 + V_3 A_{33} V_3
 \end{aligned}$$

$$\begin{aligned}
 \underline{V}^T \underline{A} \underline{V} \\
 &= (V_1 \quad V_2 \quad V_3) \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \\ V_3 \end{pmatrix}
 \end{aligned}$$

$$= V_i A_{ij} V_j$$

* Normal Modes



$$\left(\begin{array}{l} q_i \quad (i=1, \dots, N) \\ \dot{q}_i = \frac{dq_i}{dt} \end{array} \right)$$

• Special class

$$S = \int dt \left[\frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N a_{ij}(q) \dot{q}_i \dot{q}_j - V(q) \right]$$

$$a_{ij} [q_1, \dots, q_N]$$

e.g., $N=3$.

$$\begin{aligned} a_{ij} \dot{q}_i \dot{q}_j &= a_{11} \dot{q}_1^2 + a_{12} \dot{q}_1 \dot{q}_2 + a_{13} \dot{q}_1 \dot{q}_3 \\ &\quad + a_{21} \dot{q}_2 \dot{q}_1 + a_{22} \dot{q}_2^2 + a_{23} \dot{q}_2 \dot{q}_3 \\ &\quad + a_{31} \dot{q}_3 \dot{q}_1 + a_{32} \dot{q}_3 \dot{q}_2 + a_{33} \dot{q}_3^2 \end{aligned}$$

$$= \begin{pmatrix} \dot{q}_1 & \dot{q}_2 & \dot{q}_3 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{pmatrix}$$

$$= \underline{\dot{q}}^T \cdot \underline{A} \underline{\dot{q}}$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) = \frac{\partial L}{\partial q_i} \quad (i=1, \dots, N)$$

$$\frac{\partial}{\partial \dot{q}_i} (a_{ij} \dot{q}_i \dot{q}_j) \neq a_{ij} \dot{q}_j$$

$$\sum_{i=1}^N a_i b_i = \sum_{j=1}^N a_j b_j$$

$$\frac{\partial}{\partial \dot{q}_i} (a_{jk} \dot{q}_j \dot{q}_k) = a_{jk}(q) \left[\frac{\partial \dot{q}_j}{\partial \dot{q}_i} \dot{q}_k + \dot{q}_j \frac{\partial \dot{q}_k}{\partial \dot{q}_i} \right]$$

$$\frac{\partial \dot{q}_j}{\partial \dot{q}_i} = \delta_{ij}$$

$$\frac{\partial \dot{q}_k}{\partial \dot{q}_i} = \delta_{ik}$$

$$\therefore \frac{\partial}{\partial \dot{q}_i} (a_{jk} \dot{q}_j \dot{q}_k) = a_{jk} [\delta_{ij} \dot{q}_k + \delta_{ik} \dot{q}_j]$$

$$= a_{ik} \dot{q}_k + a_{ji} \dot{q}_j$$

$$= a_{ij} \dot{q}_j + a_{ij} \dot{q}_j = 2 a_{ij} \dot{q}_j$$

$$\sum_{j=1}^N a_{jk} \delta_{ij} = a_{ik}$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) = \frac{\partial L}{\partial q_i}$$

$$\frac{d}{dt} (a_{ij} \dot{q}_j) = - \frac{\partial V}{\partial q_i} + \frac{1}{2} \sum_j \sum_k \frac{\partial a_{jk}}{\partial q_i} \dot{q}_j \dot{q}_k$$

(2)

* Equilibrium Soln.s

$$\dot{q}_i = 0$$

$$\frac{\partial V}{\partial q_i} = 0 \quad \text{for } \forall i=1, \dots, N$$

↓ soln.

$$q_i^{\text{eq.}}$$

* Perturb $q_i(t) = q_{\text{eq}} + \delta q_i(t)$

$$S = \int dt \left[\frac{1}{2} a_{ij}(q_{\text{eq}} + \delta q) \delta \dot{q}_i \delta \dot{q}_j - V(q_{\text{eq}} + \delta q) \right]$$

$$\dot{q}_i = \delta \dot{q}_i$$

Keep terms in $\mathcal{L}$ to 2nd order in perturbation:

$$S = \int dt \left[\frac{1}{2} a_{ij}(q_{\text{eq}}) \delta \dot{q}_i \delta \dot{q}_j - V(q_{\text{eq}}) - \sum_{i=1}^N \left. \frac{\partial V}{\partial q_i} \right|_{q=q_{\text{eq}}} \delta q_i - \frac{1}{2} \sum_{i,j} \left. \frac{\partial^2 V}{\partial q_i \partial q_j} \right|_{q=q_{\text{eq}}} \delta q_i \delta q_j + \mathcal{O}(\delta q^3) \right]$$

$$A_{ij} = a_{ij} \quad \& \quad B_{ij} = \left. \frac{\partial^2 V}{\partial q_i \partial q_j} \right|_{q=q_{\text{eq}}}$$

$$S_{(2)} = \int dt \left[-V(q_{\text{eq}}) + \frac{1}{2} \delta \dot{q}_i A_{ij} \delta \dot{q}_j - \frac{1}{2} \delta q_i B_{ij} \delta q_j \right]$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial (\delta \dot{q}_i)} \right) = \frac{\partial \mathcal{L}}{\partial (\delta q_i)}$$

$$\frac{d}{dt} [A_{ij} \delta \dot{q}_j] = -B_{ij} \delta q_j \quad \left(\frac{\partial (\delta q_j)}{\partial (\delta q_i)} = \delta_{ij} \right)$$

$$A_{ij} \delta \ddot{q}_j = -B_{ij} \delta q_j \quad \Leftrightarrow \quad \underline{A} \delta \ddot{\underline{q}} = -\underline{B} \delta \underline{q}$$

$$m \ddot{x} = -kx$$

$$F_N = F_{\text{Hooke}}$$

$$\underline{\delta \ddot{q}}_j = -\underline{A}^{-1} \underline{B} \delta \underline{q}$$

$$= -\underline{K} \delta \underline{q}$$

Ansatz

$$\text{N.M.: } \delta \underline{q} = e^{-i\omega t} \underline{u}$$

$$\left\{ \begin{array}{l} \delta \ddot{\underline{q}} = -\omega^2 e^{-i\omega t} \underline{u} = -\omega^2 \delta \underline{q} \end{array} \right.$$

$$\omega^2 \underline{u} = \underline{K} \underline{u}$$

$$\omega^2 \rightarrow \text{Eigenval. } \underline{K}$$

Recap

$$q_1 \dots q_N \quad (i=1, \dots, N)$$

$$S = \int dt \left[\frac{1}{2} \sum_{i,j=1}^N a_{ij}(q) \dot{q}_i \dot{q}_j - V(q) \right]$$

$$\text{Equilibrium Soln.} : \frac{\partial V}{\partial q_i}(q) = 0 \rightarrow q_i^{eq}$$

* Perturb.

$$q_i = q_i^{eq} + \delta q_i$$

To quadratic order:

$$S_{(2)} = \int dt \left[\frac{1}{2} a_{ij}(q^{eq}) \dot{\delta q}_i \dot{\delta q}_j - \frac{1}{2} B_{ij} \delta q_i \delta q_j \right]$$

$$B_{ij} = \left. \frac{\partial^2 V}{\partial q_i \partial q_j} \right|_{q=q^{eq}}$$

$$\text{EOM} : \sum_{j=1}^N a_{ij} \delta \ddot{q}_j = - \sum_{j=1}^N B_{ij} \delta q_j$$

$$(\underline{A})_{ij} = a_{ij} \quad \& \quad \underline{A} \delta \ddot{\underline{q}} = - \underline{B} \delta \underline{q}$$

$$(\underline{B})_{ij} = B_{ij} = \left. \frac{\partial^2 V}{\partial q_i \partial q_j} \right|_{q=q^{eq}}$$

$$\delta \underline{q} = \begin{pmatrix} \delta q_i \\ \delta q_j \\ \delta q_k \end{pmatrix}$$

$$\text{Det}(\underline{A}) \neq 0.$$

$$\underline{A}^{-1} \underline{A} \delta \ddot{\underline{q}} = - \underline{A}^{-1} \underline{B} \delta \underline{q}$$

$$\delta \ddot{\underline{q}} = - \underline{K} \delta \underline{q} \quad \& \quad \underline{K} = \underline{A}^{-1} \underline{B}$$

(2)

* Ansatz.

$$\underline{\delta q} = e^{i\omega t} \underline{u} \quad \& \quad \underline{\delta \ddot{q}} = -\omega^2 e^{i\omega t} \underline{u}$$

$$\omega^2 \underline{u} = \underline{K} \underline{u} \quad \& \quad \underline{K} \text{ is an } N \times N \text{ matrix,}$$

↓

& $\underline{u}$ is a vector.

Eigenvalue eqn.

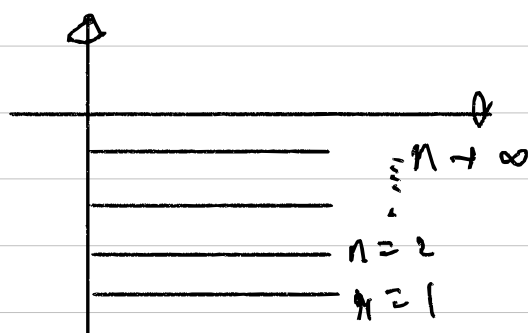
& $\omega^2 \rightarrow$ number.
 $\underline{u} \rightarrow$ e-vec. $\rightarrow$ N of them.
 $\omega \rightarrow$ e-num.

$$\underline{K} \underline{u}_\alpha = \omega_\alpha^2 \underline{u}_\alpha \quad \leftarrow \alpha = \text{e-vector index.}$$

 $\alpha = 1, \dots, N \rightarrow$ labelling the e-vectors $i, j = 1, \dots, N \rightarrow$ labelling the coordinates.

$$\hat{H} \psi_n = E_n \psi_n$$

$$\hat{H} |n\rangle = E_n |n\rangle$$



$$\psi(t) = \sum a_n \psi_n e^{-\frac{iE_n t}{\hbar}}$$

$$\text{Ansatz} : \underline{\delta q} = e^{i\omega t} \underline{u}$$

$$\underline{\delta q} = \sum_{\alpha=1}^N a_\alpha e^{i\omega_\alpha t} \underline{u}_\alpha + \sum_{\alpha=1}^N a_\alpha^\dagger e^{-i\omega_\alpha t} \underline{u}_\alpha^*$$

$$\left(\begin{array}{l} \underline{u}_\alpha^* = \underline{u}_\alpha \\ \text{Hermitian} \end{array} \right)$$

$$\underline{\delta q} = \sum_{\alpha=1}^N \left(b_\alpha \cos(\omega_\alpha t) + c_\alpha \sin(\omega_\alpha t) \right) \underline{u}_\alpha$$

$$\underline{K} \underline{u}_\alpha = \omega_\alpha^2 \underline{u}_\alpha = \omega_\alpha^2 \underline{\mathbb{1}} \underline{u}_\alpha$$

$$\begin{array}{l} \text{identity} \quad \underline{\mathbb{1}} = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix} \\ = \delta_{ij} \end{array}$$

↓

$$(\underline{K} - \omega_\alpha^2 \underline{\mathbb{1}}) \underline{u}_\alpha = 0$$

$$\det(\underline{K} - \omega_\alpha^2 \underline{\mathbb{1}}) = 0$$

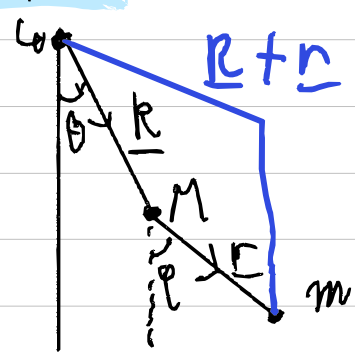
Eqn. that determines all the e-val.s.

Polynomial
solns

$$\left(\omega_\alpha^2 \right)^N + \# \left(\omega_\alpha^2 \right)^{N-1} + \dots = 0$$

(3)

Fixed



Constraints:

$$\begin{cases} C_1 = R^2 - \underline{R}^2 \\ C_2 = r^2 - \underline{r}^2 \end{cases}$$

mean that in this case, we assume lengths are const.

Equilibrium pt.: $\theta = \varphi = 0$. We can see the perturbations abt this equil. pt..

Lagrangian Set Up.

$$\textcircled{1} \text{ KE: } T = \frac{1}{2} M \dot{\underline{R}}^2 + \frac{1}{2} m (\dot{\underline{R}} + \dot{\underline{r}})^2$$

$$= \frac{1}{2} M \dot{\underline{R}}^2 + \frac{1}{2} m (\dot{\underline{R}}^2 + 2 \dot{\underline{R}} \dot{\underline{r}} \cos(\theta - \varphi) + \dot{\underline{r}}^2)$$

$$(\underline{R} = R\theta \quad \& \quad \underline{r} = r\varphi)$$

$$T = \frac{1}{2} M R^2 \dot{\theta}^2 + \frac{1}{2} m (R^2 \dot{\theta}^2 + 2 R \dot{\theta} r \dot{\varphi} \cos(\theta - \varphi) + r^2 \dot{\varphi}^2)$$

$$= \frac{1}{2} (M + m) R^2 \dot{\theta}^2 + \frac{1}{2} m (2 R r \dot{\theta} \dot{\varphi} \cos(\theta - \varphi) + r^2 \dot{\varphi}^2)$$

$$= \frac{1}{2} (M + m) R^2 \dot{\theta}^2 + \frac{1}{2} m r^2 \dot{\varphi}^2 + m R r \dot{\theta} \dot{\varphi} \cos(\theta - \varphi)$$

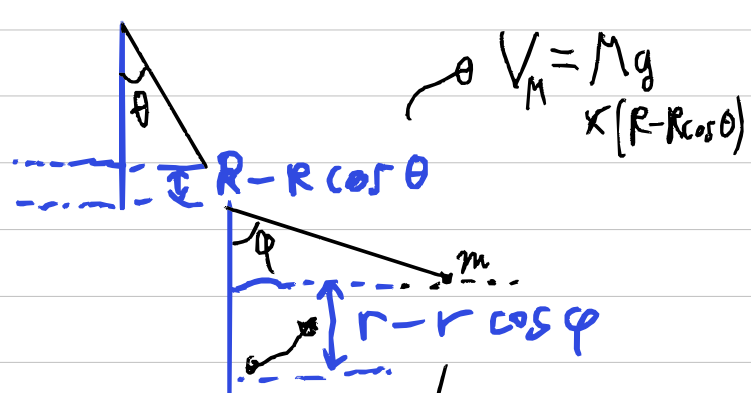
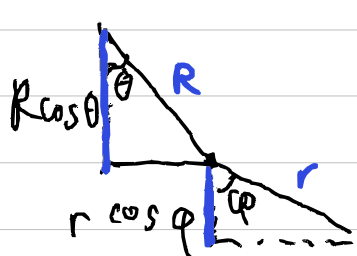
② PE:

when $\theta = 0 \rightarrow V = 0$

$$V = Mg R (1 - \cos \theta) + mg [R (1 - \cos \theta) + r (1 - \cos \varphi)]$$

For the upper M

for the lower m.



$$V_M = Mg \times [R (1 - \cos \theta)]$$

$$V_m = mg [R (1 - \cos \theta) + r (1 - \cos \varphi)]$$

At equilibrium q_i^{eq} .
($\theta = \varphi = 0$),
the potl. $V = 0$.

Expand the action to 2nd-order, e.g.,

$$\theta^2, \varphi^2, \theta\varphi, \dot{\theta}^2, \dot{\varphi}^2, \dot{\theta}\dot{\varphi}.$$

$$\therefore \text{ express: } \cos \theta = 1 - \frac{\theta^2}{2} \quad \& \quad \cos \varphi = 1 - \frac{\varphi^2}{2}$$

$$\cos(\theta - \varphi) = 1 - \frac{(\theta - \varphi)^2}{2} = 1 - \frac{\theta^2}{2} + \theta\varphi - \frac{\varphi^2}{2}$$

$$1 - \cos \theta = \frac{\theta^2}{2} \quad \& \quad 1 - \cos \varphi = \frac{\varphi^2}{2}$$

Quadratic ac.

$$S_{(2)} = \int dt \left[\frac{1}{2} (M + m) R^2 \dot{\theta}^2 + \frac{1}{2} m r^2 \dot{\varphi}^2 + m R r \dot{\theta} \dot{\varphi} \left(1 - \frac{\theta^2}{2} + \theta\varphi - \frac{\varphi^2}{2} \right) + \frac{Mg R \theta^2}{2} + mg \left(\frac{R}{2} \theta^2 + \frac{r}{2} \varphi^2 \right) \right]$$

$$\delta \underline{q} = \begin{pmatrix} \theta \\ \varphi \end{pmatrix}$$

$$\underline{A} = \begin{pmatrix} (M + m) R^2 & m R r \\ m R r & m r^2 \end{pmatrix}$$

EXAM
2x2!

$$\rightarrow \frac{1}{2} \underline{A} \dot{\underline{q}}$$

$$(\underline{B})_{ij} = \frac{\partial^2 V}{\partial q_i \partial q_j}$$

$$= \frac{1}{2} \begin{pmatrix} & \end{pmatrix} \begin{pmatrix} \dot{\theta} \\ \dot{\varphi} \end{pmatrix}$$

$$\underline{B} = \begin{pmatrix} (M + m) g R & 0 \\ 0 & -m g r \end{pmatrix}$$

* Recap on N.M.s

$$S = \int dt \left[\frac{1}{2} a_{ij}(q) \dot{q}_i \dot{q}_j - V(q) \right]$$

$$\frac{\partial V}{\partial q_i}(q_{\text{eq}}) = 0 \quad \& \quad q_i = q_i^{\text{eq}} + \delta q_i$$

$$S_{(2)} = \int dt \left[\frac{1}{2} \underline{\dot{\delta q}}^T \underline{A} \underline{\dot{\delta q}} - \frac{1}{2} \underline{\delta q}^T \underline{B} \underline{\delta q} \right]$$

compos $\underline{A}_{ij} = a_{ij}$ & $\underline{B}_{ij} = b_{ij}$.

E-L for the fluctuations: $\frac{d}{dt} \left(\frac{\partial L_{(2)}}{\partial \dot{\delta q}_i} \right) = \frac{\partial L_{(2)}}{\partial \delta q_i}$



$$a_{ij} \delta \ddot{q}_j = -b_{ij} \delta q_j$$

$$\left(\begin{array}{l} \underline{A} \text{ real, symmetric} \\ \underline{A}^T = \underline{A} \end{array} \right)$$

Cholesky decomposition



$$\underline{A} = \underline{L} \underline{L}^T$$

$$\& \quad \underline{L} = \begin{pmatrix} \# & 0 & 0 & 0 & 0 \\ \# & \# & 0 & 0 & 0 \\ \# & \# & \# & 0 & 0 \\ \# & \# & \# & \# & 0 \\ \# & \# & \# & \# & \# \end{pmatrix}$$

$$\underline{A}^T = (\underline{L} \underline{L}^T)^T$$

$$= (\underline{L}^T)^T \underline{L}^T$$

$$= \underline{L} \underline{L}^T$$

$$= \underline{A}$$

"lower triangular matrix."

(2)

$$S = \int dt \left[\frac{1}{2} \underline{\dot{q}}^T \underline{L} \underline{L}^T \underline{\dot{q}} - \frac{1}{2} \underline{\dot{q}}^T \underline{P} \underline{\dot{q}} \right]$$

Define new coords : $\underline{\delta Q} = \underline{L}^T \underline{\delta q}$

$$\underline{\delta Q}^T = \underline{\delta q}^T \underline{L}$$

$$S = \int dt \left[\frac{1}{2} \underline{\dot{\delta Q}}^T \underline{\delta Q} - \frac{1}{2} \underline{\delta Q}^T \underline{k} \underline{\delta Q} \right]$$

canonically
normalized

$$\underline{k} = (\underline{L})^{-1} \underline{P} (\underline{L}^T)^{-1}$$

$$\frac{1}{2} (\dot{\delta Q}_1^2 + \dot{\delta Q}_2^2 + \dot{\delta Q}_3^2 + \dots)$$

$$\ddot{\delta Q} = -\underline{k} \delta Q \quad \& \quad \delta Q = e^{i\omega t} \delta Q_\omega$$

$$\omega_\alpha^2 \delta Q_\alpha = \underline{k} \delta Q_\alpha$$

$$\left(\det(\omega^2 \underline{1} - \underline{k}) = 0 \rightarrow \text{eigen-freq. } \omega_\alpha \right.$$

$$\omega_\alpha^2 \underline{u}_\alpha = \underline{k} \underline{u}_\alpha$$

$$\underline{\delta Q} = \sum_{\alpha=1}^N a_\alpha e^{i\omega_\alpha t} \underline{u}_\alpha + \sum_{\alpha=1}^N a_\alpha^* e^{-i\omega_\alpha t} \underline{u}_\alpha$$

eigenvecs $\rightarrow$ should be orthonormal.
(orthogonal & normalized)

$$\underline{u}_\beta^T \underline{u}_\alpha = \delta_{\alpha\beta} : \text{Orthonormality cond.}$$

$$\int d^3x \psi_m^*(x) \psi_n(x) = \delta_{nm}$$

$$\star \text{Completeness : } \sum_{\alpha=1}^N \underline{u}_\alpha \underline{u}_\alpha^T = \underline{1}$$

$$\underline{u}_\beta^T \underline{u}_\alpha = \delta_{\alpha\beta}$$

$$\underline{u}_\beta^T \sum_{\alpha=1}^N \underline{u}_\alpha \underline{u}_\alpha^T = \sum_{\alpha} \delta_{\alpha\beta} \underline{u}_\alpha^T = \underline{u}_\beta^T$$

$$\underline{u}_\beta^T \left(\sum_{\alpha=1}^N \underline{u}_\alpha \underline{u}_\alpha^T \right) = \underline{u}_\beta^T$$

$$\omega_\alpha^2 \underline{u}_\alpha = \underline{k} \underline{u}_\alpha \rightarrow \sum_{\alpha} \omega_\alpha^2 \underline{u}_\alpha \underline{u}_\alpha^T = \sum_{\alpha} \underline{k} \underline{u}_\alpha \underline{u}_\alpha^T = \underline{k} \underline{1} = \underline{k}$$

$$\underline{k} = \sum_{\alpha} \omega_\alpha^2 \underline{u}_\alpha \underline{u}_\alpha^T$$

$$\underline{\delta Q} = \sum_{\alpha=1}^N \delta Q_\alpha \underline{u}_\alpha$$

$$S = \int dt \left[\frac{1}{2} \sum_{\alpha} (\dot{\delta Q}_\alpha)^2 - \frac{1}{2} \sum_{\alpha} \omega_\alpha^2 (\delta Q_\alpha)^2 \right]$$

$$= \sum_{\alpha} \int dt \left[\frac{1}{2} (\dot{\delta Q}_\alpha)^2 - \frac{1}{2} \omega_\alpha^2 (\delta Q_\alpha)^2 \right]$$

(3)

* Noether's Thm. ($i=1, \dots, N$)

$$S = \int dt L(q, \dot{q}) = \int dt L(q_1, q_2, \dots, q_N, \dot{q}_1, \dot{q}_2, \dots, \dot{q}_N)$$

* Continuous Global Symmetries

Symmetry tr. is labelled by a continuous param. λ ,
e.g., Lorentz tr.:

$$x' = \gamma \left(x - \frac{v}{c} t \right) \quad \& \quad \gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$v \rightarrow \delta v \quad \& \quad \lambda = v : \text{parameter}$$

$$x' - x = \delta x = - \frac{\delta v t}{c}$$

* Imagine there's a symmetry

$$\delta q_i = F_i(q, \dot{q}) \delta \lambda$$

EOMs $\rightarrow$ invariant. $\rightarrow$ The simplest way that it's true is if the Lagr. is invariant.
Global: $\dot{\delta \lambda} = 0$.

If the Lagr. is invar. $\rightarrow$ the EOMs are invar.

$$\begin{aligned} \delta L(q, \dot{q}) &= \sum_{i=1}^N \frac{\partial L}{\partial q_i} \delta q_i + \sum_{i=1}^N \frac{\partial L}{\partial \dot{q}_i} \delta \dot{q}_i \\ &= \frac{\partial L}{\partial q_i} F_i \delta \lambda + \frac{\partial L}{\partial \dot{q}_i} \dot{F}_i \delta \lambda \\ &= 0 \end{aligned}$$

* Noether's trick:

$$\delta S = \int dt \delta L = \int dt \left[\frac{\partial L}{\partial q_i} F_i \delta \lambda + \frac{\partial L}{\partial \dot{q}_i} (\dot{F}_i \delta \lambda) \right]$$

$$= \int dt \left[\underbrace{\frac{\partial L}{\partial q_i} F_i \delta \lambda + \frac{\partial L}{\partial \dot{q}_i} \dot{F}_i \delta \lambda}_{=0} + \frac{\partial L}{\partial \dot{q}_i} F_i \dot{\delta \lambda} \right]$$

$$\delta S = \int dt \left[\frac{\partial L}{\partial \dot{q}_i} F_i \dot{\delta \lambda} \right] = \int \left[- \frac{d}{dt} \left\{ \frac{\partial L}{\partial \dot{q}_i} F_i \right\} \delta \lambda \right] dt$$

IBP

$$\int u dv = uv - \int v du$$

$$= \cancel{\frac{\partial L}{\partial \dot{q}_i} F_i \delta \lambda} - \int \left[\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} F_i \right) \delta \lambda \right] dt$$

$= 0$ by EL-E & Hamil. principles.

④

$$\frac{d}{dt} \left[\frac{\partial L}{\partial \dot{q}_i} \dot{q}_i \right] = 0$$

↑
conserved quantity

En., mtn., elec. charge → all conserved
because of symmetry
↳ explained by
Noether's thm.

[Noether's Thm]

* Thm.

Given 2 Lagr.s L & L' , which differ by total deriv.:

$$L' - L = \frac{dH}{dt}(q, \dot{q})$$

the EOMs derived from L & L' are identical.

$$\begin{aligned} S' &= \int_{t_i}^{t_f} dt L'(q, \dot{q}) = \int_{t_i}^{t_f} dt \left[L + \frac{dH}{dt} \right] \\ &= S + H(q(t_f), \dot{q}(t_f), t_f) - H(q(t_i), \dot{q}(t_i), t_i) \end{aligned}$$

• Variation of the action:

$$\begin{aligned} \delta S' &= \delta S + \delta H_f - \delta H_i \\ &= \int dt \left\{ \left[\frac{\partial L}{\partial q_i} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) \delta q_i \right] + \left[\frac{\partial L}{\partial \dot{q}_i} \delta q_i \right]_{t_i}^{t_f} \right\} + \delta H_f - \delta H_i \end{aligned}$$

$$(EX) \quad L' = L + \frac{dH}{dt}$$

$$L = \frac{1}{2} m \dot{x}^2 - V(x) \quad \& \quad H = \beta x^\tau$$

$$\frac{dH}{dt} = \tau \beta \dot{x} x^{\tau-1}$$

$$L' = \frac{1}{2} m \dot{x}^2 - V(x) + \tau \beta \dot{x} x^{\tau-1}$$

$$\downarrow \text{E-L E} : \quad \frac{d}{dt} \left(\frac{\partial L'}{\partial \dot{x}} \right) = \frac{\partial L'}{\partial x}$$

(2)

$$\frac{d}{dt} \left(\frac{\partial L'}{\partial \dot{x}} \right) = \frac{\partial L'}{\partial x}$$

$$\frac{d}{dt} (m\dot{x} + \sigma \beta x^{\sigma-1}) = -\frac{\partial V}{\partial x} + \sigma(\sigma-1) \beta \dot{x} x^{\sigma-2}$$

$$m\ddot{x} + \sigma(\sigma-1) \beta \dot{x} x^{\sigma-2} = -\frac{\partial V}{\partial x} + \sigma(\sigma-1) \beta \dot{x} x^{\sigma-2}$$

* General Symmetry (Continuous Global Symmetry)

$$\lambda \rightarrow \dot{\lambda} = 0$$

$$\delta q_i = F_i(q, \dot{q}) \delta \lambda$$

$$\text{Lagr. } \delta L = \frac{dA(q, \dot{q})}{dt} \delta \lambda$$

↳ As long as this is true, the EOMs are invariant.

* Noether's Thm. Redux

Assume: symmetry (Global) → Use parameterization

$$\delta q_i = F_i \delta \lambda$$

$$\delta L = \frac{\partial L}{\partial q_i} \delta q_i + \frac{\partial L}{\partial \dot{q}_i} \delta \dot{q}_i = \left(\frac{\partial L}{\partial q_i} F_i + \frac{\partial L}{\partial \dot{q}_i} \dot{F}_i \right) \delta \lambda = \frac{dA}{dt} \delta \lambda$$

$$\text{Global symmetry} \rightarrow \therefore \delta \dot{\lambda} = 0$$

$$\frac{\partial L}{\partial q_i} F_i + \frac{\partial L}{\partial \dot{q}_i} \dot{F}_i = \frac{dA}{dt}$$

↳ Do the Noether's trick.

$\lambda = \lambda(t)$ → became a fn. of time.

$$\delta q_i = F_i \delta \lambda(t)$$

(Not allowed)

$$F_i = \frac{\delta q_i}{\delta \lambda(t)}$$

To see what happens:

$$\delta S = \int dt [\delta L] = \int dt \left[\frac{\partial L}{\partial q_i} \delta q_i + \frac{\partial L}{\partial \dot{q}_i} \delta \dot{q}_i \right]$$

$$= \int dt \left[\frac{\partial L}{\partial q_i} F_i \delta \lambda + \frac{\partial L}{\partial \dot{q}_i} \dot{F}_i \delta \lambda + \frac{\partial L}{\partial \dot{q}_i} F_i \delta \dot{\lambda} \right]$$

Using Symmetry:

$$\delta S = \int_{t_i}^{t_f} dt \left[\frac{dA}{dt} \delta \lambda + \frac{\partial L}{\partial \dot{q}_i} F_i \frac{d}{dt} (\delta \lambda) \right]$$

$$= \left[\frac{\partial L}{\partial \dot{q}_i} F_i \delta \lambda \right]_{t_i}^{t_f} + \int_{t_i}^{t_f} dt \left[\frac{dA}{dt} \delta \lambda - \delta \lambda \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} F_i \right) \right]$$

$$= \int_{t_i}^{t_f} dt \left[\frac{dA}{dt} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} F_i \right) \right] \delta \lambda$$

$$\delta \lambda(t_f) = 0$$

$$\delta \lambda(t_i) = 0$$

= 0

E-L E

by the principle of least ac.

$$(\delta L = 0)$$

(3)

$$G = \sum_{i=1}^N \frac{\partial L}{\partial \dot{q}_i} F_i - A \rightarrow \frac{dG}{dt} = 0$$

G : conserved quantity associated w/ symmetry.

The most important conserved quantity: E_n .

Time translation symmetry: $t \rightarrow t + \Delta t$.

coord.s: $q_i(t) \rightarrow q_i(t) + \Delta q_i(t) = q_i(t + \Delta t)$

$$\left\{ \begin{aligned} \delta q_i &= \dot{q}_i \delta t \\ &= \frac{\delta q_i}{\delta t} \delta t \end{aligned} \right\} \quad \& \quad \left\{ \begin{aligned} \delta \dot{q}_i &= \ddot{q}_i \delta t \end{aligned} \right\}$$

$$\delta q_i = \dot{q}_i \delta t = F_i \delta \lambda \rightarrow F_i = \dot{q}_i$$

$$\delta t \equiv \delta \lambda$$

$$\delta L = \frac{dA}{dt} \delta \lambda \rightarrow \delta L(q, \dot{q}) = \frac{\partial L}{\partial q_i} \dot{q}_i \delta t + \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i \delta t$$

$$= \frac{dL}{dt} \delta t$$

chain rule

$$\therefore A = L.$$

$$G = \frac{\partial L}{\partial \dot{q}_i} F_i - A = \frac{\partial L}{\partial \dot{q}_i} \dot{q}_i - L = p_i \dot{q}_i - L = H$$

= Energy

$$\therefore G = H.$$

* Another EX.

Lagr. $\rightarrow$ indep. of $q_k \rightarrow q_i, i=1, \dots, k, \dots, N$
 $K = \text{one } \#.$
 (Kappa)

$$\text{Just for } i=k: \frac{\partial L}{\partial q_k} = 0$$

$$\text{Symmetry} \rightarrow q_k \rightarrow q_k + \lambda \quad \& \quad \dot{q}_k \rightarrow \dot{q}_k$$

Global Symmetry

$$\dot{\lambda} = 0$$

velocity is left invariant.

$$\delta L(\dot{q}_k, q_{i \neq k}, \dot{q}_{i \neq k}) = 0 \rightarrow \therefore A = 0$$

$$= \frac{dA}{dt} \delta \lambda$$

$$\delta q_k = \delta \lambda \quad \& \quad F_i = 0 \quad \text{for } i \neq k$$

$$F_{i=k} = 1$$

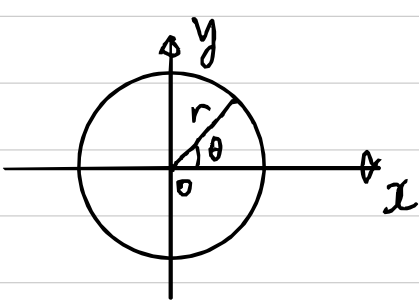
$$F_i = \delta_{ik}$$

$$A = 0 \rightarrow \therefore G = \sum_{i=1}^N \frac{\partial L}{\partial \dot{q}_i} F_i = \sum_{i=1}^N \frac{\partial L}{\partial \dot{q}_i} \delta_{ik} = \frac{\partial L}{\partial \dot{q}_k}$$

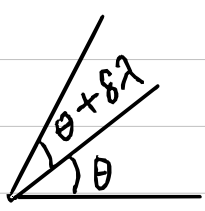
= p_k = generalized mtm.

(EX)

• A particle in 2D in a central potl.:



$$S = \int dt \left[\frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) - V(r) \right]$$



$$\theta \rightarrow \theta + \lambda \quad \& \quad \left\{ \begin{aligned} \delta \theta &= \delta \lambda \\ \delta r &= 0 \end{aligned} \right\}$$

$$\left(\begin{aligned} F_\theta &= 1, \\ F_r &= 0 \end{aligned} \right)$$

$$\delta L = 0 \rightarrow A = 0$$

$$G = \frac{\partial L}{\partial \dot{r}} \frac{\delta r}{\delta \lambda} + \frac{\partial L}{\partial \dot{\theta}} \frac{\delta \theta}{\delta \lambda} - 0 = \frac{\partial L}{\partial \dot{\theta}} = m r^2 \dot{\theta}$$

= Ang. mtm.

* Recap.

Time translation $\delta q_i(t) = \dot{q}_i(t) \delta t$

Noether "charge" $G = H = \sum_{i=1}^N \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - L$

$$\left. \begin{aligned} H &= \sum_{i=1}^N p_i \dot{q}_i - L \\ \& \quad L &= \sum p \dot{q} - H \end{aligned} \right\} : \text{Legendre Transform.}$$

* Thermodynamics

• First Law : $dU = TdS - PdV$

$U(S, V)$

$$dU = \left(\frac{\partial U}{\partial S} \right)_V dS + \left(\frac{\partial U}{\partial V} \right)_S dV$$

$$\left\{ \begin{aligned} T &= \left(\frac{\partial U}{\partial S} \right)_V \\ P &= - \left(\frac{\partial U}{\partial V} \right)_S \end{aligned} \right.$$

$F = U - T \cdot S$

"Free En."

$$\begin{aligned} dF &= dU - TdS - (dT)S \\ &= \cancel{TdS} - PdV - \cancel{TdS} - SdT \\ &= -SdT - PdV \end{aligned}$$

$$dF(V, T) = \left(\frac{\partial F}{\partial T} \right)_V dT + \left(\frac{\partial F}{\partial V} \right)_T dV$$

$$\downarrow$$

$$S = - \left(\frac{\partial F}{\partial T} \right)_V \quad \& \quad P = - \left(\frac{\partial F}{\partial V} \right)_T$$

(EX) $N=1$, $q_i = q$

Action : $S = \int dt \, L(q, \dot{q}, t)$

$$\left. \begin{aligned} H &= p \dot{q} - L \\ -H &= L - p \dot{q} \end{aligned} \right\} \longleftrightarrow \left\{ \begin{aligned} L &\equiv U \\ S &\equiv \dot{q} \\ T &\equiv P \\ -H &\equiv F \end{aligned} \right.$$

$$dL(q, \dot{q}, t) = \left(\frac{\partial L}{\partial q} \right)_{\dot{q}, t} dq + \left(\frac{\partial L}{\partial \dot{q}} \right)_{q, t} d\dot{q} + \left(\frac{\partial L}{\partial t} \right)_{q, \dot{q}} dt$$

$H = p \dot{q} - L$

$$dH = p d\dot{q} + \dot{q} dp - dL$$

$$= p \cancel{d\dot{q}} + \dot{q} dp - \frac{\partial L}{\partial q} dq - p \cancel{d\dot{q}} - \frac{\partial L}{\partial t} dt$$

$$= - \left(\frac{\partial L}{\partial q} \right)_{\dot{q}, t} dq + \dot{q} dp - \left(\frac{\partial L}{\partial t} \right)_{q, \dot{q}} dt$$

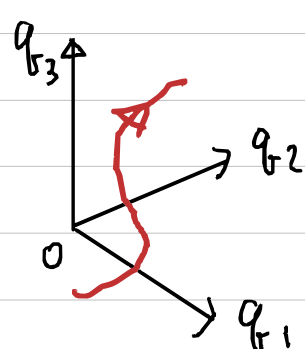
$$\begin{aligned} dH(q, p, t) &= \left(\frac{\partial H}{\partial q} \right)_{p, t} dq + \left(\frac{\partial H}{\partial p} \right)_{q, t} dp \\ &\quad + \left(\frac{\partial H}{\partial t} \right)_{q, p} dt \end{aligned}$$

$$\left(\frac{\partial H}{\partial q} \right)_{p, t} = - \left(\frac{\partial L}{\partial q} \right)_{\dot{q}, t}$$

$$\left(\frac{\partial H}{\partial p} \right)_{q, t} = \dot{q} = \left(\frac{\partial L}{\partial \dot{q}} \right)_{q, t}$$

$$\dot{q} = \left(\frac{\partial H}{\partial p} \right)_{q, t}$$

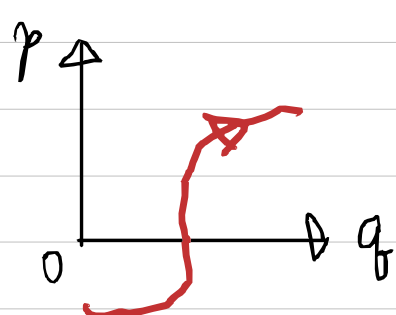
* Configuration Space



$$L(q, \dot{q}, t)$$

Trajectory $q_i(t)$
($i=1, \dots, N$)

Phase Space : $2N$ dimensional.



$$(q_i(t), p_i(t))$$

(EX) Particle in 3D

Phase space: (x, y, z, p_x, p_y, p_z)

$$m \ddot{q}_i = - \frac{\partial V}{\partial q_i}$$

1st-order DEs.

E-L:

$$S = \int dt L(q, \dot{q}, t) = \int dt \left[\sum_{i=1}^N p_i \dot{q}_i - H(q, p, t) \right]$$

Hamilton's principle ("Principle of least ac.")

$$\delta S = \int_{t_i}^{t_f} dt \left[\sum_{i=1}^N \left(p_i \delta \dot{q}_i + \delta p_i \dot{q}_i \right) - \sum_{i=1}^N \frac{\partial H}{\partial q_i} \delta q_i - \sum_{i=1}^N \frac{\partial H}{\partial p_i} \delta p_i \right]$$

$$= \left[\sum_{i=1}^N p_i \delta q_i \right]_{t_i}^{t_f} + \int_{t_i}^{t_f} dt \sum_{i=1}^N \left(-\dot{p}_i - \frac{\partial H}{\partial q_i} \right) \delta q_i$$

$$+ \int_{t_i}^{t_f} dt \sum_{i=1}^N \left(\dot{q}_i - \frac{\partial H}{\partial p_i} \right) \delta p_i$$

$$\delta S = 0 \quad \& \quad \dot{q}_i = \frac{\partial H}{\partial p_i} \quad \& \quad \dot{p}_i = - \frac{\partial H}{\partial q_i}$$

(Hamilton's Eqn.s)

• Non-rel. particle in 1D

$$S = \int dt \left[\frac{1}{2} m \dot{x}^2 - V(x) \right] \quad \& \quad p = \frac{\partial L}{\partial \dot{x}} = m \dot{x}$$

$$\& \quad \dot{x} = \frac{1}{m} p$$

$$H(x, p_x) = p_x \dot{x} - L = p_x \dot{x} - \frac{1}{2} m \dot{x}^2 + V(x)$$

$$= \frac{1}{m} p_x^2 - \frac{p_x^2}{2m} + V = \frac{p_x^2}{2m} + V = KE + PE = E_{\text{tot}}$$

$$H = \frac{p_x^2}{2m} + V(x)$$

$$\dot{x} = \frac{\partial H}{\partial p_x} = \frac{p_x}{m} \quad (\checkmark) \quad \& \quad \dot{p}_x = - \left(\frac{\partial H}{\partial x} \right)_{p_x} = - \frac{\partial V}{\partial x} \quad (\checkmark)$$

$$m \ddot{x} = - \frac{\partial V}{\partial x} \quad \& \quad \dot{x} = \frac{1}{m} p_x$$

$$\dot{p}_x = - \frac{\partial V}{\partial x}$$

③

$$E-L: S = \int dt L(q, \dot{q}, t)$$

$$\text{Hamilton: } S = \int dt \left[\sum_{i=1}^N p_i \dot{q}_i - H(q, p, t) \right]$$

Recap. Hamiltonian mechanics.

$$p_i = \frac{\partial L}{\partial \dot{q}_i} \rightarrow L(q, \dot{q}) = \sum_{i=1}^N p_i \dot{q}_i - H(q, p)$$

Legendre Tr.

$$\dot{q}_i = \frac{\partial H}{\partial p_i} \rightarrow H(q, p) = \sum_{i=1}^N p_i \dot{q}_i - L(q, \dot{q})$$

$$S[q, p] = \int_{t_i}^{t_f} dt [p_i \dot{q}_i - H(q, p, t)]$$

$$\begin{aligned} \delta S &= \int_{t_i}^{t_f} dt \left[\delta p_i \dot{q}_i + p_i \delta \dot{q}_i - \frac{\partial H}{\partial q_i} \delta q_i - \frac{\partial H}{\partial p_i} \delta p_i \right] \\ &= \left[p_i \delta q_i \right]_{t_i}^{t_f} + \int_{t_i}^{t_f} dt \left[\underbrace{\left(\dot{q}_i - \frac{\partial H}{\partial p_i} \right)}_{=0} \delta p_i + \underbrace{\left(-\dot{p}_i - \frac{\partial H}{\partial q_i} \right)}_{=0} \delta q_i \right] \end{aligned}$$

↙
• Hamilton's Eqn.s

$$\dot{q}_i = \frac{\partial H}{\partial p_i} \quad \& \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

(EX) charged non-rel. particle

$$S = \int dt \left[\frac{1}{2} m \dot{x}^2 + q A_x(t) \dot{x} - q \phi(x, t) \right]$$

$$p_x = \frac{\partial L}{\partial \dot{x}} = m \dot{x} + q A_x$$

$$\dot{x} = \frac{1}{m} (p_x - q A_x)$$

$$\begin{aligned} H = p_x \dot{x} - L &= \frac{p_x}{m} (p_x - q A_x) - \frac{1}{2} m \left(\frac{1}{m} (p_x - q A_x) \right)^2 \\ &\quad - q A_x \frac{1}{m} (p_x - q A_x) + q \phi \end{aligned}$$

$$= \frac{1}{2m} (p_x - q A_x)^2 + q \phi$$

$$\dot{x} = \frac{\partial H}{\partial p_x} = \frac{1}{m} (p_x - q A_x)$$

$$\dot{p}_x = -q \underbrace{\frac{\partial \phi}{\partial x}}_{\text{Spatial deriv. of Coulomb potl.}} + \frac{q}{m} \frac{\partial A_x}{\partial x} (p_x - q A_x)$$

Spatial
deriv.
of
Coulomb potl.

(2)

* Poisson Brackets

Consider an arbitrary fn. $F(q(t), p(t), t)$:

$$\frac{dF(q, p, t)}{dt} = \frac{\partial F}{\partial t} + \sum_{i=1}^N \frac{\partial F}{\partial q_i} \dot{q}_i - \sum_{i=1}^N \frac{\partial F}{\partial p_i} \dot{p}_i$$

Defn. Poisson Brackets

$$\{A, B\} = \sum_{i=1}^N \left(\frac{\partial A}{\partial q_i} \frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i} \frac{\partial B}{\partial q_i} \right) = 0$$

$$\text{bc } \frac{\partial}{\partial q_i} \frac{\partial}{\partial p_i} = \frac{\partial}{\partial p_i} \frac{\partial}{\partial q_i}$$

$$\frac{dF}{dt} = \frac{\partial F}{\partial t} + \{F, H\}$$

$$\dot{q}_i = \frac{dq_i}{dt} = \{q_i, H\} \quad \& \quad \dot{p}_i = \frac{dp_i}{dt} = \{p_i, H\}$$

$$\dot{q}_i = \{q_i, H\} = \sum_{j=1}^N \left(\frac{\partial q_i}{\partial q_j} \frac{\partial H}{\partial p_j} - \frac{\partial q_i}{\partial p_j} \frac{\partial H}{\partial q_j} \right)$$

$$\frac{\partial q_i}{\partial q_j} = \delta_{ij} \quad \& \quad \frac{\partial q_i}{\partial p_j} = 0$$

$$\therefore \{q_i, H\} = \sum_{j=1}^N \delta_{ij} \frac{\partial H}{\partial p_j} = \frac{\partial H}{\partial p_i} \quad (\checkmark)$$

$$\dot{p}_i = \{p_i, H\} = \sum_{j=1}^N \left(\frac{\partial p_i}{\partial q_j} \frac{\partial H}{\partial p_j} - \frac{\partial p_i}{\partial p_j} \frac{\partial H}{\partial q_j} \right)$$

$$= - \frac{\partial H}{\partial q_i}$$

(✓)

(3)

$$\{q_i, q_j\} = 0 \quad \& \quad \{p_i, p_j\} = 0$$

$$\{q_i, p_j\} = \delta_{ij} = \sum_{k=1}^N \left(\frac{\partial q_i}{\partial q_k} \frac{\partial p_j}{\partial p_k} - 0 \right)$$

$$= \sum_{k=1}^N \delta_{ik} \delta_{jk}$$

$$= \delta_{ij}$$

$$\left\{ \begin{array}{l} \frac{\partial p_i}{\partial q_k} = 0 \\ \frac{\partial q_i}{\partial p_k} = 0 \end{array} \right\}$$

$$\{q_i, p_j\} = \delta_{ij}$$

• Poisson Brackets satisfy:

*1. Leibniz rule: $\{A, BC\} = \{A, B\}C + B\{A, C\}$

$$\{A, BC\} = \sum_i \left[\frac{\partial A}{\partial q_i} \frac{\partial (BC)}{\partial p_i} - (q \leftrightarrow p) \right]$$

$$= \sum_i \left[B \frac{\partial A}{\partial q_i} \frac{\partial C}{\partial p_i} + C \frac{\partial A}{\partial q_i} \frac{\partial B}{\partial p_i} - (q \leftrightarrow p) \right]$$

$$= B \{A, C\} + \{A, B\} C$$

*2. Anti-symm.: $\{A, B\} = -\{B, A\}$

*3. Jacobi Identity

$$\{A, \{B, C\}\} + \{B, \{C, A\}\} + \{C, \{A, B\}\} = 0$$

* Heisenberg proposed:

$$q_i, p_i \rightarrow \hat{q}_i, \hat{p}_i$$

promoting the coord. & mom $\rightarrow$ ops

$$[\hat{A}, \hat{B}] = -[\hat{B}, \hat{A}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

$$[\hat{q}_i, \hat{q}_j] = 0 \quad \& \quad [\hat{p}_i, \hat{p}_j] = 0 \quad \& \quad [\hat{q}_i, \hat{p}_j] = i\hbar \delta_{ij}$$

a fn.

$\hat{F}(\hat{q}, \hat{p}, t)$ satisfies:

$$\frac{d\hat{F}}{dt} = \frac{\partial \hat{F}}{\partial t} + \frac{1}{i\hbar} [\hat{F}, \hat{H}] \quad : \text{Heisenberg EOM.}$$

(4)

(Qtm)

Commutators satisfy:

(1)

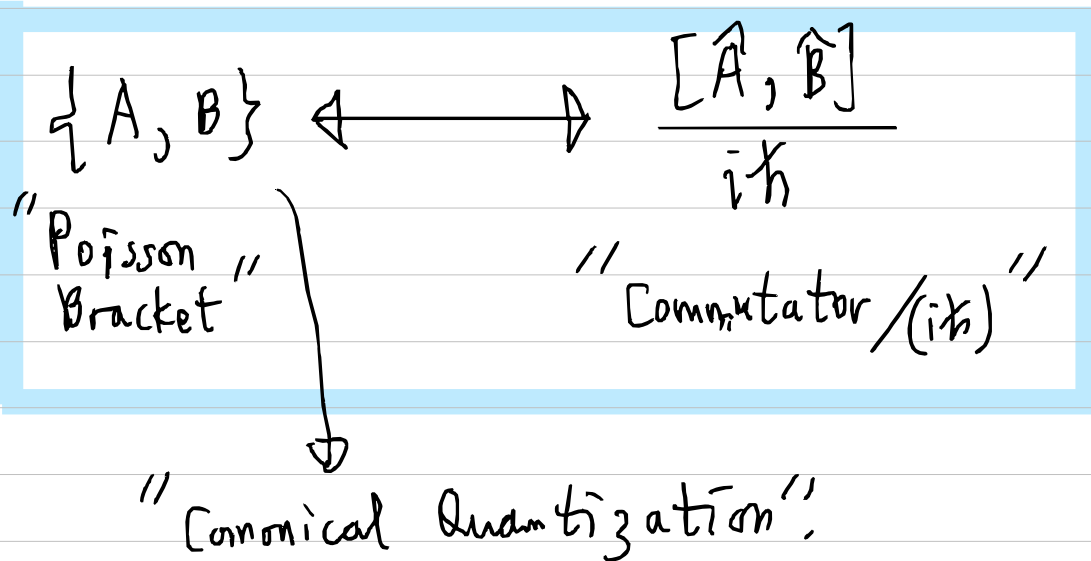
$$\begin{aligned}
 [\hat{A}, \hat{B}\hat{C}] &= \hat{A}\hat{B}\hat{C} - \hat{B}\hat{C}\hat{A} = [\hat{A}, \hat{B}]\hat{C} + \hat{B}\hat{A}\hat{C} - \hat{B}\hat{C}\hat{A} \\
 &= \hat{A}\hat{B}\hat{C} - \hat{B}\hat{A}\hat{C} + \hat{B}\hat{A}\hat{C} - \hat{B}\hat{C}\hat{A}
 \end{aligned}$$

(2)

$$[\hat{A}, \hat{B}] = -[\hat{B}, \hat{A}]$$

(3)

Jacobi Idem.: $[\hat{A}, [\hat{B}, \hat{C}]] + [\hat{B}, [\hat{C}, \hat{A}]] + [\hat{C}, [\hat{A}, \hat{B}]] = 0$



ACP L13

Poisson Brackets in QM

$$[\hat{q}_i, \hat{p}_j] = i\hbar \delta_{ij}$$

Poisson Brackets in CM (Classical Mech)

$$\{A, B\} = \frac{1}{i\hbar} [A, B]$$

Hamilton's Eq.s :

$$\frac{dF}{dt} = \frac{\partial F}{\partial t} + \{F, H\}$$

in CM.

$$\dot{q}_i = \frac{dq_i}{dt} = \{q_i, H\} \quad , \quad \dot{p}_i = \{p_i, H\}$$

$$\frac{d\hat{F}}{dt} = \frac{\partial \hat{F}}{\partial t} + \frac{1}{i\hbar} [\hat{F}, \hat{H}] \quad \text{in QM}$$

In Lagrangian form, $q'_i = q'_i(q_1, \dots, q_N)$

$$L'(q'_1, \dots, q'_N, t) = L(q_1, \dots, q_N, t)$$

(2)

§ 2.4, Canonical Transformations

To preserve the existence of Poisson Brackets,
we do the change of coord.s in phase space.
($2N - \text{dim.}$)

Then, we get a set of new coord.s : Q & P

$$\{q_1, \dots, q_N, p_1, \dots, p_N\} \rightarrow \{Q_1, \dots, Q_N, P_1, \dots, P_N\}$$

$$q \text{ \& } p \longrightarrow \begin{matrix} Q_I(q, p, t) \\ P_I(q, p, t) \end{matrix}$$

$$\{Q_I, Q_J\}_{q,p} = 0 \quad \& \quad \{P_I, P_J\}_{q,p} = 0$$

P.B.'s are left invariant.

$$\begin{aligned} \{A, B\} &= \sum_{i=1}^N \left\{ \frac{\partial A}{\partial q_i} \frac{\partial B}{\partial p_i} - (p \leftrightarrow q) \right\} \\ &= \sum_{I=1}^N \left\{ \frac{\partial A}{\partial Q_I} \frac{\partial B}{\partial P_I} - (P \leftrightarrow Q) \right\} \end{aligned}$$

canonical transformation

(3)

 $2N - \text{dim.} : \infty$ 

$$\{q_i, q_j\} = 0, \quad \{p_i, p_j\} = 0, \quad \{q_i, p_j\} = \delta_{ij}$$

$$\{Q_I, Q_J\} = 0, \quad \{P_I, P_J\} = 0, \quad \{Q_I, P_J\} = \delta_{IJ}$$

Canonical conjugates

$$\text{Check: } \{Q_I, Q_J\} = \sum_{i=1}^N \left(\frac{\partial Q_I}{\partial q_i} \frac{\partial Q_J}{\partial p_i} - \frac{\partial Q_I}{\partial p_i} \frac{\partial Q_J}{\partial q_i} \right)$$

$$= 0$$

$$\{P_I, P_J\} = \sum_{i=1}^N \left(\frac{\partial P_I}{\partial q_i} \frac{\partial P_J}{\partial p_i} - \frac{\partial P_I}{\partial p_i} \frac{\partial P_J}{\partial q_i} \right)$$

$$= 0$$

$$\{Q_I, P_J\} = \sum_{i=1}^N \left[\frac{\partial Q_I}{\partial q_i} \frac{\partial P_J}{\partial p_i} - \frac{\partial Q_I}{\partial p_i} \frac{\partial P_J}{\partial q_i} \right]$$

$$= \delta_{IJ}$$

④

The simplest possible case : $N=1$

i.e. a particle moving in 1D space dim.

$$(q, p) \longrightarrow (Q, P)$$

Transform

$$\textcircled{1} \text{ Check } \{Q_I, Q_J\} = \{Q, Q\} = 0$$

$$\textcircled{2} \text{ Check } \{P_I, P_J\} = \{P, P\} = 0$$

$$\textcircled{3} \text{ Check } \{Q_I, P_J\} = \{Q, P\} = 1 = \delta_{IJ}$$

$$= \frac{\partial Q}{\partial q} \frac{\partial P}{\partial p} - \frac{\partial Q}{\partial p} \frac{\partial P}{\partial q} \quad (I=J)$$

$$= 1 \cdot 1 - 0$$

$$= 1$$

EX1. $\begin{bmatrix} Q = q + f(p) \\ P = p \end{bmatrix} \rightarrow \frac{\partial Q}{\partial q} = 1, \frac{\partial P}{\partial p} = 1$ ⑤

$\frac{\partial Q}{\partial p} = \frac{\partial f(p)}{\partial p}, \frac{\partial P}{\partial q} = 0$

$$\therefore \{Q, P\} = 1 \times 1 - 0 = 1$$

Change frames of reference $\longleftrightarrow$ Coord. Transform

EX2. $Q = q$ & $P = p + f(q)$

$$\frac{\partial Q}{\partial q} = 1, \frac{\partial Q}{\partial p} = 0, \frac{\partial P}{\partial q} = \frac{\partial f(q)}{\partial q}, \frac{\partial P}{\partial p} = 1$$

$$\{Q, P\} = 1$$

EX3. In phase space, p 's & q 's are set on equal footing.

$$Q = p, P = -q$$

$$\frac{\partial Q}{\partial q} = 0, \frac{\partial Q}{\partial p} = 1, \frac{\partial P}{\partial q} = -1, \frac{\partial P}{\partial p} = 0$$

$$\{Q, P\} = 0 - 1 \cdot (-1) = 1$$

EX4. $Q = e^q, P = e^{-q} p$

$$\frac{\partial Q}{\partial q} = e^q, \frac{\partial Q}{\partial p} = 0, \frac{\partial P}{\partial q} = -e^{-q} p,$$

$$\frac{\partial P}{\partial p} = e^{-q}$$

$$\begin{aligned} \{Q, P\} &= e^q e^{-q} - (0) \times (-e^{-q} p) \\ &= 1 \end{aligned}$$

§ 2.4.1. Infinitesimal canonical transformations. (6)

Parameterizing by λ

$$Q_i = Q_i(q, p, \lambda)$$

$\uparrow$
 $t \rightarrow \lambda$

"Infinitesimal" $\Rightarrow$ Infinite number of infinitesimals
 $\approx$ Finite

$$\delta q_i = [Q_i(q, p, \delta\lambda) - q_i] = F_i \delta\lambda$$

$$S = \int dt (p_i \dot{q}_i - H) \xrightarrow{\text{NEW}} S' = \int dt \left[\sum_i p_i \dot{Q}_i - K \right]$$

Do the infinitesimal transformation:

$$\begin{cases} Q_i = q_i + \delta q_i \\ P_i = p_i + \delta p_i \end{cases} \rightarrow F \text{ is linear}$$

$$F = A \delta\lambda$$

$\uparrow$
"infinitesimal"

"Variational Calculus"

$$\begin{array}{l} H \rightarrow K \\ p_i \rightarrow P_i \\ q_i \rightarrow Q_i \end{array}$$

$$K = H + \delta H$$

$$\delta\lambda \frac{dA}{dt} = \sum_i p_i \underbrace{\delta \dot{q}_i}_{(1)} + \sum_i \delta p_i \underbrace{\dot{q}_i}_{(2)} - \delta H$$

$$\text{LHS} = \delta\lambda \frac{dA}{dt} = \delta\lambda \left[\frac{\partial A}{\partial t} + \sum_{i=1}^N \dot{q}_i \frac{\partial A}{\partial \dot{q}_i} + \sum_{i=1}^N \dot{p}_i \frac{\partial A}{\partial \dot{p}_i} \right]$$

$$\text{RHS} = \sum_j p_j \left[\underbrace{\frac{\partial(\delta q_j)}{\partial t}}_{(1)} + \sum_{i=1}^N \dot{q}_i \frac{\partial(\delta q_j)}{\partial \dot{q}_i} + \sum_{i=1}^N \dot{p}_i \frac{\partial(\delta q_j)}{\partial \dot{p}_i} \right]$$

$$+ \sum_i \underbrace{(\delta p_i) \dot{q}_i}_{(2)} - \delta H$$

$\dot{q}_i$'s & $\dot{p}_i$'s cancel.

$$\sum_j p_j \frac{\partial(\delta q_j)}{\partial \dot{q}_i} + \delta p_i = \delta\lambda \frac{\partial A}{\partial \dot{q}_i}$$

$$\sum_j p_j \frac{\partial(\delta q_j)}{\partial \dot{p}_i} = \delta\lambda \frac{\partial A}{\partial \dot{p}_i}$$

$$\delta H = \sum p_i \frac{\partial(\delta q_i)}{\partial t} - \frac{\partial A}{\partial t} \delta\lambda$$

Define:

$$G = \sum_j p_j \frac{\partial(\delta q_j)}{\partial \lambda} - A$$

"Noether charge"

$\parallel$
 F_j

$$\delta p_i = - \frac{\partial G}{\partial \dot{q}_i} \delta\lambda = \{p_i, G\} \delta\lambda$$

$$\delta q_i = + \frac{\partial G}{\partial \dot{p}_i} \delta\lambda = \{q_i, G\} \delta\lambda$$

$$G = \sum_i p_i \frac{\delta q_i}{\delta \lambda} - A$$

$$\begin{aligned} \{A, B\} &= \sum_i \left\{ \frac{\partial A}{\partial q_i} \frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i} \frac{\partial B}{\partial q_i} \right\} \\ &= \sum_j \left\{ \frac{\partial A}{\partial q_j} \frac{\partial B}{\partial p_j} - \frac{\partial A}{\partial p_j} \frac{\partial B}{\partial q_j} \right\} \end{aligned}$$

$$L' = L + \frac{dF}{dt}$$

$$\sum_j P_j \dot{Q}_j - K = \sum_i p_i \dot{q}_i - H + \frac{dF}{dt}$$

$$\begin{bmatrix} P_i = p_i + \delta p_i \\ Q_i = q_i + \delta q_i \\ K = H + \delta H \\ F = A \delta \lambda \end{bmatrix}$$

G = generating func.

Time Evolution

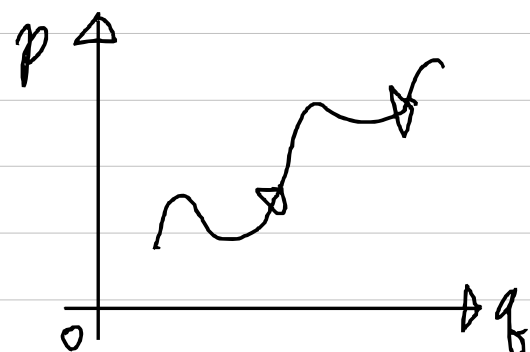
$$\delta q = \dot{q} \delta t$$

$$\delta p = \dot{p} \delta t$$

Hamilton

$$\dot{q} = \{q, H\}, \quad \dot{p} = \{p, H\}$$

$$\delta q_i = \{q_i, H\} \delta t, \quad \delta p_i = \{p_i, H\} \delta t$$



(Ex) Non-rel. particle in 2D.

In Cartesian (x, y)

$$S = \int dt \left[\frac{1}{2} m (\dot{x}^2 + \dot{y}^2) - V(r) \right]$$

$$r = \sqrt{x^2 + y^2}$$

$$\delta x = -y \delta \theta$$

$$\delta y = +x \delta \theta$$

$$\begin{bmatrix} x' = x \cos \theta - y \sin \theta \\ y' = y \cos \theta + x \sin \theta \end{bmatrix}$$

$A=0$

$$G = \sum_i p_i \frac{\delta q_i}{\delta \lambda} = p_x \frac{\delta x}{\delta \lambda} + p_y \frac{\delta y}{\delta \lambda}$$

$(\delta \lambda = \delta \theta)$

$$= \underline{x p_y - y p_x} = \text{Orbital Ang. Mtm.}$$

$\parallel$
 L_z

$$\delta x = \{x, L_z\} \delta \theta = \{x, x p_y - y p_x\} \delta \theta$$

$$= \frac{\partial x}{\partial x} \frac{\partial}{\partial p_x} (x p_y - y p_x) \delta \theta = -y \delta \theta$$

$$\delta y = \{y, L_z\} \delta \theta = \frac{\partial L_z}{\partial p_y} \delta \theta = x \delta \theta$$

$$\{A, B\} = \frac{[\hat{A}, \hat{B}]}{i\hbar}$$

(CM)

(QM)

$$\{x, p_x\} = 1$$

(QM)

$$[x, p_x] \frac{1}{i\hbar} = 1$$

$$\Rightarrow [x, p_x] = i\hbar$$

$$\Delta x = \sqrt{\langle \psi | \hat{x}^2 | \psi \rangle - \langle \psi | \hat{x} | \psi \rangle^2}$$

$$X = X(x, p_x)$$

$$P = P(x, p_x)$$

$$\Delta x \Delta p_x \geq \frac{\hbar}{2}$$

U.P.

$$\{X, P_x\} = 1 \rightarrow [\hat{X}, \hat{P}_x] = i\hbar$$

$$\Delta X \Delta P_x \geq \frac{\hbar}{2}$$

(Ex)

$$X = x + \alpha p_x$$

$$P_x = p_x$$

$$\Delta (x + \alpha p_x) \Delta p_x \geq \frac{\hbar}{2} \quad \therefore \text{U.P.}$$

$$\{X, x\} = -\alpha$$

$$\Delta (x + \alpha p_x) \Delta x \geq \frac{\hbar}{2} |\alpha|$$

(2)

Schwinger's formulation of least action

$$S = \int_{t_i}^{t_f} \left[\sum_{i=1}^N p_i \dot{q}_i - H(q, p) \right] dt$$

$$(t \rightarrow \lambda) \longrightarrow dt = \frac{dt}{d\lambda} d\lambda$$

$$S = \int_{\lambda_i}^{\lambda_f} \left(\sum_{i=1}^N \left(p_i \frac{dq_i}{d\lambda} \right) - H \frac{dt}{d\lambda} \right) d\lambda$$

Vary : $q_i \rightarrow q_i + \delta q_i$

$p_i \rightarrow p_i + \delta p_i$

$t \rightarrow t + \delta t$

$$\delta S = \int_{\lambda_i}^{\lambda_f} d\lambda \left[\sum_{i=1}^N \delta p_i \frac{dq_i}{d\lambda} + \sum_{i=1}^N p_i \frac{d(\delta q_i)}{d\lambda} - H \frac{d(\delta t)}{d\lambda} - \sum_{i=1}^N \frac{\partial H}{\partial p_i} \delta p_i \frac{dt}{d\lambda} - \sum_{i=1}^N \frac{\partial H}{\partial q_i} \delta q_i \frac{dt}{d\lambda} - \frac{\partial H}{\partial t} \delta t \frac{dt}{d\lambda} \right]$$

I b P.:

$$\delta S = \left[\sum_{i=1}^N p_i \delta q_i - H \delta t \right]_{t_i}^{t_f}$$

$$+ \int_{\lambda_i}^{\lambda_f} d\lambda \left[\sum_i \delta p_i \left(\frac{dq_i}{d\lambda} - \frac{\partial H}{\partial p_i} \frac{dt}{d\lambda} \right) + \sum_i \delta q_i \left(- \frac{dp_i}{d\lambda} - \frac{\partial H}{\partial q_i} \frac{dt}{d\lambda} \right) + \delta t \left(\frac{dH}{d\lambda} - \frac{\partial H}{\partial t} \frac{dt}{d\lambda} \right) \right]$$

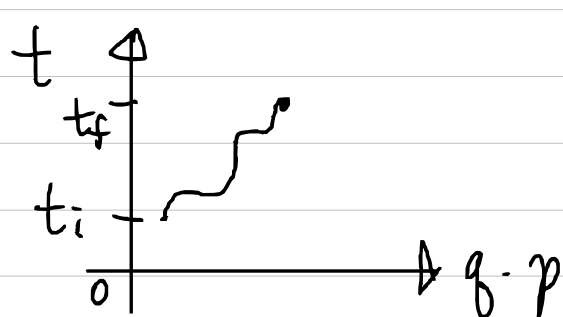
$$\downarrow \frac{d\lambda}{dt}$$

$$= \left[\sum_{i=1}^N p_i \delta q_i - H \delta t \right]_{t_i}^{t_f}$$

$$+ \int_{t_i}^{t_f} dt \left[\sum_i \delta p_i \left(\frac{dq_i}{dt} - \frac{\partial H}{\partial p_i} \cdot 1 \right) + \sum_i \delta q_i \left(- \frac{dp_i}{dt} - \frac{\partial H}{\partial q_i} \cdot 1 \right) + \delta t \left(\frac{dH}{dt} - \frac{\partial H}{\partial t} \cdot 1 \right) \right]$$

$$+ \sum_i \delta q_i \left(- \frac{dp_i}{dt} - \frac{\partial H}{\partial q_i} \cdot 1 \right) + \delta t \left(\frac{dH}{dt} - \frac{\partial H}{\partial t} \cdot 1 \right) \right]$$

$$\delta S = \delta S_{\text{boundary}} + \cancel{\delta S_{\text{bulk}}}$$



$$\delta S_{\text{bulk}} = 0$$

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = - \frac{\partial H}{\partial q_i}, \quad \frac{dH}{dt} = \frac{\partial H}{\partial t}$$

$$\frac{dF}{dt} = \frac{\partial F}{\partial t} + \{F, H\} \quad \left\{ \begin{array}{l} \text{when} \\ F=H \end{array} \right.$$

$$\frac{dH}{dt} = \frac{\partial H}{\partial t} + \{H, H\}$$

$$\delta S = \sum_i p_i(t_f) \delta q(t_f) - H(q(t_f), p(t_f), t_f) \delta t_f - \sum_i p_i(t_i) \delta q(t_i) + H(t_i, q(t_i), p(t_i)) \delta t_i$$

$$dU = TdS - PdV \quad : \text{1st Law of TD}$$

$$U(S, V)$$

Action

$$\left(\text{Varying boundaries} \right) \iff \left(\text{Varying } \delta q(t_f) \text{ \& } \delta t(t_f) \right)$$

position time

$$S = S(q(t_i), t_i; q(t_f), t_f)$$

$$\frac{\partial S}{\partial q_i(t_f)} = p_i(t_f) \quad \& \quad \frac{\partial S}{\partial t_f} = -H(t_f, q(t_f), p(t_f))$$

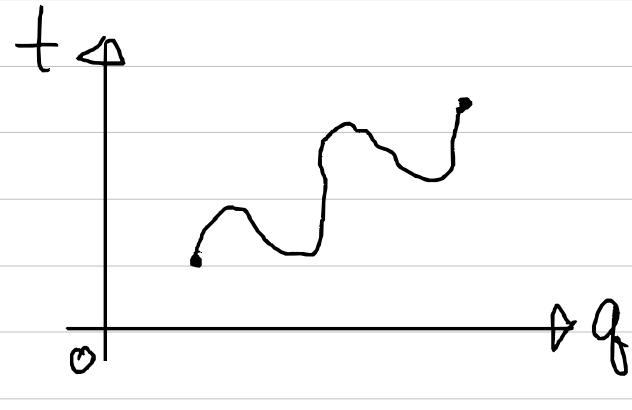
$$\frac{\partial S}{\partial q_i(t_i)} = -p_i(t_i) \quad \& \quad \frac{\partial S}{\partial t_i} = H(t_i, q(t_i), p(t_i))$$

Hamilton - Jacobi Formulation

$$S = \int_{t_i}^{t_f} dt \left[\sum_i p_i \dot{q}_i - H \right]$$

$$\delta S = \sum_i p_i(t_f) \delta q_i(t_f) - H(q(t_f), p(t_f), t_f) \delta t_f \\ - \sum_i p_i(t_i) \delta q_i(t_i) + H(q(t_i), p(t_i), t_i) \delta t_i$$

$$S = S(q(t_i), t_i; q(t_f), t_f)$$



EX $H = \sum_i \frac{1}{2m} p_i^2 + V(q)$

$$-\frac{\partial S}{\partial t_f} = \sum_i \frac{1}{2m} \left(\frac{\partial S}{\partial t_f^i} \right)^2 + V(q_{t_f})$$

Schr. Eq

Hamilton-Jacobi Eq.

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \sum_i \frac{\partial^2 \Psi}{\partial q_i^2} + V(q) \Psi$$

Sol. $\Psi = e^{iS/\hbar} \rightarrow \frac{\partial \Psi}{\partial t} = \frac{i}{\hbar} (\dot{S}) \Psi$

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\partial S}{\partial t} \Psi$$

$$\frac{\partial \Psi}{\partial q} = \frac{i}{\hbar} \frac{\partial S}{\partial q} \Psi$$

$$\frac{\partial^2 \Psi}{\partial q^2} = \frac{i}{\hbar} \frac{\partial^2 S}{\partial q^2} \Psi - \frac{1}{\hbar^2} \left(\frac{\partial S}{\partial q} \right)^2 \Psi$$

Set $\hbar = 0 \rightarrow$ Classical Case Again.

EX 1D non-rel. particle $q=x$

$$H = \frac{p^2}{2m} + V(x) \quad (\text{T.E.})$$

$\frac{\partial H}{\partial t} = 0 \rightarrow H = E$ conserved $(q(t_f), t_f)$

Legendre Transform

$$S = \tilde{S} + E t_i$$

$$\delta S = \sum_i p_i(t_f) \delta q_i(t_f) - H(q(t_f), p(t_f), t_f) \delta t_f \\ - p_i(t_i) \delta q_i(t_i) - t_i \delta E$$

$E \delta t_i = \delta(E t_i) - t_i \delta E$ P.I.

$$\frac{\partial \tilde{S}}{\partial t_f} = -H(t_f) = -E$$

Integrate $\tilde{S} = -E t_f + W(x_i, x_f)$ wrt t

$$E = \frac{1}{2m} \left(\frac{\partial \tilde{S}}{\partial x_f} \right)^2 + V(x_f)$$

$$= \frac{1}{2m} \left(\frac{\partial W}{\partial x_f} \right)^2 + V(x_f)$$

$$\left(\frac{\partial W}{\partial x_f} \right)^2 = 2m(E - V(x_f))$$

$$\frac{\partial W}{\partial x_f} = \pm \sqrt{2m(E - V(x_f))}$$

Sol.:

$$\therefore W(x_i, x_f) = \pm \int_{x_i}^{x_f} \sqrt{2m(E - V(x))} dx + C(x_i)$$

$$S = \tilde{S} + E t_i$$

$$S = -E(t_f - t_i) \pm \int_{x_i}^{x_f} \sqrt{2m(E - V(x))} dx$$

$$\tilde{S} = S - E t_i = -E t_f \pm \int_{x_i}^{x_f} \sqrt{2m(E - V(x))} dx$$

$$t_i = -\frac{\partial \tilde{S}}{\partial E} = \mp \int_{x_i}^{x_f} \frac{\partial}{\partial E} \left(\sqrt{2m(E - V(x))} \right) dx$$

$$= t_f \mp \int_{x_i}^{x_f} \frac{m}{\sqrt{2m(E - V(x))}} dx$$

$$t_f - t_i = \pm \int_{x_i}^{x_f} \frac{m}{\sqrt{2m(E - V(x))}} dx$$

$$E = \frac{p^2}{2m} + V(x) \quad \& \quad p = \pm \sqrt{2m(E - V(x))}$$

$$\frac{dx}{dt} = v = \frac{p}{m} \rightarrow \frac{dx}{dt} = \pm \frac{\sqrt{2m(E - V(x))}}{m}$$

Invert

$$\frac{dt}{dx} = \pm \frac{m}{\sqrt{2m(E - V(x))}}$$

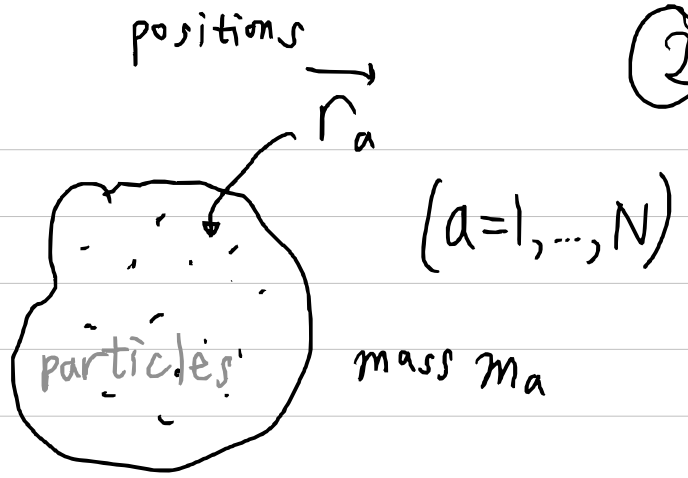
$$\therefore t_f - t_i = \int_{x_i}^{x_f} \frac{dt}{dx} dx = \int_{t_i}^{t_f} dt$$

C3. Rigid Bodies

(2)

constant

$$\frac{d}{dt} \left(|\vec{r}_a - \vec{r}_b|^2 \right) = 0$$



1st Define the COM: $\vec{R} = \frac{1}{M} \sum_{a=1}^{N_f} m_a \vec{r}_a$

(External DOF) (weighted average of the position)

$$M = \sum_{a=1}^{N_f} m_a$$

M : tot. mass

$$\vec{r}_a^* = \vec{r}_a - \vec{R}$$

(Internal DOF)

$$\frac{d}{dt} (\vec{r}_a^* \cdot \vec{r}_a^*) = 0 = 2 \vec{r}_a^* \cdot \frac{d \vec{r}_a^*}{dt}$$

bc it's internal length.

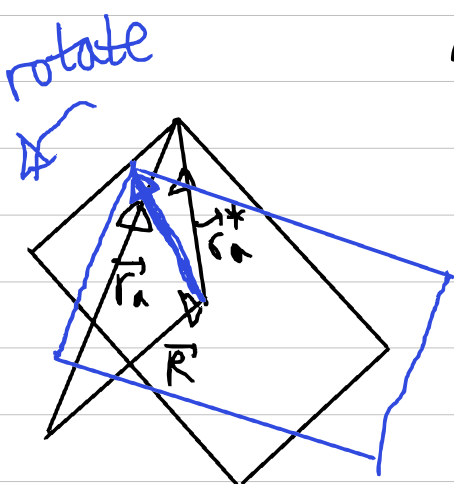
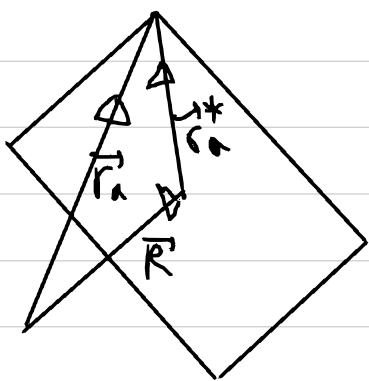
Sol.: TORQUE

$$\frac{d \vec{r}_a^*}{dt} = \vec{\omega} \times \vec{r}_a^*$$

$$(\vec{\omega} \times \vec{r}_a^*) \cdot \vec{r}_a^* = 0$$

orthogonal

$\vec{\omega}$: ang. vel.



$$v = \omega r$$

$$\omega = \frac{d\theta}{dt}$$

$$KE = \frac{1}{2} \sum_{a=1}^{N_f} m_a \dot{\vec{r}}_a^2$$

$$\text{p.I. : } \vec{r}_a = \vec{r}_a^* + \vec{R}$$

$$KE = \frac{1}{2} \sum_{a=1}^{N_f} m_a \left[\dot{\vec{R}}^2 + 2 \dot{\vec{R}} \cdot \dot{\vec{r}}_a^* + \dot{\vec{r}}_a^{*2} \right]$$

vanish bc it's internal dot external.

Easier Way:

$$\sum_a m_a \vec{r}_a^* = 0$$

$$= \sum_a m_a (\vec{r}_a - \vec{R})$$

$$= \sum_a m_a \vec{r}_a - \left(\sum_a m_a \right) \vec{R} = 0$$

$$T = \frac{1}{2} M \dot{\vec{R}}^2 + \underbrace{\sum_a \frac{1}{2} m_a (\vec{\omega} \times \vec{r}_a^*)^2}_{\frac{1}{2} I \omega^2}$$

$$S = \int dt [T - V]$$

$$= \int dt \left[\frac{1}{2} M \dot{\vec{R}}^2 + \sum_a \frac{1}{2} m_a (\vec{\omega} \times \vec{r}_a^*)^2 - V(\vec{R}) \right]$$

ACP L16

12/11/2024

①

$$T_{\text{free}} = \frac{1}{2} M \dot{\vec{R}}^2 + \sum_a \frac{1}{2} m_a (\vec{\omega} \times \vec{r}_a^*)^2$$

$$\vec{R}_{\text{com}} = \frac{\sum_a m_a \vec{r}_a}{M}$$

$$M = \sum_a m_a$$

$$= \int d^3r \rho(\vec{r})$$

$$\dot{\vec{r}}_a^* = \vec{\omega} \times \vec{r}_a^*$$

$\vec{\omega}$: ang. vel.

$$T_{\text{free}} = \frac{1}{2} M \dot{\vec{R}}^2 + \frac{1}{2} \int d^3r \rho(\vec{r}) (\vec{\omega} \times \vec{r})^2$$

Pivot $\vec{r}_p = 0$

$$T_{\text{piv}} = \frac{1}{2} \sum_a m_a (\vec{\omega} \times \vec{r}_a)^2$$

$$= \frac{1}{2} \int d^3r \rho(\vec{r}) (\vec{\omega} \times \vec{r})^2$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = \epsilon_{ijk} A_j B_k$$

(Levi-Civita tensor ϵ_{ijk})

$$(\vec{\omega} \times \vec{r}^*)_i = \sum_{j,k=1}^3 \epsilon_{ijk} \omega_j r_k^*$$

$$\epsilon_{ijk} = \begin{cases} +1, & \text{if } (i,j,k) \text{ is } (1,2,3), (2,3,1), (3,1,2) \\ -1, & \text{if } (i,j,k) \text{ is } (3,2,1), (2,1,3), (1,3,2) \\ 0, & \text{if } (i,j,k) \text{ is } (1,2,2), (1,2,1), \dots \end{cases}$$

$$(\vec{A} \times \vec{B})_i = \epsilon_{ijk} A_j B_k$$

$$(\vec{A} \times \vec{B})_1 = \epsilon_{1jk} A_j B_k = \epsilon_{123} A_2 B_3 + \epsilon_{132} A_3 B_2 = A_2 B_3 - A_3 B_2$$

$$(\vec{\omega} \times \vec{r}_a^*)^2 = (\vec{\omega} \times \vec{r}_a^*) \cdot (\vec{\omega} \times \vec{r}_a^*)$$

$$= (\vec{\omega} \times \vec{r}_a^*)_i (\vec{\omega} \times \vec{r}_a^*)_i$$

$$= (\epsilon_{ijk} \omega_j r_k^*) (\epsilon_{ilm} \omega_l r_m^*)$$

$$= \sum_{i,j,k,l,m=1}^3 \epsilon_{ijk} \epsilon_{ilm} \omega_j r_k^* \omega_l r_m^*$$

$$= \omega_j \omega_l (\delta_{jl} \delta_{km} r_k^* r_m^* - \delta_{jm} \delta_{kl} r_k^* r_m^*)$$

$$= \omega_j \omega_l (\delta_{jl} r_m^* r_m^* - r_l^* r_j^*) = \omega_j \omega_l [(\vec{r}^*)^2 \delta_{jl} - r_l^* r_j^*]$$

rank-2 tensor

I_{ij}

MOI defined w.r.t. the origin;

$$I_{ij} = \sum_a m_a (\vec{r}_a^2 \delta_{ij} - r_{ai} r_{aj})$$

MOI defined relative to the COM:

$$I_{ij}^* = \sum_a m_a (\vec{r}_a^{*2} \delta_{ij} - r_{ai}^* r_{aj}^*)$$

∴ The action for the rotating body, which is free to move ∴

$$S = \int dt \left(\frac{1}{2} M \dot{\vec{R}}^2 + \frac{1}{2} \sum_{i,j=1}^3 \omega_i I_{ij}^* \omega_j \right)$$

$$\frac{1}{2} M \dot{\vec{R}}^2$$

$$= KE_{\text{transl}}$$

$$\frac{1}{2} I \omega^2$$

$$= KE_{\text{rot}}$$

$$I_{ij} = I_{ji} \rightarrow I_{ij}^* = \int d^3r \rho(\vec{r}) \left[\delta_{ij} (\vec{r}^*)^2 - r_{ai}^* r_{aj}^* \right]$$

Let $\vec{r}^* = (x, y, z)$,

MOI w.r.t. the COM.

$$I_{ij} = \int d^3r \rho(\vec{r}) \begin{pmatrix} y^2+z^2 & -xy & -xz \\ -yx & x^2+z^2 & -yz \\ -zx & -zy & x^2+y^2 \end{pmatrix} = \text{MOI w.r.t. the ORIGIN}$$

(dropping *) Rotation $(\vec{p}_a = m \dot{\vec{r}}_a^*)$

$$\vec{L} = \sum_a \vec{r}_a \times \vec{p}_a = \sum_a m_a \vec{r}_a \times \dot{\vec{r}}_a$$

$$\vec{L} = \sum_a m_a \vec{R} \times \dot{\vec{R}} + \sum_a m_a (\vec{R} \times \dot{\vec{r}}_a^* + \vec{r}_a^* \times \dot{\vec{R}})$$

$$+ \sum_a m_a \vec{r}_a^* \times \dot{\vec{r}}_a^* \quad \sum_a m_a \vec{r}_a^* = 0$$

$$= \vec{L}_{\text{com}} + \vec{L}^*$$

$$\vec{L}^* = \vec{L}_{\text{spin}}$$

Earth $\vec{x}^{\text{com}}_0$ orbital ang. mtn (spin ang mtn)

$$\vec{L}_{\text{com}} = M \vec{R} \times \dot{\vec{R}}$$

$$\dot{\vec{r}}_a^* = \vec{\omega} \times \vec{r}_a^*$$

$$\vec{L}^* = \sum_a m_a \vec{r}_a^* \times (\vec{\omega} \times \vec{r}_a^*)$$

$$L_i^* = \sum_a m_a r_{aj}^* (\vec{\omega} \times \vec{r}_a^*)_k$$

$$= \sum_a m_a \epsilon_{ijk} \epsilon_{k\ell m} (r_a^*)_j (r_a^*)_m \omega_\ell$$

$$\epsilon_{kij} \epsilon_{k\ell m} = \delta_{i\ell} \delta_{jm} - \delta_{im} \delta_{j\ell}$$

$$= \sum_a m_a (\delta_{i\ell} \delta_{jm} - \delta_{im} \delta_{j\ell}) (r_a^*)_j (r_a^*)_m \omega_\ell$$

$$= \sum_a m_a (\delta_{i\ell} (r_a^*)_j^2 - (r_a^*)_j (r_a^*)_i (r_a^*)_j) \omega_\ell$$

$$L = \frac{1}{2} a_{ij}(q) \dot{q}_i \dot{q}_j - V(q)$$

$$p_i = \frac{\partial L}{\partial \dot{q}_i} = a_{ij}(q) \dot{q}_j$$

$$L = \frac{1}{2} M \dot{\vec{R}}^2 + \frac{1}{2} I_{ij}^* \omega_i \omega_j$$

$$L_i^* = \frac{\partial L}{\partial \omega_i} = I_{ij}^* \omega_j$$

For extended bodies, e.g. $\vec{\omega} = (0, 0, \omega)$

$$L_z^* = I_{33}^* \omega_3$$

$$= L_3^*$$

$$L_2^* = I_{23}^* \omega_3$$

$$L_1^* = I_{13}^* \omega_3$$

$$\omega_3 = \omega$$

$$\omega_1 = 0 = \omega_2$$

§3.5. Euler Angles

Rotation $O(3)$

$$\vec{r}' = \underline{\underline{M}} \vec{r}$$

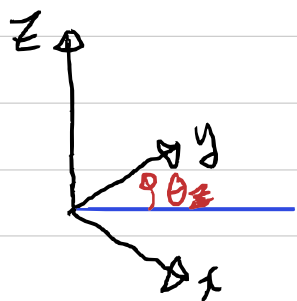
$$\underline{\underline{M}}^T = \underline{\underline{M}}^{-1} \quad \& \quad \underline{\underline{M}}^{-1} \underline{\underline{M}} = \underline{\underline{1}} = \underline{\underline{M}}^T \underline{\underline{M}}$$

ORTHOGONAL Matrix

$$\vec{r}' \cdot \vec{r}' = \vec{r} \cdot \vec{r} \quad \& \quad \vec{r}'^T \cdot \vec{r}' = \vec{r}^T \cdot \vec{r}$$

$$\vec{r}' = \underline{\underline{M}} \vec{r} \quad \& \quad (\vec{r}')^T = \vec{r}^T \underline{\underline{M}}^T$$

$$\vec{r}^T \underbrace{\underline{\underline{M}}^T \underline{\underline{M}}}_{\underline{\underline{1}}} \vec{r} = \vec{r}^T \vec{r}$$

Group $O(3)$ (Orthogonal
3x3 Lie group)

"z-rotation"

Rotation in the x-y plane

$$x' = x \cos \theta_z - y \sin \theta_z$$

$$y' = y \sin \theta_z + x \cos \theta_z$$

$$z' = z$$

$$\underline{\underline{M}}_z(\theta_z) = \begin{pmatrix} \cos \theta_z & -\sin \theta_z & 0 \\ \sin \theta_z & \cos \theta_z & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \underline{\underline{M}}_z(\theta_z) \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\underline{\underline{M}}_z^T(\theta_z) = \begin{pmatrix} \cos \theta_z & \sin \theta_z & 0 \\ -\sin \theta_z & \cos \theta_z & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\underline{\underline{M}}_z^T \underline{\underline{M}}_z = \underline{\underline{1}}$$

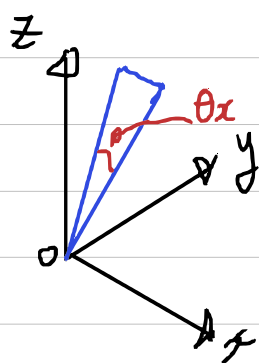
$$(\underline{\underline{M}}_1 \underline{\underline{M}}_2)^T (\underline{\underline{M}}_1 \underline{\underline{M}}_2) = \underline{\underline{M}}_2^T \underbrace{\underline{\underline{M}}_1^T \underline{\underline{M}}_1}_{\underline{\underline{1}}} \underline{\underline{M}}_2$$

$$= \underline{\underline{1}}$$

$$= \underline{\underline{M}}_2^T \underline{\underline{1}} \underline{\underline{M}}_2$$

$$= \underline{\underline{M}}_2^T \underline{\underline{M}}_2 = \underline{\underline{1}}$$

y-z plane

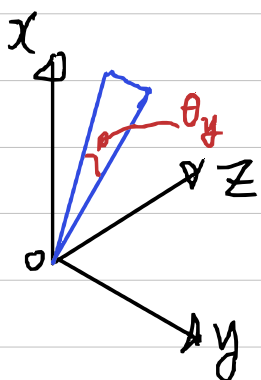


"x-rotation"

$$\underline{\underline{M}}_x(\theta_x) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_x & -\sin \theta_x \\ 0 & \sin \theta_x & \cos \theta_x \end{pmatrix}$$

$$z' = -y \sin \theta_x + z \cos \theta_x$$

x-z plane



"y-rotation"

$$\underline{\underline{M}}_y(\theta_y) = \begin{pmatrix} \cos \theta_y & 0 & \sin \theta_y \\ 0 & 1 & 0 \\ -\sin \theta_y & 0 & \cos \theta_y \end{pmatrix}$$

$$x' = -x \sin \theta_y + z \cos \theta_y$$

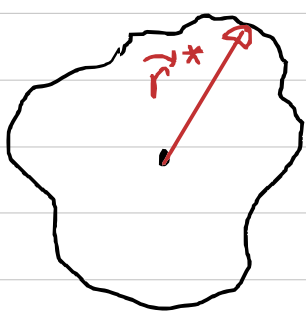
Euler angles

EX. x-rotation

$$\underline{\underline{M}}(\psi, \theta, \phi) = \underline{\underline{M}}_z(\phi) \underline{\underline{M}}_y(\theta) \underline{\underline{M}}_z(\psi)$$

the 1st & the last are the same

What's $\vec{\omega}$ in terms of ψ, θ, ϕ ?



$$\vec{r}^* = \underline{\underline{M}}_z(\phi) \underline{\underline{M}}_y(\theta) \underline{\underline{M}}_z(\psi) \vec{u}$$

$$= \underline{\underline{M}}(\phi, \theta, \psi) \vec{u}$$

$$\frac{d\vec{u}}{dt} = 0$$

$$\frac{d\vec{r}^*}{dt} = \vec{\omega} \times \vec{r}^*$$

(3)

$$\frac{d\vec{r}^*}{dt} = \frac{d\underline{\underline{M}}_z(\varphi)}{dt} \underline{\underline{M}}_y(\theta) \underline{\underline{M}}_z(\psi) \vec{u}$$

$$+ \underline{\underline{M}}_z(\varphi) \frac{d\underline{\underline{M}}_y(\theta)}{dt} \underline{\underline{M}}_z(\psi) \vec{u}$$

$$+ \underline{\underline{M}}_z(\varphi) \underline{\underline{M}}_y(\theta) \frac{d\underline{\underline{M}}_z(\psi)}{dt} \vec{u}$$

* Inverse relation

$$\vec{u} = \underline{\underline{M}}^T \vec{r}^*$$

$$\vec{u} = \underline{\underline{M}}_z^T(\gamma) \underline{\underline{M}}_y^T(\theta) \underline{\underline{M}}_z^T(\varphi) \vec{r}^*$$

$$\frac{d\vec{r}^*}{dt} = \dot{\underline{\underline{M}}}_z(\varphi) \underline{\underline{M}}_y(\theta) \underline{\underline{M}}_z(\psi) \underline{\underline{M}}_z^T(\gamma) \underline{\underline{M}}_y^T(\theta) \underline{\underline{M}}_z^T(\varphi) \vec{r}^*$$

$$+ \underline{\underline{M}}_z(\varphi) \dot{\underline{\underline{M}}}_y(\theta) \underline{\underline{M}}_z(\psi) \underline{\underline{M}}_z^T(\gamma) \underline{\underline{M}}_y^T(\theta) \underline{\underline{M}}_z^T(\varphi) \vec{r}^*$$

$$+ \underline{\underline{M}}_z(\varphi) \underline{\underline{M}}_y(\theta) \dot{\underline{\underline{M}}}_z(\psi) \underline{\underline{M}}_z^T(\gamma) \underline{\underline{M}}_y^T(\theta) \underline{\underline{M}}_z^T(\varphi) \vec{r}^*$$

$$= \underline{\underline{\Omega}} \vec{r}^*$$

$$\underline{\underline{\Omega}} = \dot{\underline{\underline{M}}}_z(\varphi) \underline{\underline{M}}_z^T(\varphi) + \underline{\underline{M}}_z(\varphi) \dot{\underline{\underline{M}}}_y(\theta) \underline{\underline{M}}_y^T(\theta) \underline{\underline{M}}_z^T(\varphi) + \underline{\underline{M}}_z(\varphi) \dot{\underline{\underline{M}}}_z(\psi) \underline{\underline{M}}_z^T(\psi) \underline{\underline{M}}_z^T(\varphi)$$

$$\underline{\underline{M}}_z(\varphi) = \begin{pmatrix} \cos \varphi & -\sin \varphi & 0 \\ \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\dot{\underline{\underline{M}}}_z(\varphi) = \dot{\varphi} \begin{pmatrix} -\sin \varphi & -\cos \varphi & 0 \\ \cos \varphi & -\sin \varphi & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$\frac{d}{dt}$

$$\dot{\underline{\underline{M}}}_z(\varphi) \underline{\underline{M}}_z^T(\varphi)$$

$$= \dot{\varphi} \begin{pmatrix} -\sin \varphi & -\cos \varphi & 0 \\ \cos \varphi & -\sin \varphi & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \cos \varphi & +\sin \varphi & 0 \\ -\sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \dot{\varphi} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

(4)

$$\left(\dot{\underline{M}}_{\underline{z}}(\varphi) \underline{M}_{\underline{z}}^T(\varphi) \right)_{12} = -\dot{\varphi} = -\epsilon_{123} \dot{\varphi} = \Omega_{12}$$

$$\underline{\Omega}_{ij} = -\sum_{k=1}^3 \epsilon_{ijk} \omega_k \quad (\omega = \dot{\varphi})$$

$$\underline{\underline{\Omega}}^T = -\underline{\underline{\Omega}}$$

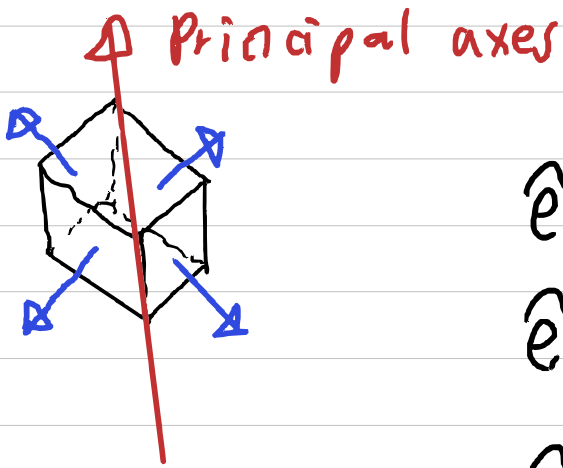
In 3D, anti-symm. 2-tensor is a vector.

$$\begin{aligned} \frac{d\vec{r}_i^*}{dt} &= \Omega_{ij} \vec{r}_j^* = -\epsilon_{ijk} \omega_k \vec{r}_j^* \\ &= +\epsilon_{ijk} \omega_k \vec{r}_k^* \\ &\equiv (\vec{\omega} \times \vec{r}^*)_i \end{aligned}$$

$$\vec{\omega} = \begin{pmatrix} \dot{\psi} \cos \theta \sin \varphi - \dot{\theta} \sin \varphi \\ \dot{\psi} \sin \theta \cos \varphi + \dot{\theta} \cos \varphi \\ \dot{\varphi} + \dot{\psi} \cos \theta \end{pmatrix}$$

$$\hat{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \hat{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \hat{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Principal axes themselves are rotating.



$$\hat{e}_1 = M(\psi, \theta, \varphi) \hat{i}$$

$$\hat{e}_2 = M(\psi, \theta, \varphi) \hat{j}$$

$$\hat{e}_3 = M(\psi, \theta, \varphi) \hat{k}$$

$$\hat{e}_3 = \begin{pmatrix} \cos \varphi \sin \theta \\ \sin \varphi \sin \theta \\ \cos \theta \end{pmatrix}$$

$$\hat{e}_i \cdot \hat{e}_j = \delta_{ij}$$

Useful to write $\vec{\omega}$ in terms of non-inertial frame:

$$\vec{\omega} = \begin{pmatrix} -\dot{\psi} \cos \theta \sin \varphi + \dot{\theta} \sin \varphi \\ \dot{\psi} \sin \theta \sin \varphi + \dot{\theta} \cos \varphi \\ \dot{\varphi} + \dot{\psi} \cos \theta \end{pmatrix}$$

Free $S = \int dt \left[\frac{1}{2} M \dot{\vec{R}}^2 + \frac{1}{2} \omega_i I_{ij}^* \omega_j + \dots \right]$

$$I_{ij}^* = \sum_a m_a (\vec{r}_a^{*2} \delta_{ij} - r_i^* r_j^*)$$

$$\equiv \int d^3\vec{r} \rho(\vec{r}) (\vec{r}^{*2} \delta_{ij} - r_i^* r_j^*)$$

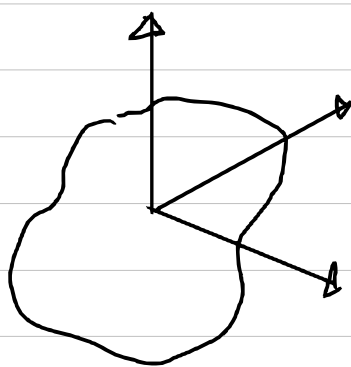
$$I_{ij} \equiv \int d^3\vec{r} \rho(\vec{r}) (\vec{r}^2 \delta_{ij} - r_i r_j)$$

§3.6. Principal Axes of Inertia

$I_{ij} \rightarrow$ 2-tensor
"MOI" tensor

$\underline{\underline{I}} \rightarrow$ real
& symmetric

$$\underline{\underline{I}}^T = \underline{\underline{I}}$$



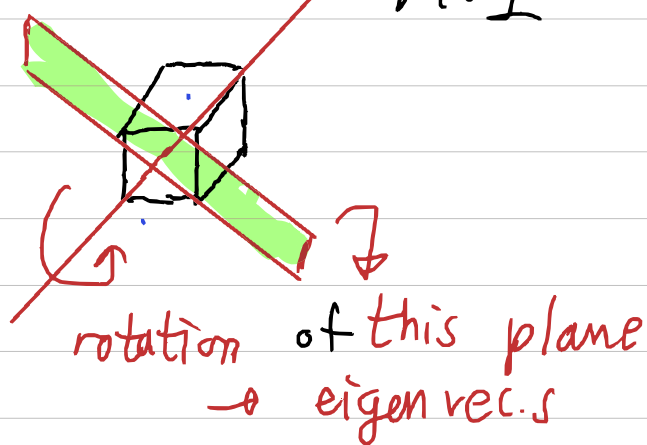
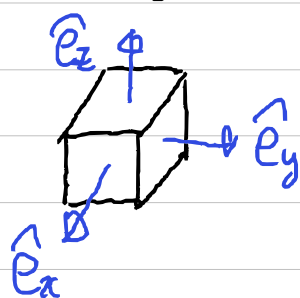
* Eigenvalue / Eigenvector

$$\underline{\underline{I}} \hat{e}_\alpha = I_\alpha \hat{e}_\alpha$$

$\uparrow$
 α labelling eigenvector

$\hat{e}_\alpha$: Principal axes

$\hat{e}_\alpha$ (principal)
MOI



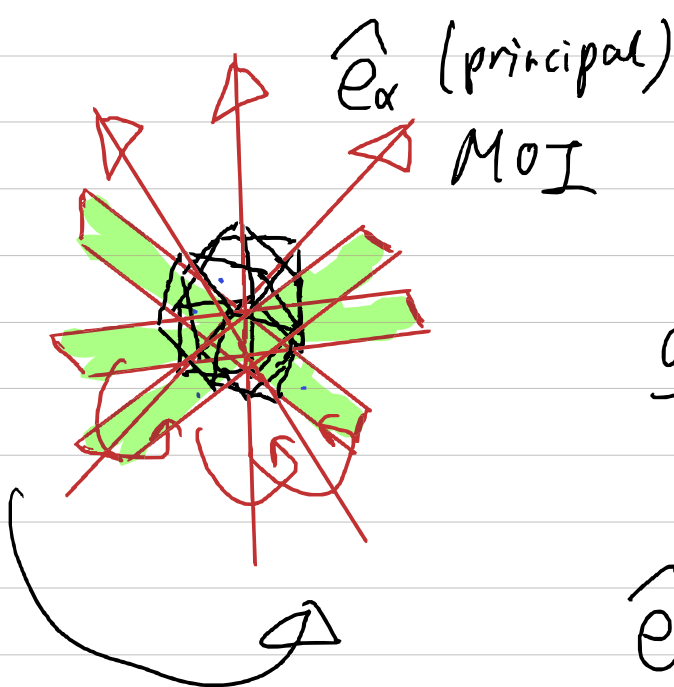
plane $\perp$ orthogonal
to $\hat{e}_\alpha$

$$\hat{e}_\beta^T \underline{\underline{I}} \hat{e}_\alpha = I_\alpha \hat{e}_\beta^T \hat{e}_\alpha$$

$$= (\underline{\underline{I}} \hat{e}_\beta)^T \hat{e}_\alpha = I_\beta \hat{e}_\beta^T \hat{e}_\alpha$$

$$\left. \begin{array}{l} I_\alpha \neq I_\beta, \quad \hat{e}_\beta^T \hat{e}_\alpha = 0 \\ I_\alpha = I_\beta, \quad \hat{e}_\beta^T \hat{e}_\alpha = 1 \end{array} \right\} \underline{\underline{\hat{e}_\beta^T \hat{e}_\alpha = \delta_{\alpha\beta}}}$$

(2)

Totally Symm. : $I_1 = I_2 = I_3$ Symmetric : $I_1 = I_2 \neq I_3$ Cyl.
→ (symm.)

$$\frac{d\hat{e}_\alpha}{dt} = \vec{\omega} \times \hat{e}_\alpha$$

$$\hat{e}_1 = M_z(\varphi) M_y(\theta) M_z(\psi) \hat{i}$$

$$\hat{e}_2 = M_z(\varphi) M_y(\theta) M_z(\psi) \hat{j}$$

$$\hat{e}_3 = M_z(\varphi) M_y(\theta) M_z(\psi) \hat{k}$$

In Cartesian / Inertial frm,

$$\begin{aligned} \underline{\omega} = & (\dot{\psi} \cos \theta \sin \varphi - \dot{\theta} \sin \varphi) \hat{i} \\ & + (\dot{\psi} \sin \theta \cos \varphi + \dot{\theta} \cos \varphi) \hat{j} \\ & + (\dot{\psi} \cos \theta + \dot{\varphi}) \hat{k} \end{aligned}$$

$$\underline{\omega} = \sum_{\alpha} \omega_{\alpha} \hat{e}_{\alpha} \quad (\text{non-inertial / rotating frm})$$

$$\begin{aligned} \underline{\omega} = & (-\dot{\varphi} \cos \psi \sin \theta + \dot{\theta} \sin \psi) \hat{e}_1 \\ & + (\dot{\varphi} \sin \psi \sin \theta + \dot{\theta} \cos \psi) \hat{e}_2 \\ & + (\dot{\psi} + \cos \theta \dot{\varphi}) \hat{e}_3 \end{aligned}$$

$$\underline{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k} = \sum_{\alpha} A_{\alpha} \hat{e}_{\alpha}$$

$$\begin{aligned} \left(\frac{dA}{dt} \right)_I &= \dot{A}_x \hat{i} + \dot{A}_y \hat{j} + \dot{A}_z \hat{k} \\ &= \sum_{\alpha} \dot{A}_{\alpha} \hat{e}_{\alpha} + \sum_{\alpha} A_{\alpha} \dot{\hat{e}}_{\alpha} \\ &= \sum_{\alpha} \dot{A}_{\alpha} \hat{e}_{\alpha} + \sum_{\alpha} A_{\alpha} \underline{\omega} \times \hat{e}_{\alpha} \end{aligned}$$

$$\left(\frac{dA}{dt} \right)_I = \left(\frac{dA}{dt} \right)_R + \underline{\omega} \times \underline{A}$$

Earth rotating & we living on it

→ all Non-inertial frm.

(3)

$$L_i^* = I_{ij}^* \omega_j \rightarrow \text{Ang. Mtm. w.r.t. COM}$$

$$\underline{L}^* = \underline{I}^* \underline{\omega} = \underline{I}^* \left(\sum_{\alpha} \omega_{\alpha} \hat{e}_{\alpha} \right)$$

$$= \sum_{\alpha} \omega_{\alpha} \underline{I}^* \hat{e}_{\alpha} = \sum_{\alpha} \omega_{\alpha} I_{\alpha}^* \hat{e}_{\alpha}$$

$$\underline{L}^* = \sum_{\alpha} L_{\alpha} \hat{e}_{\alpha} \quad \left(L_{\alpha}^* = \omega_{\alpha} I_{\alpha}^* \right. \\ \left. \uparrow \right. \\ \left. \text{NOT } ij! \right)$$

$$I_{ij} \longrightarrow I_{\alpha}$$

* Pivoted

$$S = \int dt \frac{1}{2} \omega_i I_{ij} \omega_j$$

$$= \int dt \frac{1}{2} \underline{\omega}^T \underline{I} \underline{\omega}$$

$$\left(\underline{\omega} = \sum_{\alpha} \omega_{\alpha} \hat{e}_{\alpha} \right) = \int dt \frac{1}{2} \underline{\omega}^T \sum_{\alpha} I_{\alpha} \omega_{\alpha} \hat{e}_{\alpha}$$

$$= \int dt \frac{1}{2} \sum_{\alpha=1}^3 I_{\alpha} \omega_{\alpha}^2$$

$$\left(\underline{\omega}^T = \sum_{\alpha} \omega_{\alpha} \hat{e}_{\alpha}^T \right)$$

$$\underline{\omega}^T \cdot \hat{e}_{\alpha} = \sum_{\beta} \omega_{\beta} \hat{e}_{\beta}^T \cdot \hat{e}_{\alpha} = \omega_{\alpha}$$

$$S = \int dt \frac{1}{2} \sum_{\alpha=1}^3 I_{\alpha} \omega_{\alpha}^2$$

$$= \int dt \frac{1}{2} \left[I_1 (-\dot{\varphi} \cos \psi \sin \theta + \dot{\theta} \sin \psi)^2 \right. \\ \left. + I_2 (\dot{\phi} \sin \psi \sin \theta + \dot{\theta} \cos \psi)^2 \right. \\ \left. + I_3 (\dot{\psi} + \cos \theta \dot{\varphi})^2 \right]$$

Typically Ang Mtm $\rightarrow$ conserved unless there's an external torque.

$$\left(\frac{d\underline{L}}{dt} \right)_{\underline{I}} = \underline{\tau} \quad : \text{torque}$$

$\uparrow$
Inertial

"Tariff"

$$\underline{I} = \sum_a \vec{r}_a \times \vec{r}_a$$

$$\left(\frac{d\underline{L}}{dt} \right)_R + \underline{\omega} \times \underline{L} = \underline{\tau}$$

$$\underline{L} = \sum_{\alpha} I_{\alpha} \omega_{\alpha} \hat{e}_{\alpha}$$

$$\left(\frac{d\underline{L}}{dt} \right)_R = \sum_{\alpha} I_{\alpha} \dot{\omega}_{\alpha} \hat{e}_{\alpha} \quad (\dot{I}_{\alpha} = 0)$$

$$(\underline{\omega} \times \underline{L})_i = \epsilon_{ijk} \omega_j L_k$$

$$(\underline{\omega} \times \underline{L})_{\alpha} = \sum_{\beta, \gamma} \epsilon_{\alpha\beta\gamma} \omega_{\beta} L_{\gamma} = \sum_{\beta, \gamma} \epsilon_{\alpha\beta\gamma} \omega_{\beta} I_{\gamma} \omega_{\gamma}$$

④

$$I_\alpha \frac{d\omega_\alpha}{dt} + \sum_{\beta, \gamma} \epsilon_{\alpha\beta\gamma} \omega_\beta I_\gamma \omega_\gamma = \tau_\alpha$$

($\beta, \gamma = 1, 2, 3$)

$$I_1 \dot{\omega}_1 + (I_3 - I_2) \omega_2 \omega_3 = \tau_1$$

$$I_2 \dot{\omega}_2 + (I_1 - I_3) \omega_3 \omega_1 = \tau_2$$

$$I_3 \dot{\omega}_3 + (I_2 - I_1) \omega_1 \omega_2 = \tau_3$$

$$\underline{\tau} = \sum_\alpha \tau_\alpha \hat{e}_\alpha$$



$$\left. \begin{aligned} \omega_1 &= 0 \\ \omega_2 &= 0 \\ \omega_3 &= \omega_0 \end{aligned} \right\} \text{P.I.}$$

$$\underline{\tau} = 0$$

Let's add perturbation.

$$\begin{aligned} \omega_1 &= 0 + \delta\omega_1 \\ \omega_2 &= 0 + \delta\omega_2 \\ \omega_3 &= \omega_0 + \delta\omega_3 \end{aligned}$$

$$I_1(\delta\dot{\omega}_1) + (I_3 - I_2)\delta\omega_2 \omega_0 = \tau_1$$

$$I_2\delta\dot{\omega}_2 + (I_1 - I_3)\omega_0\delta\omega_1 = \tau_2$$

These 2 are sufficient.

$$I_3\delta\dot{\omega}_3 + (I_2 - I_1)\delta\omega_1\delta\omega_2 = \tau_3$$

$$I_1\delta\ddot{\omega}_1 + (I_3 - I_2)\omega_0\delta\dot{\omega}_2 = 0$$

$$I_1\delta\ddot{\omega}_1 - \frac{1}{I_1}(I_3 - I_2)(I_1 - I_3)\omega_0^2\delta\omega_1 = 0$$

$$\delta\ddot{\omega}_1 = -\frac{1}{I_1 I_1}(I_3 - I_2)(I_3 - I_1)\omega_0^2\delta\omega_1$$



Mechanics

Field Theory

 t
 t
 i
 $q_i \rightarrow x$
 x
 q
 ϕ
 $\sum_i$
 $\int dx$
 m
 μ

e.g.

$$S = \int dt \sum_i \frac{1}{2} m \dot{q}_i^2 + \dots$$

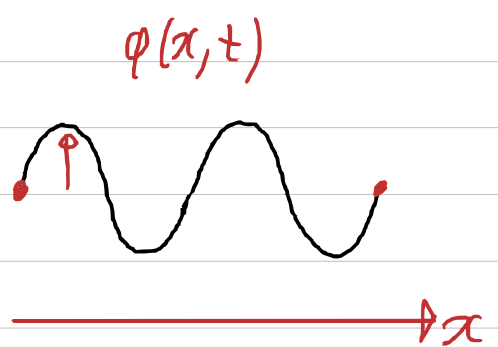
$$\downarrow$$

$$\int dt \int_0^{l_0} dx \frac{1}{2} \mu \left(\frac{\partial \phi}{\partial t} \right)^2 + \dots$$

= spacetime volume

$\phi(t, x)$

Classical Fld Theory

 $(1+1)\text{-dim}$
 ϕ


$$q_i = \phi(x)$$

$$q \rightarrow \phi$$

$$i \rightarrow x, y, z$$

$$m \rightarrow \mu$$

$$\dot{q}_i \rightarrow \dot{\phi}$$

$$L \rightarrow \mathcal{L}_\mu \text{ (Lagrangian Density)}$$

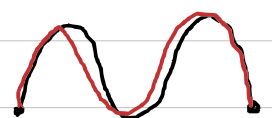
$$S = \int dt \int_0^{l_0} dx \left[\frac{1}{2} \mu \left(\frac{\partial \phi}{\partial t} \right)^2 - \frac{1}{2} k \left(\frac{\partial \phi}{\partial x} \right)^2 \right]$$

$\mathcal{L}(\phi, \partial_t \phi, \partial_x \phi)$

Principle of least action.

$$\delta S = 0$$

$$\phi(t, x) \Rightarrow \phi(t, x) + \delta \phi(t, x)$$



$$\delta S = \int dt \int_0^{l_0} dx \left[\frac{1}{2} \mu \left(2 \frac{\partial \phi}{\partial t} \frac{\partial (\delta \phi)}{\partial t} \right) - \frac{k}{2} \left(2 \frac{\partial \phi}{\partial x} \frac{\partial (\delta \phi)}{\partial x} \right) \right]$$

$$= \left(\frac{\partial (\phi + \delta \phi)}{\partial t} \right)^2 - \left(\frac{\partial \phi}{\partial t} \right)^2 - \left(\frac{\partial (\phi + \delta \phi)}{\partial x} \right)^2 + \left(\frac{\partial \phi}{\partial x} \right)^2$$

$$= \cancel{\left(\frac{\partial \phi}{\partial t} \right)^2} + 2 \frac{\partial \phi}{\partial t} \frac{\partial (\delta \phi)}{\partial t} + \left(\frac{\partial (\delta \phi)}{\partial t} \right)^2 - \cancel{\left(\frac{\partial \phi}{\partial t} \right)^2} - \left(\frac{\partial \phi}{\partial x} \right)^2 + 2 \frac{\partial \phi}{\partial x} \frac{\partial (\delta \phi)}{\partial x} + \left(\frac{\partial (\delta \phi)}{\partial x} \right)^2$$

②

Now,

Neglect time boundary & Track space boundary terms.

$$\delta S = \int dt \int_0^{l_0} dx \left[-\mu \frac{\partial^2 \varphi}{\partial t^2} \delta \varphi + k \left(\frac{\partial^2 \varphi}{\partial x^2} \delta \varphi \right) \right]$$

$$+ \int dt \left[-k \frac{\partial \varphi}{\partial x} \delta \varphi \right]_0^{l_0}$$

$$-\mu \frac{\partial^2 \varphi}{\partial t^2} + k \left(\frac{\partial^2 \varphi}{\partial x^2} \right) = 0$$

$$E - LE$$

$$\frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = \left(\frac{\partial^2 \varphi}{\partial x^2} \right) \quad \& \quad L = \sqrt{\frac{k}{\mu}}$$

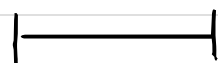
$$BC. : \left. \frac{\partial \varphi}{\partial x} \delta \varphi \right|_{\text{boundary}} = 0$$

$$\text{Dirichlet Cond. : } \left. \frac{\partial \varphi}{\partial x} \right|_{x=0, l_0} = \text{Newton BC.}$$

$$\delta \varphi = 0$$

time boundary $\rightarrow$ limit (t_0, t_f)

space boundary $\rightarrow$ actual boundary



(3)

$$(1+3)\text{-dim.} \longrightarrow x^\mu = (t, \underline{r})$$

$$x^0 = t$$

$$x^1 = x$$

$$x^2 = y$$

$$x^3 = z$$

Contravariant
(Up stairs)

$$c = \hbar = 1$$

* Partial derivs

$$\partial_\mu = \frac{\partial}{\partial x^\mu} \quad \& \quad \partial^\mu = \frac{\partial}{\partial x_\mu}$$

$$= \begin{pmatrix} \frac{\partial}{\partial t} \\ \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix}$$

* 4×4 Tensor

$$\partial_\mu \partial_\nu \varphi$$

$$(\mu, \nu = 0, 1, 2, 3)$$

$$\partial_\mu \partial^\mu = \square$$

* Cartesian (3×3) Tensor

$$\underline{I}_{ij}$$

$$(i, j = 1, 2, 3)$$

* Spacetime Volume Integral

$$\int d^4x = \int dx^0 \int_{-\infty}^{\infty} dx^1 \int_{-\infty}^{\infty} dx^2 \int_{-\infty}^{\infty} dx^3 = \int dt \int dV$$

$$S = \int dt \mathcal{L} = \int dt \int dV(\mathcal{L})$$

$$= \int d^4x (\mathcal{L})$$

(4)

* Scalar field theory in (3+1)-dim.

$$\hookrightarrow \varphi(x) = \varphi(x^0, x^1, x^2, x^3)$$

↑
no indices $\rightarrow$ scalar

generic form : $S = \int d^4x [\mathcal{L}(\varphi, \partial_\mu \varphi)]$

EX. K-G Action

$$(\square + m^2) \varphi = 0$$

$$= (\partial^2 + m^2) \varphi$$

$$= (\partial_\mu \partial^\mu + m^2) \varphi = 0$$

$$S = \int d^4x \left[\frac{1}{2} \mu \left(\frac{\partial \varphi}{\partial t} \right)^2 - \frac{k}{2} \left[\left(\frac{\partial \varphi}{\partial x} \right)^2 + \left(\frac{\partial \varphi}{\partial y} \right)^2 + \left(\frac{\partial \varphi}{\partial z} \right)^2 \right] - \frac{1}{2} m^2 \varphi^2 \right]$$

$$\Leftrightarrow (\partial_\mu \partial^\mu + m^2) \varphi = 0$$

$$S = \int d^4x \mathcal{L}(\varphi, \partial_\mu \varphi) \quad \partial_\mu (\delta \varphi)$$

$$\delta S = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \varphi} \delta \varphi + \sum_{\mu=0}^3 \frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi)} \delta (\partial_\mu \varphi) \right]$$

[1 term + 4 terms]

$$= \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \varphi} - \sum_{\mu=0}^3 \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi)} \right) \right] \delta \varphi$$

$$= 0$$

$$\frac{\partial \mathcal{L}}{\partial \varphi} = \sum_{\mu=0}^3 \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi)} \right)$$

$$= \partial_t \left(\frac{\partial \mathcal{L}}{\partial (\partial_t \varphi)} \right) + \partial_x \left(\frac{\partial \mathcal{L}}{\partial (\partial_x \varphi)} \right)$$

$$+ \partial_y \left(\frac{\partial \mathcal{L}}{\partial (\partial_y \varphi)} \right) + \partial_z \left(\frac{\partial \mathcal{L}}{\partial (\partial_z \varphi)} \right)$$

$$\text{LHS} = -m^2 \varphi$$

$$\text{RHS} = \partial_t \left(\mu \frac{\partial \varphi}{\partial t} \right) + \partial_x \left[-k \left(\frac{\partial \varphi}{\partial x} \right) \right]$$

$$+ \partial_y \left(-k \frac{\partial \varphi}{\partial y} \right) + \partial_z \left(-k \frac{\partial \varphi}{\partial z} \right)$$

$$-m^2 \varphi = \mu \ddot{\varphi} - k \left(\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial^2 \varphi}{\partial z^2} \right) \varphi$$

(= ∇^2)

$$= \mu \ddot{\varphi} - k \nabla^2 \varphi$$

$$\frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} - \nabla^2 \varphi = -\frac{m^2}{k} \varphi$$

$$\frac{\mu}{k} = \frac{1}{c^2}$$

describes spin 1 particles

$$\left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \right) \varphi = -\frac{m^2}{k} \varphi$$

Recap.

$$S = \int d^4x \underbrace{\mathcal{L}(\varphi, \partial_\mu \varphi)}_{\text{Lagr. density}}$$

$$E-L E: \quad \frac{\partial \mathcal{L}}{\partial \varphi} = \sum_{\mu=0}^3 \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi)} \right)$$

I: interaction
(classical)

$$\int d^4x = \int dx^0 \int dx^3 = \int dt \int dV$$

$$x^\mu = (t, x, y, z) = (x^0, x^1, x^2, x^3)$$

$$\partial_\mu = \frac{\partial}{\partial x^\mu} = (\partial_0, \partial_1, \partial_2, \partial_3) = (\partial_t, \partial_x, \partial_y, \partial_z)$$

★ Real scalar fld $\varphi(x)$

$$\mathcal{L} = -\frac{1}{2} \sum_{\mu=0}^3 \sum_{\nu=0}^3 \eta^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi - V(\varphi)$$

$$\eta^{\mu\nu} = \begin{pmatrix} -\frac{1}{c^2} & & 0 \\ & 1 & \\ 0 & & 1 & \\ & & & 1 \end{pmatrix} \quad \& \quad \eta_{\mu\nu} = \begin{pmatrix} -c^2 & & 0 \\ & 1 & \\ 0 & & 1 & \\ & & & 1 \end{pmatrix}$$

$$\mathcal{L} = \frac{1}{2c^2} (\partial_t \varphi)^2 - \underbrace{\frac{1}{2} (\partial_i \varphi)^2}_{\text{II}} - V(\varphi)$$

$$- \frac{1}{2} (\partial_x \varphi)^2 - \frac{1}{2} (\partial_y \varphi)^2 - \frac{1}{2} (\partial_z \varphi)^2$$

$$\frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi)} = \frac{\partial}{\partial (\partial_\mu \varphi)} \left(-\frac{1}{2} \eta^{\alpha\beta} \partial_\alpha \varphi \partial_\beta \varphi \right)$$

$$= -\frac{1}{2} \eta^{\alpha\beta} \frac{\partial (\partial_\alpha \varphi)}{\partial (\partial_\mu \varphi)} \partial_\beta \varphi$$

$$- \frac{1}{2} \eta^{\alpha\beta} \partial_\alpha \varphi \frac{\partial (\partial_\beta \varphi)}{\partial (\partial_\mu \varphi)}$$

$$\frac{\partial (\partial_\alpha \varphi)}{\partial (\partial_\mu \varphi)} = \delta^\mu_\alpha = \begin{cases} 0, & \text{if } \mu \neq \alpha \\ 1, & \text{if } \mu = \alpha \end{cases}$$

$$- \frac{1}{2} \eta^{\alpha\beta} \delta^\mu_\alpha \partial_\beta \varphi - \frac{1}{2} \eta^{\alpha\beta} \partial_\alpha \varphi \delta^\mu_\beta$$

$$= -\frac{1}{2} \eta^{\mu\beta} \partial_\beta \varphi - \frac{1}{2} \eta^{\mu\alpha} \partial_\alpha \varphi = - \sum_{\alpha=0}^3 \eta^{\mu\alpha} \partial_\alpha \varphi$$

$$\frac{\partial \mathcal{L}}{\partial \varphi} = -\frac{\partial V}{\partial \varphi} \quad \Rightarrow \quad \partial_\mu (-\eta^{\mu\alpha} \partial_\alpha \varphi) = -\frac{\partial V}{\partial \varphi}$$

$$\square = \eta^{\mu\alpha} \partial_\mu \partial_\alpha \quad \Rightarrow \quad \square \varphi = \frac{\partial V}{\partial \varphi}$$

$$\square = -\frac{1}{c^2} \frac{\partial^2}{\partial t^2} + \nabla^2$$

d'Alembertian

* Complex Scalar Fld $\varphi(x) \neq \varphi^*(x)$

(2)

$$S = \int d^4x \left[-\eta^{\mu\nu} \partial_\mu \varphi^* \partial_\nu \varphi - V(\varphi^* \varphi) \right]$$

Abelian Higgs model

* Global Symmetry:

$$\varphi \rightarrow \varphi' = e^{i\lambda} \varphi$$

$$\varphi^* \rightarrow \varphi^{*'} = e^{-i\lambda} \varphi^*$$

$$\delta\varphi = \varphi' - \varphi = i(\delta\lambda)\varphi$$

λ : real parameter

$$\delta\varphi^* = -i(\delta\lambda)\varphi^*$$

Noether's Thm.

* Continuous global symmetry

$$\delta\varphi_I = F_I(\varphi_I, \partial_\mu \varphi_I) \delta\lambda$$

Assume symmetry: $\delta S = 0$ (up to boundary terms)

$$\delta\mathcal{L} = \partial_\mu A^\mu \delta\lambda$$

$$= (\partial_0 A^0 + \partial_1 A^1 + \partial_2 A^2 + \partial_3 A^3) \delta\lambda$$

(symmetry)

$$\delta\mathcal{L} = \sum_I \frac{\partial\mathcal{L}}{\partial\varphi_I} \delta\varphi_I + \sum_I \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi_I)} \partial_\mu(\delta\varphi_I)$$

$$\sum_I \frac{\partial\mathcal{L}}{\partial\varphi_I} F_I + \sum_I \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi_I)} \partial_\mu F_I = \partial_\mu A^\mu$$

$$\delta\varphi_I = F_I \delta\lambda(x)$$

$$\delta S = \int d^4x \left[\sum_I \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi_I)} F_I \delta\lambda(x) + \sum_I \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi_I)} \left((\partial_\mu F_I) \delta\lambda + F_I \partial_\mu(\delta\lambda) \right) \right]$$

(because of the symmetry)

$$= \int d^4x \left[\partial_\mu A^\mu \delta\lambda + \sum_I \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi_I)} F_I \partial_\mu(\delta\lambda) \right]$$

$$\delta S = \text{boun. term} + \int d^4x \left[\partial_\mu A^\mu - \partial_\mu \left(\sum_I \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi_I)} F_I \right) \right] \delta\lambda$$

$= 0$ by Principle of least action.

$$J^\mu = \sum_I \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi_I)} F_I - A^\mu$$

$$\& \partial_\mu J^\mu = 0$$

J^μ : 4-vector ("current")

(3)

$$J^\mu = (\underbrace{J^0}_{J^0 = \rho}, \underbrace{J^1, J^2, J^3}_{\underline{J} \text{ or } \vec{J}})$$

$$\partial_\mu J^\mu = \frac{\partial \rho}{\partial t} + \underline{\nabla} \cdot \underline{J} = 0 \quad \text{: charge cons. eq.}$$

Abelian Higgs

$$\delta \varphi = i \delta \lambda \varphi$$

$$\delta \varphi^* = -i \delta \lambda \varphi^*$$

$$\left[\begin{array}{l} F_\varphi = i \varphi \\ F_{\varphi^*} = -i \varphi^* \end{array} \right]$$

$$\left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi)} = -\eta^{\mu\nu} \partial_\nu \varphi^* \right]$$

$$\frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi^*)} = -\eta^{\mu\nu} \partial_\nu \varphi \quad (A^\mu = 0)$$

$$J^\mu = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi)} F_\varphi + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi^*)} F_{\varphi^*} - 0$$

$$= i \left[-\eta^{\mu\nu} (\partial_\nu \varphi^*) \varphi + \eta^{\mu\nu} \varphi^* \partial_\nu \varphi \right]$$

$$J^{\mu*} = J^\mu$$

Sp.-Tm. Translation: $x^\mu \rightarrow x^\mu + a^\mu$

$$\delta \varphi = \sum_{\mu=0}^3 \partial_\mu \varphi \delta a^\mu$$

$$\delta \mathcal{L} = \partial_\nu \mathcal{L} \delta a^\nu = \partial_\mu (\delta^\mu_\nu \mathcal{L}) \delta a^\nu$$

$$A^\mu_{(\nu)} = \delta^\mu_\nu \mathcal{L} \rightarrow \delta \mathcal{L} = (\partial_\mu A^\mu) \delta a^\nu$$

$$J^\mu_{(\nu)} = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi)} \partial_\nu \varphi - A^\mu_\nu$$

$$T^\mu_\nu = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi)} \partial_\nu \varphi - \delta^\mu_\nu \mathcal{L}$$

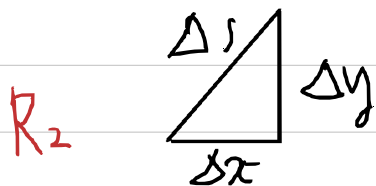
Generalization of the Hamil. in fld theory

: Stress-Energy Tensor

$$(\underline{\eta})_{\mu\nu} = \eta_{\mu\nu} = \begin{pmatrix} -c^2 & & & \\ & 1 & & \\ & & 0 & \\ & & & 1 \\ & & & & 1 \end{pmatrix}$$

(in Cartesian coord. η)

2D:



$$\Delta s^2 = \Delta x^2 + \Delta y^2$$

3D:



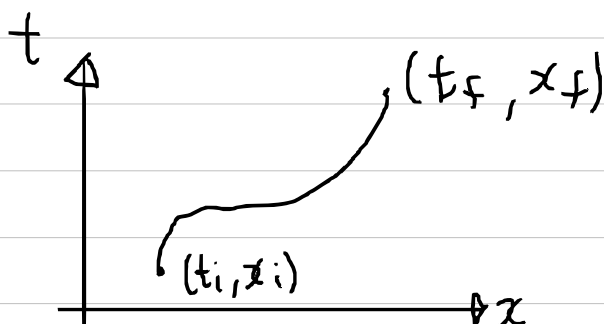
$$\Delta s^2 = \Delta x^2 + \Delta y^2 + \Delta z^2$$

4D Minkowski:

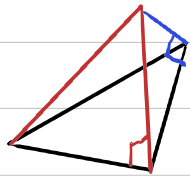
$$\Delta s^2 = \Delta x^2 + \Delta y^2 + \Delta z^2 - c^2 \Delta t^2$$

SR

$$x^{\mu} = \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix} = \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix}$$



(2)



Pythagoras 3 times
to get
 $\Delta s^2_{(TD)}$

$$c = m/s$$

Check LN.

t in m because $c=1$

$$\begin{aligned}\Delta s^2 &= \sum_{\mu=0}^3 \sum_{\nu=0}^3 \eta_{\mu\nu} \Delta x^\mu \Delta x^\nu \\ &= \eta_{00} (\Delta x^0)^2 + \eta_{01} \Delta x^0 \Delta x^1 + \dots \\ &\quad + \dots\end{aligned}$$

$$= -c^2 (\Delta t)^2 + \Delta x^2 + \Delta y^2 + \Delta z^2$$

(- + + +)

Proper time $\tau \longrightarrow \frac{d\tau}{dt} \rightarrow$ lapse fn.

$$d\tau^2 = - \frac{ds^2}{c^2} = dt^2 \left(1 - \frac{\dot{x}^2}{c^2} - \frac{\dot{y}^2}{c^2} - \frac{\dot{z}^2}{c^2} \right) \text{ (time travel)}$$

infinitesimal $\rightarrow d$, not Δ .

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$$

$$\tau = \int dt \frac{1}{\gamma} = \int dt \sqrt{1 - \frac{v^2}{c^2}}$$

(infinitesimal proper time)

In Classical F.T. language, existence of
global symmetry is implied by SR.

Symmetry transformation = map between
inertial frames

(3)

Symmetry 'rotation' + translation

$$\Delta x' = \Delta x \cos \theta - \Delta y \sin \theta$$

$$\Delta y' = \Delta y \cos \theta + \Delta x \sin \theta$$

$$\Delta x'^2 + \Delta y'^2 = \Delta x^2 + \Delta y^2$$

Global Symmetry $\longleftrightarrow$ OR SR

Poincaré Group

$$x'^{\mu} = \underbrace{\Lambda^{\mu}_{\nu}}_{\text{(rotation)}} x^{\nu} + \underbrace{a^{\mu}}_{\text{(transl.)}}$$

$$(ex) \quad x'^0 = \Lambda^0_0 x^0 + \Lambda^0_1 x^1 + \Lambda^0_2 x^2 + \Lambda^0_3 x^3 + a^0$$

Individual Coord. Transfrm.

$$dx'^{\mu} = \underbrace{\Lambda^{\mu}_{\nu}}_{\substack{4 \times 4 \\ \text{matrix}}} dx^{\nu} \quad \& \quad da^{\mu} = 0 \quad (\text{transl. doesn't change})$$

$$\text{Assume: } ds^2 = \eta_{\mu\nu} dx'^{\mu} dx'^{\nu}$$

$$= \eta_{\mu\nu} (\Lambda^{\mu}_{\alpha} dx^{\alpha}) (\Lambda^{\nu}_{\beta} dx^{\beta})$$

$$= \eta_{\mu\nu} \Lambda^{\mu}_{\alpha} \Lambda^{\nu}_{\beta} dx^{\alpha} dx^{\beta}$$

$$= \Lambda^{\nu}_{\alpha} \Lambda^{\nu}_{\beta} dx^{\alpha} dx^{\beta}$$

$$= \eta_{\alpha\beta} dx^{\alpha} dx^{\beta}$$

$$\eta_{\alpha\beta} = \Lambda^{\nu}_{\alpha} \Lambda^{\nu}_{\beta}$$

$$\eta_{\mu\nu} \Lambda^{\mu}_{\alpha} \Lambda^{\nu}_{\beta} = \eta_{\alpha\beta}$$

$$= \Lambda^{\mu}_{\alpha} \eta_{\mu\nu} \Lambda^{\nu}_{\beta} \rightarrow \underline{\underline{\Lambda^T \eta \Lambda = \eta}} \quad \begin{array}{l} \text{(matrix-y)} \\ \text{lang.} \end{array}$$

$$(\underline{\Lambda})^{\mu}_{\alpha} = \Lambda^{\mu}_{\alpha}$$

$$(\underline{\eta})_{\mu\nu} = \eta_{\mu\nu}$$

Group $\begin{array}{cc} \text{sp.} & \text{tm.} \\ \downarrow & \downarrow \\ \mathcal{O}(3,1) \end{array}$ or $\mathcal{O}(1,3)$ all Λ that satisfy these relations $\in \mathcal{O}(3,1)$ Λ^{μ}_{ν} : Lorentz GroupLorentz Grp

$$a^\mu = \begin{pmatrix} a^0 \\ a^1 \\ a^2 \\ a^3 \end{pmatrix} \rightarrow \begin{matrix} 4 \text{ distinct translations} \\ 4 \text{ symmetries} \end{matrix} \quad (4)$$

In 3D, 3 distinct Rotations

$$(xz, xy, yz) \rightarrow 3 \text{ planes}$$

In 4D, 6 planes $\rightarrow$ 6 distinct rotations

$$6 = 3 + 3 \quad \begin{matrix} \uparrow & \uparrow \\ \text{Lorentz} & \text{rotations} \\ \text{boosts} & \end{matrix} \quad \begin{pmatrix} xy, xz, wx, wy, wz, \\ yz \end{pmatrix}$$

$$\Lambda^\mu{}_\nu = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \begin{pmatrix} \underline{\underline{M}} \end{pmatrix} \\ 0 & \\ 0 & \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & M_{xx} & M_{xy} & M_{xz} \\ 0 & M_{yx} & M_{yy} & M_{yz} \\ 0 & M_{zx} & M_{zy} & M_{zz} \end{pmatrix}$$

$$\eta_{ij} = \delta_{ij}$$

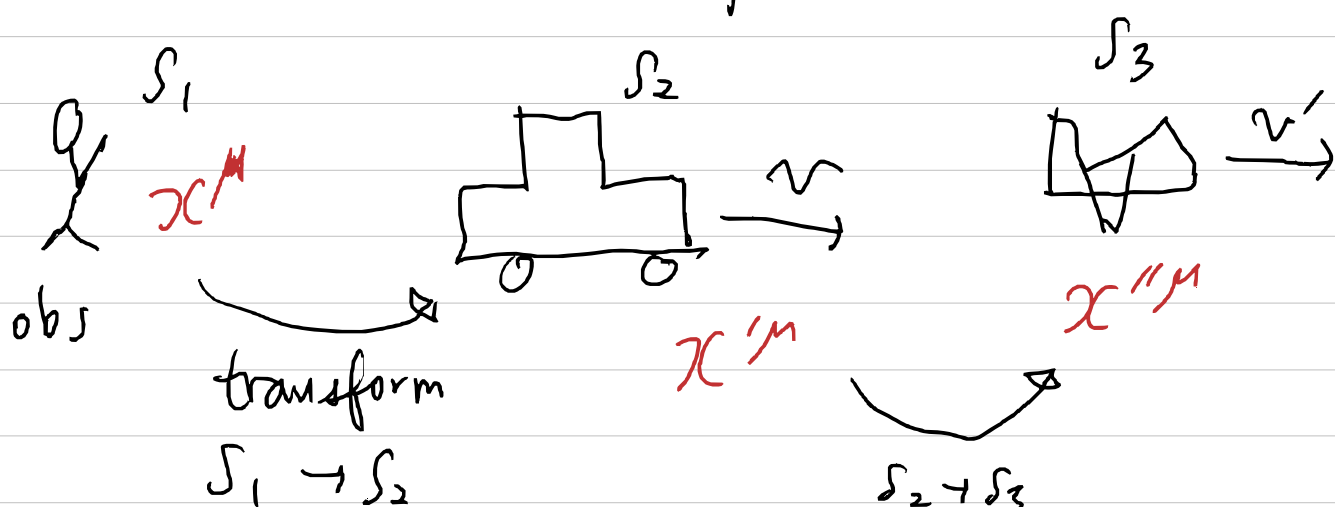
$$\Lambda^\mu{}_\nu = \begin{pmatrix} \gamma & -\frac{\gamma v}{c^2} & 0 & 0 \\ -\gamma v & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{aligned} \text{L.T.: } x' &= \gamma(x - vt) \\ t' &= \gamma\left(t - \frac{vx}{c^2}\right) \end{aligned}$$

Lorentz
boost / transform
in x-dir.

$$\begin{aligned} \text{Lorentz factor } \gamma &= \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \\ &= \frac{1}{\sqrt{1 - \beta^2}} \end{aligned}$$

$$\Lambda^\mu{}_\nu = \begin{pmatrix} \gamma & 0 & -\frac{\gamma v}{c^2} & 0 \\ 0 & 1 & 0 & 0 \\ -\gamma v & 0 & \gamma & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$



$$S_1 \rightarrow S_2 : x'^\mu = \Lambda^\mu{}_\nu x^\nu + a_1^\mu$$

$$S_2 \rightarrow S_3 : x''^\mu = \Lambda^\mu{}_\nu x'^\nu + a_2^\mu$$

$$S_1 \rightarrow S_3 : x''^\mu = \underbrace{\Lambda^\mu{}_\nu}_{\text{composite}} x^\nu + a_{1,2}^\mu$$

$$x''^\mu = \Lambda^\mu{}_\nu (\Lambda^\nu{}_\alpha x^\alpha + a_1^\nu) + a_2^\mu$$

$$\text{So, } \Lambda^\mu{}_{\alpha} = \Lambda^\mu{}_\nu \Lambda^\nu{}_\alpha$$

$$a_{1,2}^\mu = \Lambda^\mu{}_\nu a_1^\nu + a_2^\mu$$

Poincaré group

$$x'^{\mu} = \underbrace{\Lambda^{\mu}_{\nu} x^{\nu}}_{\text{Lorentz} = \text{rotation} + \text{boost}} + \underbrace{a^{\mu}}_{\text{Translations } T^4 \text{ or } \mathbb{R}^4}$$

$$S \xrightarrow{\text{Poincaré}} S'$$

Use

$$c=1$$

$$(\Lambda^T)_{\alpha}{}^{\mu} = \Lambda^{\mu}{}_{\alpha}$$

$$\eta_{\mu\nu} \Lambda^{\mu}{}_{\alpha} \Lambda^{\nu}{}_{\beta} = \eta_{\alpha\beta}$$

$$(\Lambda^T)_{\alpha}{}^{\mu} \eta_{\mu\nu} (\Lambda)^{\nu}{}_{\beta} = \eta_{\alpha\beta}$$

$$O(3,1)$$

$$= SO(3,1)$$

$$(\det \Lambda = 1)$$

$$\underline{\underline{\Lambda^T}} \underline{\underline{\eta}} \underline{\underline{\Lambda}} = \underline{\underline{\eta}}$$

$$\det(\underline{\underline{\Lambda^T}} \underline{\underline{\Lambda}}) \eta = \det(\underline{\underline{\Lambda}})^2 \det(\underline{\underline{\eta}}) = \det(\underline{\underline{\eta}})$$

$$\det(\Lambda) = \pm 1$$

 $\Lambda \in \text{Lorentz grp.}$

Rot.:

$$\Lambda = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \begin{pmatrix} M \end{pmatrix} \\ 0 & \\ 0 & \end{pmatrix}$$

$M \in SO(3)$ $(\det M = 1)$

Boosts:

$$\Lambda = \begin{pmatrix} \gamma & -\gamma v & 0 & 0 \\ \gamma v & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

(2)

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \quad (c=1)$$

$$= \frac{1}{\sqrt{1 - v^2}} \quad ; \text{ Lorentz factor}$$

$$S \xrightarrow[\text{LT}]{\Lambda_1} S' \xrightarrow[\text{LT}]{\Lambda_2} S''$$

2 LT's

$$\underline{\underline{\Lambda_3}} = \underline{\underline{\Lambda_2}} \underline{\underline{\Lambda_1}}$$

$$\Lambda_{1 \circ 2} = \Lambda_3 \quad :$$

$$(\Lambda_3)^\mu{}_\nu$$

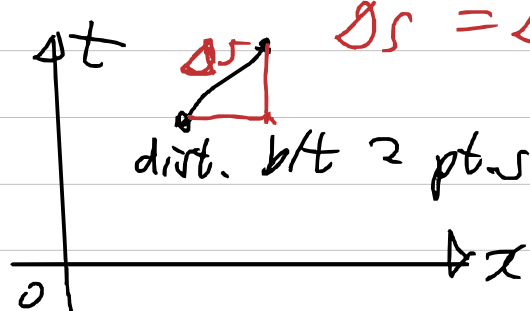
$$= (\Lambda_2)^\mu{}_\alpha (\Lambda_1)^\alpha{}_\nu$$

* lowering the indices

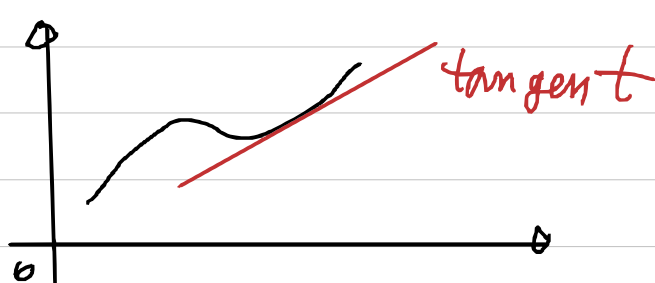
$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$$

(-+++)

Minkowski S.T.



$$ds^2 = \Delta x^2 - \Delta t^2$$



$$dx^\nu = \text{vec. in tangent sp.} \begin{pmatrix} dx^0 \\ dx^1 \\ dx^2 \\ dx^3 \end{pmatrix}$$

$$\eta_{\mu\nu} dx^\nu \equiv dx_\mu$$

downstairs = covariant

up = contravariant

$$dx_0 = \eta_{00} dx^0 + \cancel{\eta_{01} dx^1} + \dots$$

(-+++)

$$dx_i = dx^i \quad (\text{spatial } i=1,2,3)$$

$$dx_0 = -dx^0$$

$$dx_\mu = (dx_0, dx_1, dx_2, dx_3)$$

$$= (-dx^0, dx^1, dx^2, dx^3)$$

$$ds^2 = dx_\mu dx^\mu \quad (\mu = 0, 1, 2, 3) \quad (4D)$$

$$= (dx_0 \ dx_1 \ dx_2 \ dx_3) \begin{pmatrix} dx^0 \\ dx^1 \\ dx^2 \\ dx^3 \end{pmatrix}$$

$$= -(dx^0)^2 + (dx^1)^2 + (dx^2)^2 + (dx^3)^2$$

(3)

Inverse : itself!

$$\underline{\eta} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \quad \& \quad \underline{\eta}^{-1} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$dx_\mu = \eta_{\mu\nu} dx^\nu$$

$$\begin{aligned} (\eta^{-1})^{\alpha\mu} dx_\mu &= (\eta^{-1})^{\alpha\mu} \eta_{\mu\nu} dx^\nu \\ &= \delta^\alpha_\nu dx^\nu \\ &= dx^\alpha \end{aligned}$$

$$\underline{\eta^{\mu\nu}} = (\eta^{-1})^{\mu\nu}$$

$$dx^\mu = \eta^{\mu\nu} dx_\nu$$

$$dx_\mu = \eta_{\mu\nu} dx^\nu$$

raising of indices
lowering of indices

$$dx'^\mu = \Lambda^\mu_\nu dx^\nu$$

Using lowering $\eta_{\mu\nu}$

$$\begin{aligned} dx'_\mu &= \eta_{\mu\nu} dx'^\nu = \eta_{\mu\nu} \Lambda^\nu_\alpha dx^\alpha \\ &\quad \text{(Bob's Frame)} \quad \text{(Alice's Frame)} \\ &= \eta_{\mu\nu} \Lambda^\nu_\alpha \eta^{\alpha\beta} dx_\beta \end{aligned}$$

$$\eta_{\mu\nu} \Lambda^\mu_\alpha \Lambda^\nu_\beta \eta^{\alpha\sigma} = \eta^{\alpha\sigma} \eta_{\alpha\beta}$$

$$\left(\eta_{\mu\nu} \Lambda^\mu_\alpha \eta^{\alpha\sigma} \right) \Lambda^\nu_\beta = \delta^\sigma_\beta$$

$$\Lambda_{\nu}{}^\sigma \Lambda^\nu_\beta = \delta^\sigma_\beta$$

$$\begin{aligned} \left(\Lambda_{\nu}{}^\sigma \Lambda^\nu_\beta \right) (\Lambda^{-1})^\beta_\alpha &= \delta^\sigma_\beta (\Lambda^{-1})^\beta_\alpha \\ &= (\Lambda^{-1})^\sigma_\alpha \end{aligned}$$

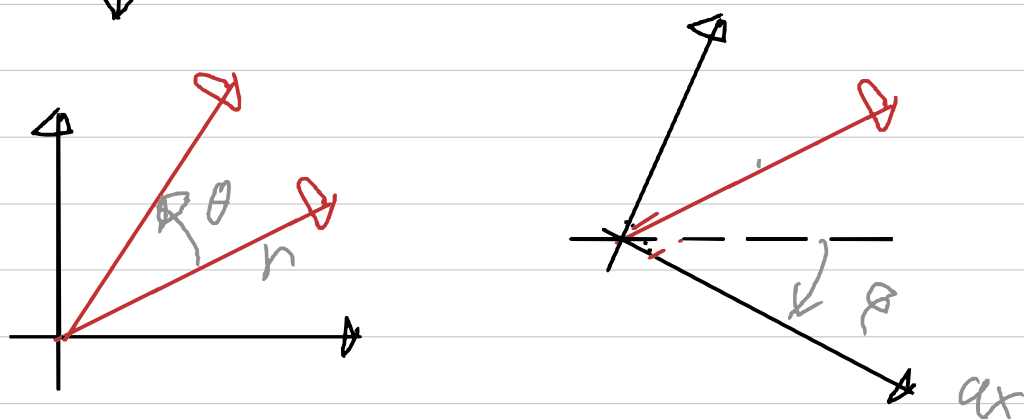
$$\underline{\Lambda_{\alpha}{}^\sigma = (\Lambda^{-1})^\sigma_\alpha}$$

$$\underline{dx'_\mu = \Lambda^\mu_\beta dx_\beta = (\Lambda^{-1})^\mu_\beta dx_\beta}$$

In QM

$$\langle \Psi_2 | \Psi_2 \rangle = \int \Psi_2^*(x) \Psi_2(x) dx$$

$$= |\Psi_2|^2$$

* Active or Passive Transformations

$$V'^{\mu} = \Lambda^{\mu}_{\nu} V^{\nu} \quad \begin{array}{l} \text{(Bob's)} \\ \text{Contravariant} \\ \text{(Alice's)} \end{array}$$

$$V'_{\mu} = \Lambda_{\mu}^{\nu} V_{\nu} \quad \begin{array}{l} \text{(Bob's)} \\ \text{Covariant} \\ \text{(Alice's)} \end{array}$$

$$= (\Lambda^{-1})^{\nu}_{\mu} V_{\nu}$$

$$(n, m) \text{ tensor} : T^{\mu_1 \mu_2 \dots \mu_m}_{\nu_1 \nu_2 \dots \nu_n}$$

Simplest EX : $(2, 0)$ tensor $T^{\mu_1 \mu_2}$
(Alice's)

(2, 0)

$$T^{\mu_1 \mu_2} = \Lambda^{\mu_1}_{\nu_1} \Lambda^{\mu_2}_{\nu_2} T^{\nu_1 \nu_2}$$

(Bob's)

rank $-(n, 0)$:

$$T^{\mu_1 \mu_2 \dots \mu_n} = \Lambda^{\mu_1}_{\nu_1} \Lambda^{\mu_2}_{\nu_2} \dots \Lambda^{\mu_n}_{\nu_n} T^{\nu_1 \nu_2 \dots \nu_n}$$

(Bob's)

$$(0, 2) : T_{\mu_1 \mu_2} = \Lambda^{\nu_1}_{\mu_1} \Lambda^{\nu_2}_{\mu_2} T_{\nu_1 \nu_2}$$

$$= (\Lambda^{-1})^{\nu_1}_{\mu_1} (\Lambda^{-1})^{\nu_2}_{\mu_2} T_{\nu_1 \nu_2}$$

 (n, m) :

$$T^{\mu_1 \dots \mu_n}_{\nu_1 \dots \nu_m} = \Lambda^{\mu_1}_{\alpha_1} \dots \Lambda^{\mu_n}_{\alpha_n} \Lambda^{\beta_1}_{\nu_1} \dots \Lambda^{\beta_m}_{\nu_m} T^{\alpha_1 \dots \alpha_n}_{\beta_1 \dots \beta_m}(x)$$

(Ex) scalar fld

a pt. ~~*~~

⑤

$$\underset{\text{(Alice's)}}{x} = \underset{\text{(Bob's)}}{x'}$$

$$\varphi(x) = \varphi'(x')$$

02/12/2024

ACP L23

Alice's frame $\longrightarrow$ Bob's frame x^μ

$$x'^\mu = \Lambda^\mu_\nu x^\nu + a^\mu$$

 x'^μ

$$A'^\mu = \Lambda^\mu_\nu A^\nu, \quad B'_\mu = \Lambda_\mu^\nu B_\nu = (\Lambda^{-1})^\nu_\mu B_\nu$$

Bob's frame

$$A'^\mu B'_\mu = \Lambda^\mu_\nu A^\nu \Lambda_\mu^\omega B_\omega$$

$$= (\Lambda^\mu_\nu \Lambda_\mu^\omega) A^\nu B_\omega = ((\Lambda^{-1})^\omega_\mu \Lambda^\mu_\nu) A^\nu B_\omega$$

$$= (\Lambda^{-1} \Lambda)^\omega_\nu A^\nu B_\omega = \delta^\omega_\nu A^\nu B_\omega = \delta^\mu_\nu A^\nu B_\mu$$

$$= A^\mu B_\mu \longrightarrow \text{Scalar under Lorentz transformations.}$$

Alice's frame

$$\text{Remember: } \Lambda_\mu^\nu = \eta_{\mu\alpha} \Lambda^\alpha_\beta \eta^{\beta\nu} \neq \Lambda^\mu_\nu \\ \neq \Lambda^\nu_\mu$$

$$\Lambda_\mu^\nu = (\Lambda^{-1})^\nu_\mu$$

$$\Lambda \in SO(3,1)$$

Inner Product:

$$(A, B) = A^\mu B_\mu = A_\mu B^\mu$$

$$= A^0 B_0 + A^1 B_1 + A^2 B_2 + A^3 B_3$$

$$= -A^0 B^0 + A^1 B^1 + A^2 B^2 + A^3 B^3$$

$$= -A^0 B^0 + \vec{A} \cdot \vec{B}$$

$$\eta_{\mu\nu} = (-, +, +, +)$$

When in the index notation, everything is a number.
(Classical)

(2)

For tensors:

$$\text{Similarly to } A'^{\mu} B'_{\mu} = A^{\mu} B_{\mu},$$

$$T'^{\mu\nu} S'_{\mu\nu} = T^{\mu\nu} S_{\mu\nu}$$

$$T'^{\mu_1 \dots \mu_N} S'_{\mu_1 \dots \mu_N} = T^{\mu_1 \dots \mu_N} S_{\mu_1 \dots \mu_N}$$

Action for a relativistic free particle.

$$S = \int dt \left(\vec{p} \cdot \frac{d\vec{r}}{dt} - H \right)$$

Massive particle

• m

$$\rightarrow H = mc^2$$

$$\left\{ \frac{d\vec{r}}{dt} = \text{vel.} = 0 \right.$$

$$dt^2 = -\frac{ds^2}{c^2} = -\frac{\eta_{\mu\nu} dx^{\mu} dx^{\nu}}{c^2}$$

$$\tau = \int \sqrt{-\frac{\eta_{\mu\nu} dx^{\mu} dx^{\nu}}{c^2}} = \int \sqrt{dt^2 - \frac{d\vec{r}^2}{c^2}} = \int dt \sqrt{1 - \frac{v^2}{c^2}}$$

Proper time $\tau \rightarrow$ At Rest: $\tau = t$.

Lorentz scalar

pbc τ is a Lorentz scalar

$$S = \int dt (0 - mc^2) = \int d\tau (-mc^2) = -mc^2 \int d\tau$$

The action in the ref. frm. when the particle is at rest.

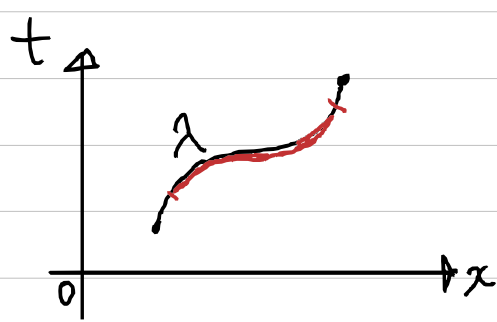
Set $c=1$.

$$S = -m \int \sqrt{-\eta_{\mu\nu} dx^{\mu} dx^{\nu}}$$

In parameterized form,

$$S = -m \int d\lambda \sqrt{-\eta_{\mu\nu} \frac{dx^{\mu}}{d\lambda} \frac{dx^{\nu}}{d\lambda}}$$

→ Like in the Fermat's principle.



$$\frac{d}{d\lambda} \left(\frac{\partial L}{\partial \left(\frac{dx^{\mu}}{d\lambda} \right)} \right) = \frac{\partial L}{\partial x^{\mu}} \rightarrow 4 \text{ eq.s } (x^{\mu})$$

$$S = \int d\lambda L \left(x^{\mu}, \frac{dx^{\mu}}{d\lambda} \right)$$

$$L = -m \sqrt{-\eta_{\alpha\beta} \frac{dx^{\alpha}}{d\lambda} \frac{dx^{\beta}}{d\lambda}} \rightarrow \frac{\partial L}{\partial x^{\mu}} = 0$$

$$\frac{\partial L}{\partial \left(\frac{dx^{\mu}}{d\lambda} \right)} = -\frac{1}{2} \frac{m}{\sqrt{-\eta_{\alpha\beta} \frac{dx^{\alpha}}{d\lambda} \frac{dx^{\beta}}{d\lambda}}} \left[-\eta_{\alpha\beta} \frac{\partial \left(\frac{dx^{\alpha}}{d\lambda} \frac{dx^{\beta}}{d\lambda} \right)}{\partial \left(\frac{dx^{\mu}}{d\lambda} \right)} \right]$$

$$\frac{\partial \left(\frac{dx^{\alpha}}{d\lambda} \right)}{\partial \left(\frac{dx^{\mu}}{d\lambda} \right)} = \delta^{\alpha}_{\mu} = \begin{cases} 0, & \alpha \neq \mu \\ 1, & \alpha = \mu \end{cases}$$

(3)

$$\frac{\partial L}{\partial \left(\frac{dx^\mu}{d\lambda}\right)} = \frac{m \eta_{\mu\nu} \frac{dx^\nu}{d\lambda}}{\sqrt{-\eta_{\alpha\beta} \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda}}}$$

$$\frac{\partial L}{\partial \left(\frac{dx^\mu}{d\lambda}\right)} = \frac{m \eta_{\mu\nu} \frac{dx^\nu}{d\lambda}}{\sqrt{-\eta_{\alpha\beta} \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda}}} = m \eta_{\mu\nu} \frac{dx^\nu}{d\tau} = p_\mu$$

$$p_1 = m \frac{dx^1}{d\tau} = \frac{m}{\sqrt{1-\frac{v^2}{c^2}}} \frac{dx^1}{dt}$$

$$\vec{p} = \frac{m\vec{v}}{\sqrt{1-\frac{v^2}{c^2}}} = \gamma m \vec{v}$$

$$E-L \rightarrow \frac{d\lambda}{d\tau} \frac{d}{d\lambda} \left(m \eta_{\mu\nu} \frac{dx^\nu}{d\tau} \right) = 0$$

$$m \frac{d^2 x^\mu}{d\tau^2} = 0$$

If we choose 'gauge' $\lambda = t$, then

$$\begin{aligned} S &= -m \int dt \sqrt{-\eta_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt}} \\ &= -mc^2 \int dt \sqrt{1 - \frac{v^2}{c^2}} \end{aligned}$$

For a charged relativistic particle:

$$A_\mu(x) = (-\phi, \vec{A})$$

↑
Coulomb potential

$$\begin{aligned} S &= -m \int d\lambda \sqrt{-\eta_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} \\ &\quad + q \int d\lambda A_\mu(x) \frac{dx^\mu}{d\lambda} \\ &= \int d\lambda L \end{aligned}$$

$$L = -m \sqrt{-\eta_{\alpha\beta} \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda}} + q A_\alpha(x) \frac{dx^\alpha}{d\lambda}$$

electric charge

(4)

$$\frac{d}{d\lambda} \left(m \eta_{\mu\nu} \frac{dx^\mu}{d\tau} + q A_\mu(x) \right) = q \frac{\partial A_\alpha(x)}{\partial x^\mu} \frac{dx^\alpha}{d\lambda}$$

Doing $\times \frac{d\lambda}{d\tau}$ on both sides:

$$\frac{d\lambda}{d\tau} \frac{d}{d\lambda} \left(m \eta_{\mu\nu} \frac{dx^\mu}{d\tau} + q A_\mu(x) \right) = q \frac{\partial A_\alpha(x)}{\partial x^\mu} \frac{dx^\alpha}{d\lambda} \frac{d\lambda}{d\tau}$$

$$m \eta_{\mu\nu} \frac{d^2 x^\mu}{d\tau^2} + q \frac{dA_\mu(x)}{d\tau} = q \partial_\mu A_\nu \frac{dx^\nu}{d\tau}$$

$$\frac{dA_\mu(x)}{d\tau} = \frac{dx^\nu}{d\tau} \frac{\partial A_\mu}{\partial x^\nu} = \frac{dx^\nu}{d\tau} \partial_\nu A_\mu$$

$$m \eta_{\mu\nu} \frac{d^2 x^\nu}{d\tau^2} = q F_{\mu\nu} \frac{dx^\nu}{d\tau}$$

$$m \frac{d^2 x^\nu}{d\tau^2} = q F^\mu{}_\nu \frac{dx^\nu}{d\tau} \quad \text{for a charged } [q] \text{ relativistic particle.}$$

$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$: "field strength"
(or Faraday tensor)

$$F_{0i} = \partial_0 A_i - \partial_i A_0 = \partial_0 A_i - \partial_i \phi = -E_i$$

$$\vec{E} = -\vec{\nabla} \phi - \frac{\partial \vec{A}}{\partial t}$$

Mag. Fld.

$$F_{ij} = \partial_i A_j - \partial_j A_i = \epsilon_{ijk} B^k$$

$$F_{\mu\nu} = -F_{\nu\mu} \text{ : anti-symm.}$$

For $\mu = i$ (spatial)
 $= 1, 2, 3$

$(x^0 = t)$

$$\frac{d}{d\tau} (p^i) = q F^i{}_0 \frac{dx^0}{d\tau} + q F^i{}_j \frac{dx^j}{d\tau}$$

$$\frac{dp^i}{dt} = q F^i{}_0 \frac{dx^0}{dt} + q F^i{}_j \frac{dx^j}{dt} \quad \tau \rightarrow t$$

$$= q F^i{}_0 + q F^i{}_j = q E^i + q (\vec{v} \times \vec{B})^i$$

Lorentz force

⑤

All we did was adding a simple coupling

$q A_\mu \frac{dx^\mu}{d\lambda}$ to the action.

Recap. Action for relativistic charged particles

$c=1$

$$S = -m \int d\tau + q \int A_\mu dx^\mu$$

$$= -m \int \left(\frac{d\tau}{d\lambda} \right) d\lambda + q \int A_\mu \frac{dx^\mu}{d\lambda} d\lambda$$

$$= \int d\lambda \left[-m \sqrt{-\eta_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} + q A_\mu \frac{dx^\mu}{d\lambda} \right]$$

Lorentz

$$\frac{dp^\mu}{d\tau} = q F^\mu{}_\nu \frac{dx^\nu}{d\tau}$$

$$p^\mu = m \frac{dx^\mu}{d\tau} \quad \left(\begin{array}{l} \text{e.g. } p^i = m \frac{dx^i}{d\tau} \\ = \frac{m v^i}{\sqrt{1-v^2}} \\ = \gamma m v^i \end{array} \right)$$

EX

$\lambda = t$

$$S = \int dt \left[-m \sqrt{1-v^2} + q A_0 + q \vec{A} \cdot \frac{d\vec{r}}{dt} \right]$$

$$A_\mu = \begin{pmatrix} -\phi \\ A_1 \\ A_2 \\ A_3 \end{pmatrix}, \quad A^\mu = \begin{pmatrix} +\phi \\ A_1 \\ A_2 \\ A_3 \end{pmatrix} = \eta^{\mu\nu} A_\nu$$

ϕ : Coulomb potl.

Lorentz: $\frac{dp^\mu}{d\tau} = q F^\mu{}_\nu \frac{dx^\nu}{d\tau}$

$$p^\mu = m \frac{dx^\mu}{d\tau} \quad \text{e.g.} \quad p^i = \frac{dr^i}{d\tau} = \frac{m v^i}{\sqrt{1-v^2}}$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = -F_{\nu\mu}$$

$$= \underset{3}{\text{E.F.}} + \underset{3}{\text{M.F.}}$$

$$F_{0i} = \partial_0 A_i - \partial_i A_0 = \partial_0 A_i + \partial_i \phi^i = -E_i$$

$$F^0{}_i = +E_i \quad \& \quad F_{ij} = \partial_i A_j - \partial_j A_i = \epsilon_{ijk} B_k \quad (\text{M.F.})$$

Faraday Tensor

(2)

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix}$$

Gauge Invariance / Local Symmetry

$$\tilde{A}_\mu = A_\mu + \partial_\mu X$$

$\uparrow$ shifted $\downarrow$ vector

$$\tilde{F}_{\mu\nu} = \partial_\mu \tilde{A}_\nu - \partial_\nu \tilde{A}_\mu = \partial_\mu (A_\nu + \partial_\nu X) - \partial_\nu (A_\mu + \partial_\mu X)$$

$$= F_{\mu\nu} + \cancel{\partial_\mu \partial_\nu X} - \cancel{\partial_\nu \partial_\mu X}$$

$$= F_{\mu\nu}$$

F : gauge invariant observable

4 - A 's & 6 - F 's
 A_0, A_1, A_2, A_3 $\uparrow$
 red herring

* Maxwell's Eq.s

Non-Rel. Gauss's Law $\nabla \cdot \underline{E} = \frac{\rho_e}{\epsilon_0}$ 1 Eq.

No mag. monopole charges. ($\rho_m = 0$) $\nabla \cdot \underline{B} = 0$ 1 Eq.

Ampère's:

$$\nabla \times \underline{B} = \mu_0 \epsilon_0 \frac{\partial \underline{E}}{\partial t} + \mu_0 \underline{J}$$

3 Eq.s

So

$$\nabla \cdot \underline{E} = \mu_0 \rho_e$$

$$\epsilon_0 = \frac{1}{\mu_0}$$

here

μ_0 : permeability of free sp.

$$\mu_0 \epsilon_0 = \frac{1}{c^2} = 1$$

Faraday's $\nabla \times \underline{E} = - \frac{\partial \underline{B}}{\partial t}$ 3 Eq.s

$\downarrow$
 $\underline{J}_M = 0$
 doesn't exist.

(3)

$$\partial_\alpha F_{\mu\nu} = \partial_\alpha \partial_\mu A_\nu - \partial_\alpha \partial_\nu A_\mu$$

$(6 \times 4) = 24$ indep. components

cyclic permutations,

$$\partial_\alpha F_{\mu\nu} + \partial_\mu F_{\nu\alpha} + \partial_\nu F_{\alpha\mu} = 0 = \bigcirc_{\alpha\mu\nu} F_{\mu\nu}$$

$$= \cancel{\partial_\alpha (\partial_\mu A_\nu)} - \cancel{\partial_\alpha \partial_\nu A_\mu} + \cancel{\partial_\mu \partial_\nu A_\alpha} - \cancel{\partial_\mu \partial_\alpha A_\nu} + \cancel{\partial_\nu \partial_\alpha A_\mu} - \cancel{\partial_\nu \partial_\mu A_\alpha}$$

$$= 0$$

EX. $0, i, j$
time + 2 sp.

$$\partial_0 F_{ij} + \partial_i F_{j0} + \partial_j F_{0i} = 0$$

Faraday's Law $\left(\begin{array}{l} = \frac{\partial B_z}{\partial t} + \partial_x E_y - \partial_y E_x \\ = 0 \end{array} \right)$

$$\left(\begin{array}{l} \hat{r} = x \\ \hat{v} = y \end{array} \right)$$

$$\nabla \times \underline{E} = - \frac{\partial \underline{B}}{\partial t}$$

EX. (x, y, z)
 $\hat{i}, \hat{j}, \hat{k}$
(3D sp.)

$$\partial_x F_{yz} + \partial_y F_{zx} + \partial_z F_{xy} = 0$$

$\downarrow$

$$\partial_x B_x + \partial_y B_y + \partial_z B_z = 0$$

$$= \nabla \cdot \underline{B} = 0$$

$$\partial_\alpha F_{\mu\nu} + \partial_\mu F_{\nu\alpha} + \partial_\nu F_{\alpha\mu} = 0$$

Its $\rightarrow$ Soln.: $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$

How can we construct a vec. & 1st-deriv. of F from tensor?
($F_{\mu\nu}$)

WRONG: $\partial_\mu F_{\mu\nu}$

CORRECT: $\partial^\mu F_{\mu\nu}$ or $\partial_\mu F^\mu{}_\nu$

$$\text{down } \partial_\mu = \frac{\partial}{\partial x^\mu} \text{ up}$$

(4)

Time compo. $0 = \nu$

$$\begin{aligned}\partial_\mu F^\mu_0 &= \partial_t F^0_0 + \partial_i F^i_0 \\ &= -\partial_t F_{00} + \partial_i E^i = \underline{\nabla} \cdot \underline{E} \\ &\text{used } (\eta_{\mu\nu}) \quad = \mu_0 \rho_e\end{aligned}$$

 $\nu = i$

$$\begin{aligned}\partial_\mu F^\mu_i &= \partial_t F^0_i + \partial_j F^{ij} \\ &= \frac{\partial E^i}{\partial t} + \epsilon^i_{jk} \partial_j B^k \\ &= \left(\frac{\partial \underline{E}}{\partial t} - \underline{\nabla} \times \underline{B} \right)_i \\ &= \mu_0 \vec{J}_i\end{aligned}$$

+ current

$$\vec{J}_\mu = \begin{pmatrix} -\rho_e \\ J_x \\ J_y \\ J_z \end{pmatrix}$$

$$\partial_\mu F^\mu_\nu = -\mu_0 J_\nu$$

The real form of Maxwell's Eqs.

Take

 $\nabla \cdot$ (div.):

$$\partial_\mu F^{\mu\nu} = -\mu_0 J^\nu$$

$$\partial_\nu \partial_\mu F^{\mu\nu} = -\mu_0 \partial_\nu J^\nu \quad (\text{used } \eta^{\mu\nu})$$

 $= 0$

$$(\underbrace{F^{\mu\nu} = -F^{\nu\mu}}_{\text{antisym.}} \longrightarrow \partial_\mu J^\mu = 0 \quad (\text{Noether's Thm}))$$

$$\frac{\partial \rho}{\partial t} + \underline{\nabla} \cdot \underline{j} = 0 \quad : \text{continuity Eq.}$$

"conserv. of current"

⑤

Mathematically simpler form using gauge symmetry :
(Fixing a gauge) (or choosing a gauge)

EX. Lorenz gauge $\partial_\mu A^\mu = 0$

$$\partial_\mu \tilde{A}^\mu = \partial_\mu A^\mu + \partial_\mu \partial^\mu \chi = \partial_\mu A^\mu + \square \chi$$

($\square$) "box"

Solve

$$\begin{cases} \partial_\mu A^\mu + \square \chi = 0 \\ \square \chi = -\partial_\mu A^\mu \end{cases}$$

Klein-Gordon Eq.

$$(\square + m^2) \phi = 0$$

χ = gauge fixing fn.

In Lorenz gauge, $\partial_\mu A^\mu = 0$.
 $\downarrow$
 holds.

$$\partial_\mu (\partial^\mu A^\nu - \partial^\nu A^\mu) = -\mu_0 J^\nu$$

If $\nu = i$: $\nabla \times \underline{B} = \mu_0 \epsilon_0 \frac{\partial \underline{E}}{\partial t} + \mu_0 \underline{J}$

If $\nu = 0$: $\nabla \cdot \underline{E} = \mu_0 \rho_e = \frac{\rho_e}{\epsilon_0}$

$$\partial_\mu \partial^\mu A^\nu = -\mu_0 J^\nu = \square A^\nu$$

where $\partial_\mu \partial^\nu A^\mu = \partial^\nu \partial_\mu A^\mu = 0$

$$\rightarrow \square A_{\text{hom}}^\nu = 0$$

$$A^\nu = A_{\text{homogeneous}}^\nu - \mu_0 \frac{1}{\square} J^\nu$$

$$\square A^\nu = \cancel{\square A_{\text{hom}}^\nu} - \mu_0 \square \frac{1}{\square} J^\nu = 0$$

$= -\mu_0 J^\nu$

Maxwell Eq.s Recap

$$\partial_\mu F_{\nu\omega} + \partial_\nu F_{\omega\mu} + \partial_\omega F_{\mu\nu} = 0 \quad \left(\text{or } \partial^\mu \tilde{F}_{\mu\nu} = 0 \right)$$

↓

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad \& \quad A_\mu = (-\varphi, \underline{A})$$

$$\partial_\mu F^{\mu\nu} = -\mu_0 J^\nu \quad \& \quad \partial_\mu F^{\mu}{}_\nu = -\mu_0 J_\nu$$

$$J_\mu = (-\rho, \underline{J})$$

$$\partial^\mu F_{\mu\nu} = -\mu_0 J_\nu \quad \partial^\mu F_\mu{}^\nu = -\mu_0 J^\nu$$

L.T. properties

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu + a^\mu$$

$$dx'^\mu = \Lambda^\mu{}_\nu dx^\nu$$

$$\partial'_\mu = \Lambda_\mu{}^\nu \partial_\nu$$

$$= (\Lambda^{-1})^\nu{}_\mu \partial_\nu$$

$$A'_\mu = \Lambda_\mu{}^\nu A_\nu = (\Lambda^{-1})^\nu{}_\mu A_\nu$$

$$F'_{\mu\nu} = \partial'_\mu A'_\nu - \partial'_\nu A'_\mu = \Lambda_\mu{}^\alpha \Lambda_\nu{}^\beta F_{\alpha\beta}$$

$$F'^{\mu\nu} = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta F^{\alpha\beta} \quad \Downarrow \quad \text{Partial Deriv.}$$

$$\partial'_\mu F'^{\mu\nu} = (\Lambda_\mu{}^\alpha \partial_\alpha) (\Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma F^{\rho\sigma})$$

$$= \Lambda_\mu{}^\alpha \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma \partial_\alpha F^{\rho\sigma}$$

$$= (\Lambda^{-1})^\alpha{}_\mu \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma \partial_\alpha F^{\rho\sigma}$$

$$= (\underbrace{\Lambda^{-1} \Lambda}^\delta)^\alpha{}_\rho \Lambda^\nu{}_\sigma \partial_\alpha F^{\rho\sigma}$$

$$= \delta^\alpha{}_\rho \Lambda^\nu{}_\sigma \partial_\alpha F^{\rho\sigma}$$

$$= \Lambda^\nu{}_\sigma \partial_\alpha F^{\alpha\sigma} = \text{LHS}$$

$$d\varphi = dx^\mu \partial_\mu \varphi$$

(2)

$$\text{RHS} = -\mu_0 J'^{\nu} = -\mu_0 \overset{\text{rot.}}{\Delta^{\nu}} \tau J^{\tau}$$

$$\cancel{\Delta^{\nu}} \tau \partial_{\alpha} F^{\alpha\tau} = -\mu_0 \cancel{\Delta^{\nu}} \tau J^{\tau}$$

$$\partial_{\alpha} F^{\alpha\tau} = -\mu_0 J^{\tau}$$

* Gauge symmetry / Local symmetry

$$\tilde{A}_{\mu} = A_{\mu} + \partial_{\mu} \chi$$

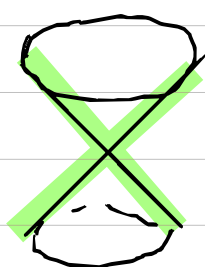
$$\tilde{F}_{\mu\nu} = F_{\mu\nu} \rightarrow \text{Invariant Faraday Tensor}$$

* Choice of gauge

• Lorenz gauge $\rightarrow$ invariant in Lorentz Tr.
 (Danish Lorenz) $\downarrow$ (Dutch Hendrik Lorentz)
 $\partial_{\mu} A^{\mu} = 0$

• Coulomb gauge $\rightarrow \nabla \cdot \underline{A} = 0$

• Light-cone gauge $\rightarrow \eta^{\mu} A_{\mu} = 0$



$$\eta^{\mu} \eta_{\mu} = 0$$

(null-vector)

• Weyl gauge $\rightarrow A_0 = 0$

• Fock-Schwinger gauge $\rightarrow x^{\mu} A_{\mu} = 0$

(EX) $x^{\mu} A_{\mu} \neq 0$

Find χ s.t. $x^{\mu} \tilde{A}_{\mu} = 0$

$$x^{\mu} A_{\mu} + x^{\mu} \partial_{\mu} \chi = 0$$

$$-x^{\mu} A_{\mu} = [x^{\mu} \partial_{\mu}] \chi$$

$$\chi = -\frac{1}{[x^{\mu} \partial_{\mu}]} x^{\mu} A_{\mu} \quad (\text{if we can solve this } \chi)$$

* Lorenz gauge

$$\partial_{\mu} \tilde{A}^{\mu} = 0 = \partial_{\mu} A^{\mu} + \square \chi = 0$$

$$\text{Box } \square \chi = -\partial_{\mu} A^{\mu}$$

$$\partial_{\mu} F^{\mu\nu} = -\mu_0 J^{\nu} \rightarrow \square A^{\nu} = -\mu_0 J^{\nu}$$

$$= (\partial_{\mu} \partial^{\mu} A^{\nu} - \partial_{\mu} \partial^{\nu} A^{\mu})$$

$$\square = \eta^{\mu\nu} \partial_{\mu} \partial_{\nu} = -\frac{1}{c^2} \frac{\partial^2}{\partial t^2} + \nabla^2$$

$$A^{\mu}(x) = A^{\mu}_{\text{homogen}}(x) - \frac{1}{\square} \mu_0 J^{\mu}(x)$$

$$\therefore \square A^{\mu}_{\text{homog}}(x) = 0$$

Define Green's Fn. : $\square G(x-x') = \delta^{(4)}(x-x')$
 $= \delta(t-t') \delta^{(3)}(\underline{x}-\underline{x}')$

$\hookrightarrow$ Then, $G(x, x') = \frac{\delta^{(4)}(x-x')}{\square}$

(3)

$$\therefore A^\mu(x) = A^\mu_{\text{homog}}(x)$$

$$-\mu_0 \int G(x, x') J^\mu(x') d^4x'$$

$$\square A^\mu(x) = 0 - \mu_0 \int d^4x' (\square G(x, x')) J^\mu(x')$$

$$= -\mu_0 \int d^4x' (\delta^{(4)}(x-x')) J^\mu(x')$$

$$= -\mu_0 J^\mu(x)$$

* Retarded Green's fn. $G_R(x, x') = 0$

Causality

↑
If $x^0 < x'^0$

↑
PAST (in time)

(Heaviside fn.)

* Theta fn.

$$G_R(x, x') = \theta(x^0 - x'^0) \Delta(x - x')$$

$$\theta(x^0) = \begin{cases} 1 & \text{if } x^0 > 0 \\ 0 & \text{if } x^0 < 0 \end{cases}$$

Theta Fn.

Let

$$\square \Delta(x - x') = 0 \rightarrow \text{then}$$

$$\square G_R = \left(-\frac{\partial^2}{\partial (x^0)^2} + \nabla^2 \right) \theta(x^0 - x'^0) \Delta(x - x')$$

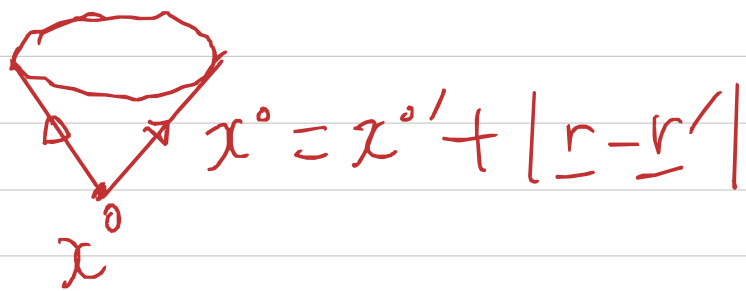
$$= -\frac{\partial^2}{\partial (x^0)^2} \theta(x^0 - x'^0) \Delta(x - x')$$

$$+ \delta(x^0 - x'^0) \frac{\partial \Delta(x - x')}{\partial x^0}$$

$$\therefore \Delta = -\frac{1}{4\pi |\underline{r} - \underline{r}'|} \delta(x^0 - x'^0 - |\underline{r} - \underline{r}'|)$$

$$G_R(x, x') = -\frac{\theta(x^0 - x'^0)}{4\pi |\underline{r} - \underline{r}'|} \delta(x^0 - x'^0 - |\underline{r} - \underline{r}'|)$$

(Light-cone)



$$A^\mu(x) = A^\mu_{\text{homog.}}(x)$$

$$+ \mu_0 \frac{1}{4\pi} \int_{-\infty}^{x^0} dx'^0 \int d^3\underline{r}' \frac{\delta(-x'^0 + \underline{x}^0 - |\underline{r} - \underline{r}'|)}{|\underline{r} - \underline{r}'|}$$

$$\times J^\mu(x'_0, \underline{r}')$$

$$= A^\mu_{\text{homog}}(x) + \frac{\mu_0}{4\pi} \int d^3\underline{r}' \frac{J^\mu(x^0 - |\underline{r} - \underline{r}'|, \underline{r}')}{|\underline{r} - \underline{r}'|}$$

*Action for Maxwell?

(4)

$$-\frac{1}{2} \partial_\mu \phi \partial^\mu \phi$$

$$\partial^\mu \phi \Rightarrow F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

↑
Scalar

$$S = \int d^4x \left[-\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} \right]$$

$$= \int d^4x \left[-\frac{1}{4\mu_0} \left((\partial_\mu A_\nu - \partial_\nu A_\mu) (\partial^\mu A^\nu - \partial^\nu A^\mu) \right) \right]$$

$$= \int d^4x \left(-\frac{1}{4\mu_0} \right) \left(\partial_\mu A_\nu \partial^\mu A^\nu - \partial_\mu A_\nu \partial^\nu A^\mu - \partial_\nu A_\mu \partial^\mu A^\nu + \partial_\nu A_\mu \partial^\nu A^\mu \right)$$

$$= \int d^4x \left(-\frac{1}{2\mu_0} \right) \left(\partial_\mu A_\nu \partial^\mu A^\nu - \cancel{\partial_\mu A_\nu \partial^\nu A^\mu} \right)$$

because ↗

$$\partial^\nu A_\nu \partial_\mu A^\mu = 0 \quad \text{in Lorenz gauge}$$
$$= \partial_\mu A_\nu \partial^\nu A^\mu$$

$$\text{Then, } S = \int d^4x \left(-\frac{1}{2\mu_0} \right) \left[\partial_\mu A_\nu \partial^\mu A^\nu \right]$$

Maxwell : $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$

$$\partial^\mu F_{\mu\nu} = -\mu_0 J_\nu$$

$$S = \int d^4x \left[-\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} + A_\mu J^\mu \right]$$

$$A_\mu \rightarrow A_\mu + \delta A_\mu$$

$$S = \int d^4x \left[-\frac{1}{4\mu_0} \eta^{\mu\alpha} \eta^{\nu\beta} F_{\mu\nu} F_{\alpha\beta} + J^\mu A_\mu \right]$$

$$\delta S = \int d^4x \left[-\frac{1}{4\mu_0} \eta^{\mu\alpha} \eta^{\nu\beta} (\delta F_{\mu\nu}) F_{\alpha\beta} \right. \\ \left. - \frac{1}{4\mu_0} \eta^{\mu\alpha} \eta^{\nu\beta} F_{\mu\nu} \delta(F_{\alpha\beta}) + J^\mu \delta A_\mu \right]$$

$\nearrow \alpha, \beta \rightarrow \mu, \nu$

$$= \int d^4x \left[-\frac{1}{2\mu_0} F^{\mu\nu} \delta(F_{\mu\nu}) + J^\mu \delta A_\mu \right]$$

$$\delta F_{\mu\nu} = \partial_\mu (\delta A_\nu) - \partial_\nu (\delta A_\mu)$$

$$F^{\mu\nu} \delta F_{\mu\nu} = F^{\mu\nu} \partial_\mu (\delta A_\nu) - F^{\nu\mu} \partial_\nu (\delta A_\mu)$$

$$= F^{\mu\nu} \partial_\mu (\delta A_\nu) + F^{\mu\nu} \partial_\mu (\delta A_\nu)$$

$$= 2 F^{\mu\nu} \partial_\mu (\delta A_\nu)$$

$$F_{\mu\nu} = -F_{\nu\mu}$$

$$\delta S = \int d^4x \left[-\frac{1}{\mu_0} F^{\mu\nu} \partial_\mu (\delta A_\nu) + J^\nu \delta A_\nu \right]$$

$$= \text{Boundary term} + \int d^4x \left[-\frac{1}{\mu_0} \partial_\mu F^{\mu\nu} + J^\nu \right] \delta A_\nu$$

$\parallel$
 0

$$\therefore \partial_\mu F^{\mu\nu} = -\mu_0 J^\nu$$

(2)

Single massive charged particle:

$$S = -m \int d\lambda \sqrt{-\eta_{\mu\nu} \frac{dX^\mu}{d\lambda} \frac{dX^\nu}{d\lambda}}$$

$$+ q \int d\lambda A_\mu(x) \frac{dX^\mu}{d\lambda}$$

Linear in A_μ

Writing the particle in the lang. of fld. theory:

x^μ World Line (1D curve in Sp-Tm.)
 $x^\mu(\lambda)$

Unity

$$\text{Unit op. : } 1 = \int d^4x \delta^{(4)}(x - X(\lambda))$$

$$= \int d^4x \delta(x^0 - X^0(\lambda)) \delta^{(3)}(x^i - X^i(\lambda))$$

($i = 1, 2, 3$)

Action for a Maxwell coupled to a charged particle.

$$S = \int d^4x \left[-\frac{1}{4\mu_0} F_{\mu\nu}(x) F^{\mu\nu}(x) \right.$$

$$- m \int \delta^{(4)}(x - X(\lambda)) \sqrt{-\eta_{\mu\nu} \frac{dX^\mu}{d\lambda} \frac{dX^\nu}{d\lambda}} d\lambda$$

q "picks out" $x = X(\lambda)$

$$+ q \int d\lambda \delta^{(4)}(x - X(\lambda)) A^\mu(x) \frac{dX^\mu}{d\lambda} \Big]$$

$$= \int d^4x \left[-\frac{1}{4\mu_0} F_{\mu\nu}(x) F^{\mu\nu}(x) \right.$$

$$- m \int \delta^{(4)}(x - X(\lambda)) \sqrt{-\eta_{\mu\nu} \frac{dX^\mu}{d\lambda} \frac{dX^\nu}{d\lambda}} d\lambda$$

q "picks out" $x = X(\lambda)$

$$+ q A^\mu(x) \int d\lambda \delta^{(4)}(x - X(\lambda)) \frac{dX^\mu}{d\lambda} \Big]$$

Current den. $J^\mu(x)$ is
 Linear in $A_\mu(x)$.

$$\hookrightarrow J^\mu(x) = q \int d\lambda \delta^{(4)}(x - X(\lambda)) \frac{dX^\mu}{d\lambda}$$

(3)

For many particles,

$$J^\mu(x) = \sum_I q_I \int d\lambda \delta^{(4)}(x - X_I(\lambda)) \frac{dX_I^\mu}{d\lambda}$$

Gauge choice $\lambda = x^0 = t$

$$J^0 = \rho_e = \sum_I q_I \int dt \delta(t - X_I(t)) \times \delta^{(3)}(\underline{r} - X_I(\underline{r}))$$

$$\rho_e = J^0 = \sum_I q_I \delta^{(3)}(\underline{r} - X_I(t))$$

$$S = \int d^4x \left[-\frac{1}{4\mu_0} F_{\mu\nu}(x) F^{\mu\nu}(x) \right.$$

$$- \sum_I q_I \int d\lambda \delta^{(4)}(x - X_I(\lambda)) \sqrt{-\eta_{\mu\nu} \frac{dX_I^\mu}{d\lambda} \frac{dX_I^\nu}{d\lambda}} d\lambda$$

$$+ \sum_I q_I A^\mu(x) \int d\lambda \delta^{(4)}(x - X_I(\lambda)) \frac{dX_I^\mu}{d\lambda} \Big]$$

↓ "picks out" $x = X_I(\lambda)$

* Abelian Higgs Model

Charged spin-0 particles (1st quantized)

= complex scalar flds (2nd quantized)

$$\Phi = |\Phi| e^{i \text{Arg}(\Phi)}$$

↑
argument

Under gauge transf. $A_\mu \rightarrow A_\mu + \partial_\mu \chi(x)$

$$\Phi \rightarrow \Phi e^{i q \chi}$$

↑

$$Z^* Z = 1$$

Unitary grp: $U(1)$

④

Maxwell Theory $\rightarrow$ Gauge theory of the gauge group $U(1)$

Electroweak $\rightarrow$ $SU(2) \times U(1)$
 W, Z bosons (Weak force) $\uparrow$ photon (EM force)

Standard Model $\rightarrow$ $SU(3) \times SU(2) \times U(1)$
 Strong Force (nucleons)

$$\Phi(x) \rightarrow \Phi(x) e^{iq\chi(x)}$$

$$\partial_\mu \Phi(x) \rightarrow e^{iq\chi(x)} \partial_\mu \Phi(x) + iq \partial_\mu \chi(x) e^{iq\chi(x)} \Phi(x)$$

Covariant Deriv. : $D_\mu \Phi = \partial_\mu \Phi - iq A_\mu \Phi$

$$D_\mu \Phi(x) \rightarrow \partial_\mu (e^{iq\chi(x)} \Phi(x)) - iq (A_\mu + \partial_\mu \chi) (e^{iq\chi(x)} \Phi(x))$$

$$= e^{iq\chi} \partial_\mu \Phi + \cancel{iq \partial_\mu \chi e^{iq\chi} \Phi} - iq A_\mu e^{iq\chi} \Phi - \cancel{iq \partial_\mu \chi e^{iq\chi} \Phi}$$

$$= e^{iq\chi} D_\mu \Phi(x)$$

(5)

* Action for Abelian-Higgs model
(-+++)

$$S = \int d^4x \left[-\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} - |D_\mu \Phi|^2 - V(\Phi^* \Phi) \right]$$

↓

Action is Real $\in \mathbb{R}$

↻
Real

↑
($\square \Phi^2$)

$$|D_\mu \Phi|^2 = (D_\mu \Phi)^* (D_\mu \Phi)$$

$$\Phi \rightarrow e^{iqx} \Phi \quad \& \quad \Phi^* \rightarrow e^{-iqx} \Phi^*$$

∴ $\Phi^* \Phi$ is invariant

$$\Phi^* \Phi \rightarrow e^{-iqx} e^{iqx} \Phi^* \Phi = \Phi^* \Phi$$

Covariant Deriv,

$$D_\mu \Phi \rightarrow e^{iqx} D_\mu \Phi$$

$$(D_\mu \Phi)^* \rightarrow e^{-iqx} (D_\mu \Phi)^*$$

$$(D_\mu \Phi)^* (D_\mu \Phi) \rightarrow (D_\mu \Phi)^* (D_\mu \Phi) \quad \text{is invariant}$$

↳ Simple Generalization

$$S = \int d^4x \left[\frac{f(\Phi^* \Phi)}{4\mu_0} F_{\mu\nu} F^{\mu\nu} - g(\Phi^* \Phi) |D_\mu \Phi|^2 - V(\Phi^* \Phi) \right]$$

↳ Modified Theory

NON-Renormalizable

↳ gravity is

non-renormalizable

Renormalizability in QFT

$$V(\Phi^* \Phi) = m^2 \Phi^* \Phi + \lambda (\Phi^* \Phi)^2$$

SM EFT (Std Model EFT)